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Immunodominant T cell epitopes preferentially targeted in multiple individuals are the critical element of successful vaccines and targeted immunotherapies. However, the underlying principles of this “convergence” of adaptive immunity among different individuals remain poorly understood. To quantitatively describe epitope immunogenicity, here we propose a supervised machine learning framework generating probabilistic estimates of immunogenicity, termed “immunogenicity scores,” based on the numerical features computed through sequence-based simulation approximating the molecular scanning process of peptides presented onto major histocompatibility complex (MHC) by the human T cell receptor (TCR) repertoire. Notably, overlapping sets of intermolecular interaction parameters were commonly utilized in MHC-I and MHC-II prediction. Moreover, a similar simulation of individual TCR-peptide interaction using the same set of interaction parameters yielded correlates of TCR affinity. Pathogen-derived epitopes and tumor-associated epitopes with positive T cell reactivity generally had higher immunogenicity scores than non-immunogenic counterparts, whereas thymically expressed self-epitopes were assigned relatively low scores regardless of their immunogenicity annotation. Immunogenicity score dynamics among single amino acid mutants delineated the landscape of position- and residue-specific mutational impacts. Simulation of position-specific immunogenicity score dynamics detected residues with high escape potential in multiple epitopes, consistent with known escape mutations in the literature. This study indicates that targeting of epitopes by human adaptive immunity is to some extent directed by defined thermodynamic principles. The proposed framework also has a practical implication in that it may enable to more efficiently prioritize epitope candidates highly prone to T cell recognition in multiple individuals, warranting prospective validation across different cohorts.

Keywords: T cell epitope, T cell receptor repertoire, immunogenicity, escape mutation, machine learning

INTRODUCTION

T cell epitopes bound to major histocompatibility complex [MHC; also called the human leukocyte antigen (HLA) in humans] molecules activate T cells to initiate subsequent immunological orchestration (1–3). MHC class I (MHC-I) molecules typically present 8- to 11-aa peptides generated through proteasomal cleavage of intracellular proteins to activate CD8⁺ cytotoxic T lymphocytes (CTLs), whereas MHC class II (MHC-II) molecules with an open-ended binding groove accommodate peptides of more variable length derived from endocytosed proteins to

activate CD4⁺ T helper (Th) cells (4). Evidence suggests that not all peptides presented on MHC molecules are immunogenic, i.e., trigger functional T cell activation (5–7). T cells recognize peptide-MHC (pMHC) complexes by their TCRs, most predominantly via complementarity determining region 3 (CDR3) loops (8–10). However, the determinants of epitope immunogenicity in association with their recognition by T cells remain poorly understood. Given the fact that different individuals have different TCR repertoires, in theory, epitope immunogenicity should differ between individuals. However, there are several examples of immunodominant epitopes that are targeted by the adaptive immunity of different individuals (11–15). Indeed, immunodominant epitopes have already been clinically utilized, for example, in the interferon-gamma release assay, a clinically available peripheral blood assay to determine if the subject has previously been sensitized by *Mycobacterium tuberculosis* (Mtb) (16, 17). To explain this phenomenon, it is plausible to hypothesize that those immunodominant epitopes share some intrinsic patterns which render them more prone to be recognized by the T cell immunity of multiple individuals. Because TCR-epitope interaction is governed by the physicochemical principles like other protein-protein interactions, more immunodominant epitopes are expected to have a higher chance of stronger interaction when scanned by a large set of TCRs. In this scenario, we could utilize TCR sequences as “baits” to probe highly immunodominant epitopes. We assumed that commonly shared TCR sequences among multiple individuals (referred to “public TCR repertoire” hereafter) more likely reflect the hidden patterns for immunodominant epitopes. Indeed, contrary to the long-standing view that TCR repertoire is highly stochastic and individualized, accumulating evidence suggests that the biased generation of TCR repertoire due to non-random rearrangement and positive/negative thymic selections leads to an extensive utilization of a limited subset of possible sequences and larger-than-expected overlaps of repertoires across individuals (18–25). Moreover, it has been shown that even distinct TCR repertoires can convergently recognize a limited set of pathogen-derived immunodominant epitopes (12).

In this context, here we propose that a computational framework mimicking the thermodynamic interactions between pMHC complexes and public TCR clonotypes, termed TCR-peptide contact potential profiling (CPP), generates probabilistic estimates of immunogenicity which effectively recapitulate essential characteristics of T cell immunity and enable quantitative assessment of potential mutational impacts on the dynamics of immunogenicity in a position-specific context. Datasets and in-house codes to reproduce the entire work are made available as the package *Repitope* (<https://github.com/masato-ogishi/Repitope>).

RESULTS

The Concept of Repertoire-Wide TCR-Peptide Contact Potential Profiling

The hypothesis underlying the overall study is that public TCR repertoire is biased toward immunodominant epitopes

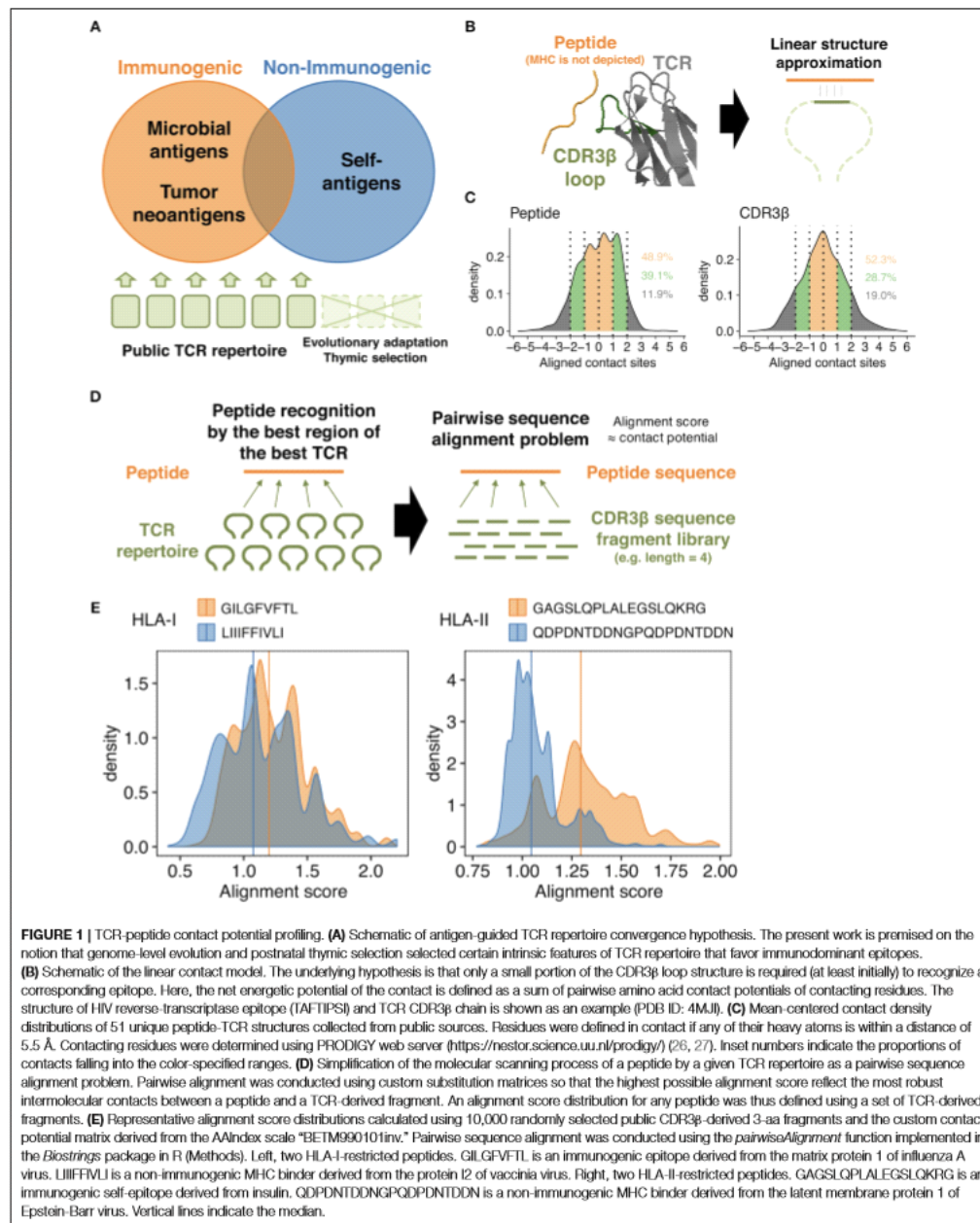
through evolutionary adaptation and negative selection at thymus, and therefore could be used as “baits” to probe peptides with high immunogenic potential (Figure 1A). Because of the sequence-level and structural diversity of TCR and pMHC, it is unfeasible to comprehensively characterize every single TCR-pMHC interaction experimentally. To circumvent this problem, we instead designed a sequence-based simulation framework designed to mimic molecular scanning of pMHCs by TCR repertoire, which we termed repertoire-wide TCR-peptide contact potential profiling (CPP).

The concept of CPP relies on some simplifications. First, we focused on TCR β chain CDR3 (CDR3 β) loops, since this region has the highest genetic variability and is primarily responsible for the interactions with peptides presented onto the MHC grooves, whereas more conserved CDR1 and CDR2 loops typically interact with MHC α -helices (2, 3, 28, 29). Second, we assumed that the recognition of MHC-presented peptides requires only a small portion of the CDR3 β loop, at least initially, that could be approximated to linear structure (Figure 1B). This assumption is based on the relatively low affinities ($K_D = 0.1 \sim >500 \mu\text{M}$) of TCR-peptide interactions compared to other immunoglobulin-like molecules including antibodies, the relatively flat interfaces observed in known TCR-peptide-MHC structures, and the substantial conformational changes induced upon recognition of pMHC complexes (30–33). Moreover, analysis of known TCR-pMHC complex structures using PRODIGY (<https://nestor.science.uu.nl/prodigy/>) (26, 27) revealed that around 50% and more than 80% of the intermolecular contacts are limited within the 3-aa and 5-aa ranges, respectively, in both peptides and CDR3 β regions (Figure 1C and Supplementary Data Sheet 1).

Based on these premises, we first defined the energetic potential of intermolecular contacts between a portion of a peptide and a fragment of a TCR CDR3 β region as the sum of amino acid pairwise contact potentials (Figure 1B). Next, we approximated the “best-match” problem between a peptide and the entire TCR repertoire to the pairwise sequence alignment problem, where a set of fragments derived from a given TCR repertoire are pairwise-aligned to the peptide sequence to identify the “best-aligned” positions that maximize their alignment scores (Figure 1D). Usually, a pairwise sequence alignment algorithm attempts to maximize the alignment score that reflects sequence homology. However, we utilize custom substitution matrices during the alignment process so that the alignment scores reflect the energetic potentials of intermolecular contacts (Methods). To put it briefly, higher alignment scores are considered a hallmark of epitope immunogenicity, as exemplified in Figure 1E. For more accurate immunogenicity prediction through machine learning, we defined CPP features as a set of descriptive statistics (e.g., mean and standard deviation) of the alignment score distribution (Methods).

Datasets and Analysis Workflow

We compiled epitope datasets comprising 21,162 8- to 11-aa HLA-I-restricted and 31,693 11- to 30-aa HLA-II-restricted peptide sequences from various sources (Figure 2, Supplementary Figure 1 and Supplementary Data Sheet 2) (34–42). At this point, we did not filter peptide sequences based on their sequence-level homologies, because even a single



substitution could considerably affect their immunogenicity. Annotations of qualitative T cell assay results (i.e., positive or negative) were retrieved from source databases and coalesced by individual peptides. For those obtained from the Immune Epitope Database (IEDB), for which more detailed annotations about the evidence of T cell reactivity were available, only functional cell-based assay results were considered (Methods). Peptides retrieved from the Los Alamos National Laboratory (LANL) HIV and HCV databases and the TANTIGEN database were assumed to be positive *a priori*. In this manner, 21,162 and 31,693 HLA-I and HLA-II-restricted peptides were identified, respectively, of which 1,873 (8.9%) and 4,505 (14.2%) had contradicting annotations on immunogenicity. In such cases, peptides with at least one positive annotation were considered immunogenic, given that coexistence of negative assay results does not necessarily preclude the possibility of being recognized by any of the TCRs in the public TCR repertoire. Eventually, 6,957 (32.9%) and 16,642 (52.5%) HLA-I and HLA-II-restricted peptides were considered immunogenic, respectively. Collected peptide sequences and annotations are summarized in **Supplementary Data Sheet 2**.

Next, we extracted 23,006,555 unique CDR3 β sequences from TCR repertoire datasets derived from healthy individuals using MiXCR software (43). We obtained 191,326 unique CDR3 β sequences identified in at least 22 out of 206 (~11%) different datasets. Note that due to limited annotation we were not able to strictly select one repertoire dataset per one donor. However, the aim of collecting CDR3 β sequences from multiple datasets is to delineate a public TCR repertoire and use that repertoire as a “bait” for epitope immunogenicity. Therefore, we proceeded to the computation of CPP features using the 191,326 CDR3 β sequences.

TCR-Peptide Interaction Parameters Explaining Epitope Immunogenicity and TCR Affinity

We computed CPP features as well as other features based on peptide physicochemical properties. The 32 and 27 most reproducibly selected features for MHC-I and MHC-II predictions, respectively, were identified (**Figure 2**, **Supplementary Figure 2** and **Supplementary Data Sheet 3**). Only nine of 32 and none of 27 features were derived from peptide physicochemical descriptors, highlighting the indispensable contributions of CPP features on immunogenicity prediction. Common parameter usage patterns for CPP features were observed in MHC-I and MHC-II (**Figures 3A–C**). Notably, features derived from short fragments (i.e., 3-aa and 4-aa) and the longest fragment (i.e., 8-aa and 11-aa in MHC-I and MHC-II, respectively) appeared heavily weighted (**Figure 3A**). Meanwhile, skewness- and kurtosis-derived features showed markedly higher importance values, indicating that distinct repertoire-wide contact potential distributions are the hallmark of immunogenicity (**Figure 3B**). The inverted version of BETM990101 (termed “BETM990101inv”) (44) and other six AAIndex scales

of the highest importance for both MHC-I and MHC-II (**Figure 3C**).

To test if the selected AAIndex scales reflect thermodynamic aspects of TCR-peptide interactions, we next analyzed 82 TCR-peptide complexes with experimentally determined affinities collected from literature (**Supplementary Data Sheet 2**) through a modified framework termed single-TCR contact potential profiling (sCPP). We found that BETM990101inv and other six AAIndex scales important for immunogenicity prediction also generated sCPP features correlating with affinities (**Figures 3D,E** and **Supplementary Table 1**). As expected, multivariate regression did not find any correlation between affinity and sCPP features computed if either peptide or TCR sequences were permuted before sCPP feature computation, indicating the importance of sequence-intrinsic hidden patterns (**Figure 3F**). Collectively, these observations support the idea that our CPP framework recapitulates the essential thermodynamic properties of TCR recognition of pMHC complex which governs both MHC-I and MHC-II systems.

Probabilistic Estimation of Epitope Immunogenicity

We applied machine learning techniques to convert these most predictive features into a unidimensional scale of immunogenicity by averaging the probability estimates from five times repeated 5-fold cross-validations (CVs). The estimates were reasonably consistent across CVs; the normalized standard deviations distributed with medians and interquartile ranges (IQRs) of 8.3% (2.3 to 14.3%) and 5.0% (1.4 to 8.5%) in MHC-I and MHC-II, respectively. We first tested several settings for machine learning. Addition of MHC binding prediction results as features for machine learning only marginally improved the predictive performance for MHC-I and had almost no effect for MHC-II. Reduction of the number of features utilized during machine learning and removal of peptides with high sequence-level homology (i.e., 80% or higher) from the dataset before machine learning resulted in only a modest decrease in predictive performance (**Figure 4A** and **Supplementary Figure 3**). Based on these observations, we decided to utilize the averaged probabilistic estimates derived without MHC-associated features hereafter (termed as “immunogenicity scores”) (**Supplementary Data Sheet 4**). Next, we examined whether immunogenicity scores were applicable to the entire dataset. Indeed, there was an apparent separation of immunogenic epitopes and non-immunogenic MHC binders across various subsets of the datasets. This was indeed the case even when we focused on the epitope data obtained from *ex vivo* assays using human subject-autologous antigen-presenting cells and T cells (**Figure 4B**). Then, we examined the applicability and implication of the immunogenicity score system in various biological contexts. First, immunogenicity score framework was extrapolated to non-human peptide datasets by applying models trained from the human peptide datasets. Overall, the classification performance of extrapolated immunogenicity scores less efficiently distinguished immunogenic epitopes from non-immunogenic binders (**Figure 4C** and **Supplementary Figure 4**). Second, thymically expressed self-epitopes obtained from a previous study (45)

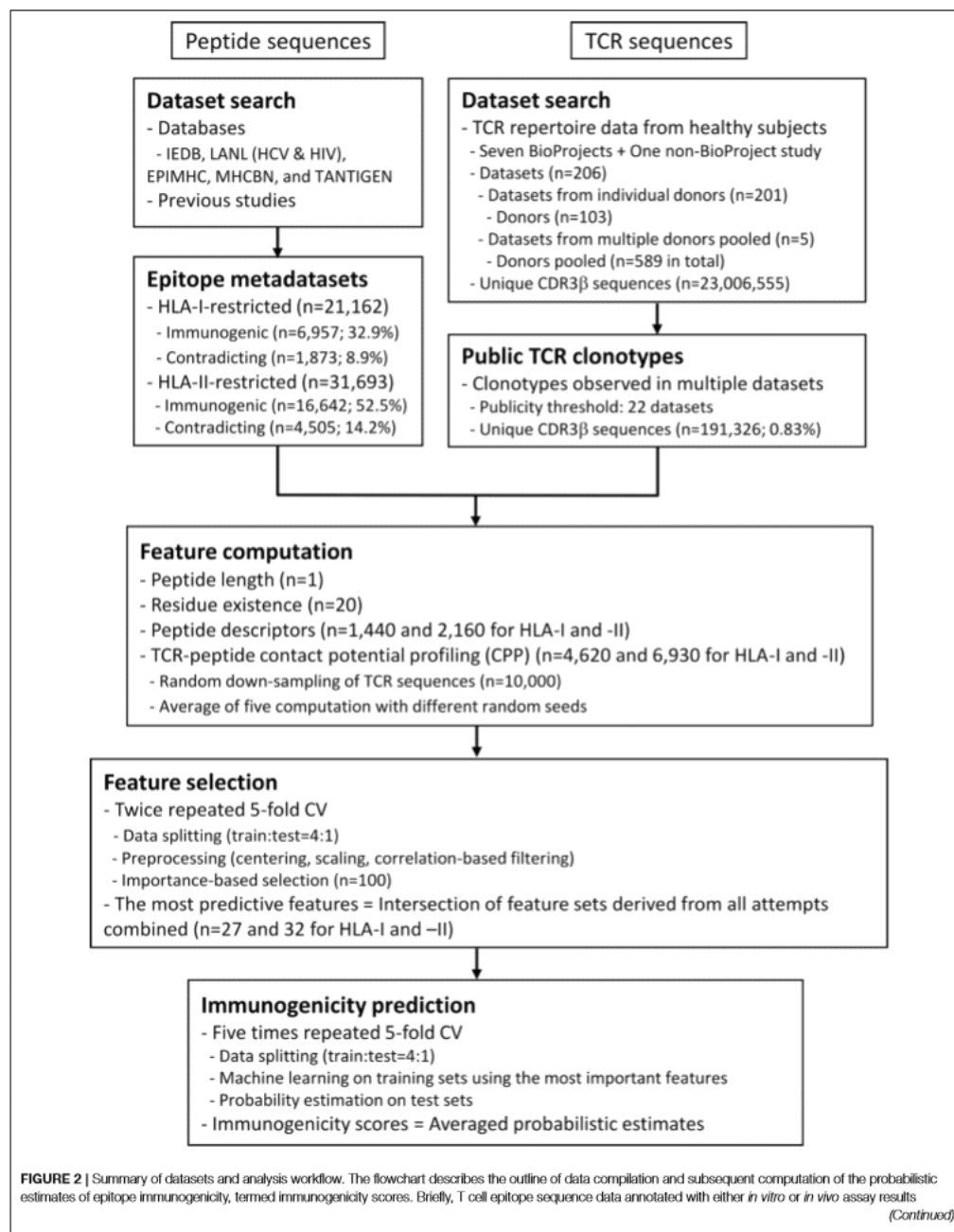
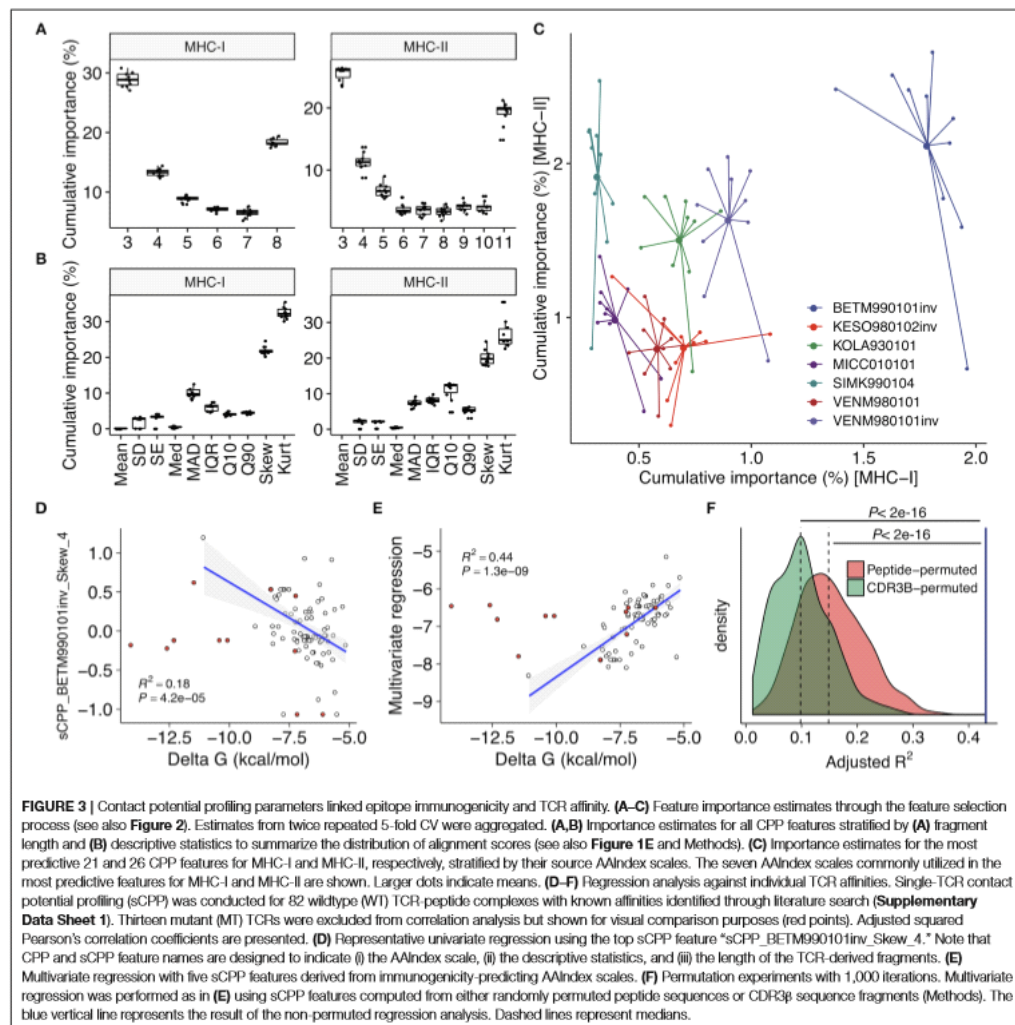


FIGURE 2 | (positive or negative) were retrieved from various public databases. Peptide data were not necessarily accompanied by the restricting MHC metadata. In the meantime, public TCR sequence data were extracted from previous studies. A set of features were computed for each of the peptides, including the CPP features (see also **Figure 1** and Methods). Minimal sets of the most predictive features were selected through the feature selection process through twice repeated 5-fold cross-validation (CV). In other words, the entire dataset was split into five chunks, and machine learning was conducted on a leave-one-chunk-out basis. Feature importance was estimated, and the top 100 features were retained. The consensus of the five feature sets was retained. The entire process was repeated twice, and the final consensus of the feature sets was identified as the most predictive features. Finally, probabilistic estimates of immunogenicity were computed through five times repeated 5-fold CV only using the most predictive features. This procedure ensures that an immunogenicity score for any peptide is an average of five machine-learned classifiers trained without using the target peptide itself.



were compared to the rest of the peptides. Interestingly, based on database annotation, the ratio of immunogenic epitopes was lower among thymic self-peptides than the rest of the peptide data. In addition, even peptides annotated as immunogenic had overall lower scores than non-thymus immunogenic epitopes, but comparable scores to thymically expressed non-immunogenic peptides (Figures 4D,E). Third, immunogenic peptides from various organisms generally had higher scores than non-immunogenic counterparts, with some exceptions such as MHC-I peptides derived from human and Mtb. Meanwhile, immunogenicity scores fit particularly well into IEDB-derived HIV and HCV peptides (Figures 4F,G), and this was also the case in the epitopes from the Los Alamos HIV and HCV immune epitope databases (Figure 4H). Likewise, MHC-I-restricted tumor-associated epitopes obtained from TANTIGEN database also exhibited higher scores than non-immunogenic peptides from various origins (Figure 4I). Tumor-associated antigens yielding peptides of high predicted immunogenicity included CTG1B and MAGB1 (Figure 4J). As a comparison, we tested the MHC-I immunogenicity prediction tool currently available in IEDB (<http://tools.iedb.org/immunogenicity/>) (34) using the same MHC-I epitope dataset but only found marginal predictive power (Supplementary Figure 5). Note that there is no tool available for MHC-II immunogenicity prediction to compare. Collectively, immunogenicity scores are the quantitative metric of the probability of functional T cell recognition, and would provide another layer of criteria for robust prioritization of MHC-I and MHC-II epitopes with high expected immunogenicity.

Dynamics of Epitope Immunogenicity by Single Amino Acid Alterations

Single amino acid mutations can affect peptide immunogenicity bidirectionally, namely, acquisition and loss of immunogenicity, in sequence space (Figure 5A). We termed the discordance of annotated immunogenicity between neighbors, i.e., single-aa mutants, as “immune transition,” and defined peptides with evidence of immune transition as “transitional.” Note that we cannot confidently define “non-transitional” peptides solely based on our datasets, because the lack of evidence of immune transition could simply be the lack of experimental data. We identified 1,360 and 976 transitional peptides for MHC-I and MHC-II, respectively. We noted that immunogenicity scores were less applicable to transitional peptides (Figures 5B,C and Supplementary Figure 6), which may result from suboptimal machine learning due to the dearth of transitional peptide data. To test if the integration of neighbor information improves immunogenicity prediction for transitional peptides, we next constructed neighbor networks for transitional peptides using all possible single-aa mutants simulated computationally ($N = 232,723$ and $292,437$ for MHC-I and MHC-II, respectively), and compute immunogenicity scores by extrapolation. We termed this process as “*in silico* mutagenesis.” As expected, the averaged extrapolated immunogenicity scores more accurately predicted the immunogenicity of transitional peptides (Figures 5B,C). To further characterize the position- and residue-specific mutational impacts, we next focused on the immunogenicity dynamics between neighbor pairs. We screened our datasets to

identify 6,179 and 3,076 MHC-I and MHC-II-restricted single-aa mutated peptide pairs. Immunogenicity scores changed more dynamically in transitional pairs (Figure 5D). This observation was also the case when we focused on the neighbor pairs of the same organismal origins (Figure 5E). We also noted that scores changed more eminently among MHC-I-restricted neighbor pairs with mutations in their anchor residues (Figure 5F). This is an interesting observation since our framework does not utilize any positionally defined features. To gain insights into the residue-specific mutational impacts, we next analyzed 2,580,890 and 3,101,092 neighbor pairs from all simulated single-aa mutants of transitional MHC-I- and MHC-II-restricted peptides, respectively (Figures 5G,H). We utilized simulated peptide data because neighbor pairs identified in experimentally tested epitope datasets were too sparsely distributed and likely biased. Heatmap clustering analysis revealed residue-specific impacts on immunogenicity that are partially interpretable with known physicochemical properties of amino acid residues [e.g., hydrophobic/aliphatic (V, L, and I), negative (D and E), and aromatic (F and W)]. However, some exceptions were also notable [e.g., positive (K and R) in MHC-I]. Collectively, our systematic characterization delineates the landscape of bidirectional effects of single-aa mutations on immunogenicity in a position- and residue-dependent context.

Inference of Escape Mutations From Simulated Neighbor Network Architecture

Escape mutations are the principal obstacle for vaccine development. Neither reactivity of the specific T cell clones tested *in vitro* against target epitopes or immune activation after administration *in vivo* does not guarantee a sustainable immune response in the real-world setting because of the possibility of deleterious escape mutations. We tackled this fundamental problem by quantifying position-specific “escape potentials” from neighbor network architecture. Figure 6A shows two representative neighbor networks constructed from observed neighbors of immunodominant influenza A virus epitopes GILGFVFTL (GIL) and SRYWAIRTR (SRY). Apparently, GIL seems more robustly immunogenic than SRY because the Cluster4 in the SRY network was enriched with non-immunogenic peptides, suggesting a path for escaping. *In silico* mutagenesis followed by neighbor network clustering analysis revealed position-specific dynamics of predicted immunogenicity (Figure 6B). P6 and P4 were indicated to have the most substantial impact on escaping for GIL and SRY, respectively. Intriguingly, P6 of GIL peptide has been shown to be involved in two hydrogen bonds with two distinct TCR clones F50 and JM22 (46). Furthermore, five P4 mutants of SRY peptide have been shown to result in an undetectable level of cytotoxicity by at least three CTL clones (47). Encouraged by these observations, we next examined two MHC-II-restricted epitopes. GAGSLQPLALEGSLQKRG (GAG) is a self-epitope derived from insulin. Our simulation indicated that P8-11 and P15-16 were required for its immunogenicity. Strikingly, immunodominant epitope LALEGLQK has been localized previously, showing a complete match with the prediction (48). It is worth mentioning that LALEGLQK is usually degraded proteolytically during the maturation of the insulin molecule.

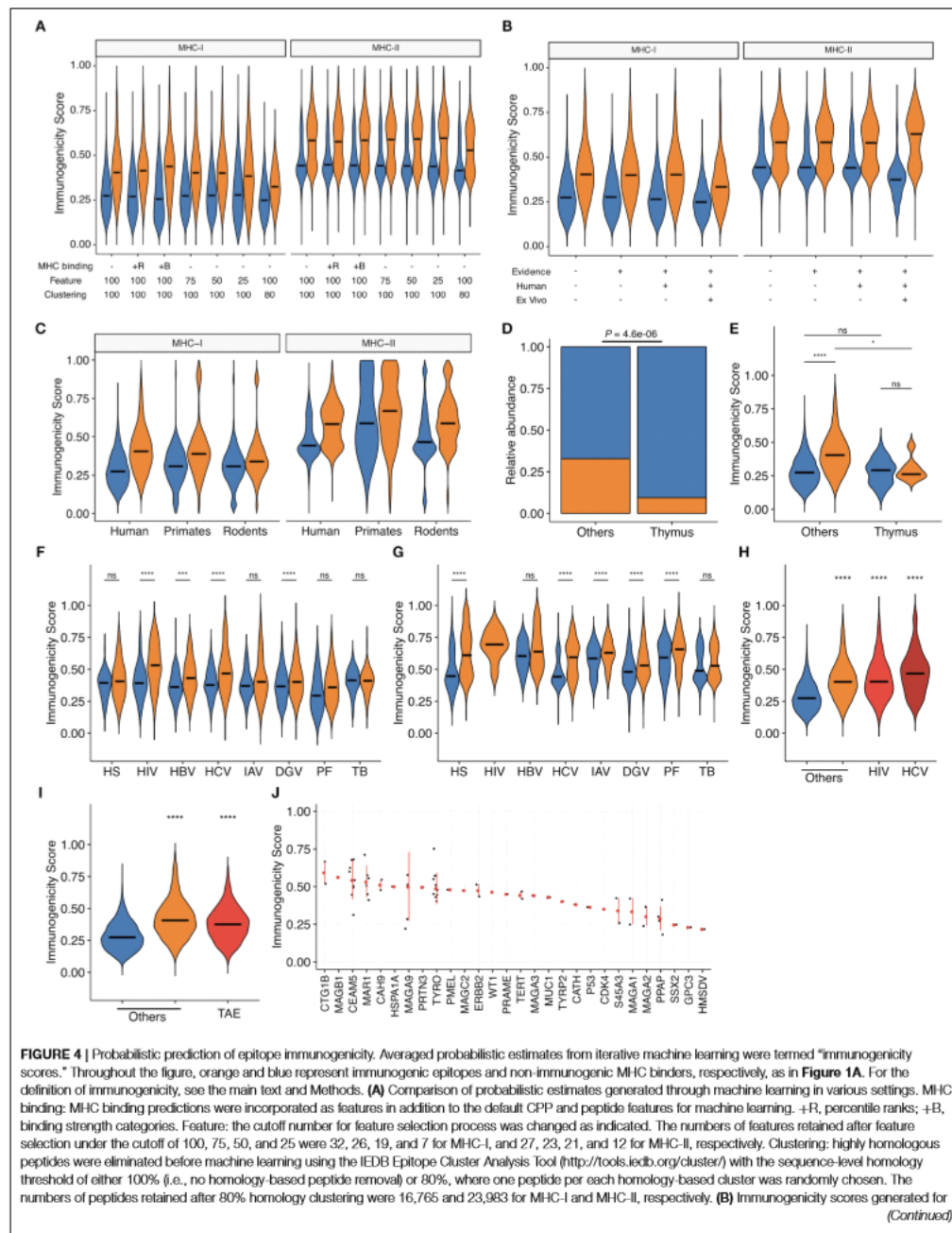


FIGURE 4 | various subsets of our epitope datasets. Machine learning was conducted with 32 and 27 pre-selected features for MHC-I and MHC-II, respectively, but without predicted MHC binding information. Evidence: IEDB-derived peptides with functional T cell assay evidence (i.e., epitopes from non-IEDB sources lacking T cell assay annotation were excluded); Human: epitopes tested in human (i.e., epitopes tested only in transgenic animal models were excluded); ex vivo: IEDB-derived peptides annotated from direct ex vivo assays using autologous effector and antigen-presenting cells in human. The numbers of peptides included after applying each filter were 21,162, 13,058, 11,084, and 6,048 for MHC-I, and 31,693, 31,693, 30,188, and 2,023 for MHC-II, respectively. **(C)** Immunogenicity scores of human epitopes and extrapolated scores for primate and rodent epitopes ($N = 411$ and 8,756 for MHC-I and 76 and 8,445 for MHC-II, respectively). **(D,E)** Distributions of **(D)** annotated immunogenicity and **(E)** immunogenicity scores for the human thymus MHC-I peptidome (45). **(F,G)** Immunogenicity scores among IEDB-derived **(F)** MHC-I- and **(G)** MHC-II-restricted peptides of various origins. HS, *homo sapiens*; HIV, human immunodeficiency virus type 1; HBV, hepatitis B virus; HCV, hepatitis C virus; IAV, influenza A virus; DGV, dengue virus; PF, *Plasmodium falciparum*; TB, *Mycobacterium tuberculosis*. **(H)** Immunogenicity scores among MHC-I-restricted peptides from various sources. HIV, Los Alamos HIV Epitope Database; HCV, Los Alamos HCV Epitope Database. **(I,J)** Immunogenicity scores of MHC-I-restricted tumor-associated epitopes (TAEs) identified from TANTIGEN database. In **(J)**, TAEs were grouped by their source tumor antigens. Red points and bars represent median and interquartile ranges, respectively. In **(A-D)** and **(F-H)**, bars represent medians. In **(E-I)**, ns, not significant; * $P < 0.05$; *** $P < 0.001$; **** $P < 0.0001$.

Since our framework does not integrate factors affecting intracellular peptide processing and presentation, appropriate peptide presentation onto MHC should be considered a prerequisite for performing immunogenicity score-based analysis. PKGQTGEPLAGFKGEQGP (PKG) is another self-epitope derived from collagen. The simulation indicated that P10, P11, P13, and less evidently, P14 and P16 were required for its immunogenicity. Indeed, mutations at P10, P13, and P16 have been shown to almost diminish the T cell reactivity collected from pre-immunized transgenic mice, and mutations at P14 partially impaired the response (49).

One possibility is that the epitope- and position-specific dynamics of simulated immunogenicity scores originated from overfitting of machine-learned models to the single amino acid variants of the target peptide of interest in the source dataset. To test this possibility, we repeated the entire analysis from machine learning to the computation of immunogenicity scores to neighbor network simulation using datasets in which both the target peptide of interest and all of its neighbors were removed. Indeed, the most prominent escape-prone positions for every peptide tested, namely, P6 of GIL, P4 of SRY, P8-P14 of GAG, and P10/P13 of PKG, were still identifiable regardless of the presence or absence of neighbor peptides in the source datasets (Figure 7). Meanwhile, it is also notable that some positions such as P7 of SRY and P16 of PKG became less prominent after removal of neighbor peptides from the source datasets, indicating that epitope-specific patterns could also be incorporated from their neighbors as well as some pan-epitope principles of immunogenicity learned from a general set of epitopes.

The methodology to computationally prioritize epitopes of high immunogenicity and low escape potential would have enormous practical implication for developing immunotherapies applicable to global populations. To this end, we introduced another metric termed “escape potential,” which was defined as the maximal difference of cluster-averaged immunogenicity scores between the cluster of interest (containing the target peptide of interest) and other clusters in the neighbor network constructed from all possible single amino acid variants. A negative escape potential means that the target peptide belongs to the least likely immunogenic cluster. Immunogenicity score-escape potential (IS-EP) two-dimensional plots indicated that, although there was a moderate correlation between IS and EP, considerable variation also existed (Figures 8A,B). Of note, HIV and dengue virus (DGV)-derived immunogenic

epitopes had higher EP than human epitopes compared, and this trend was not observed in non-immunogenic binders (Figure 8C). Two representative HIV-derived MHC-I epitopes with high (GGKKYKL) and low (ITTESIVW) EPs were shown in Figure 8D. Interestingly, there are known naturally occurring escape mutations at P3, P5, and P7 of GGKKYKL (50), of which P3 and P5 appeared consistent with our analysis (Figure 8D). Although we did not find information on escape variants of ITTESIVW from the literature, this epitope represents the most common sequence found in the circulating HIV-1 clade B population in the Los Alamos HIV Sequence Database, suggesting that this epitope may be less prone to undergo mutations (51). Likewise, two representative MHC-II epitopes with high and low Eps derived from *P. falciparum* and two from Mtb were shown in Figures 8E,F, respectively. Indeed, the prevalence of T cell reactivity to the low-EP epitope GKLLSTGLVQNFPTIISK was reported to be 25% in a Kenyan cohort (16). Moreover, 29.4% of South African patients with either latent or active Mtb infection were shown to be positive by ELISpot assay against the epitope VRVAESHGVAAVLFAATAA, C-terminus of which is identical to the low-EP peptide AESHGVAAVLFAATAA (17). Collectively, these observations indicate that the proposed framework would serve as a valuable tool, along with other bioinformatics tools such as intracellular antigen processing prediction and MHC binding prediction, to screen a large set of epitope candidates before experimental validation based on the two newly introduced metrics, namely, the immunogenicity score and escape potential.

DISCUSSION

This study provides a novel viewpoint that targeting of epitopes by human adaptive immunity is actually to some extent predictable by defined thermodynamic principles recapitulating the thermodynamic interaction profiles between MHC-presented peptides and the host TCR repertoire. Although previous studies have suggested the relevance of the physicochemical property of TCR-contact residues of the peptides and population-level frequency of peptide-specific T cell clones to epitope immunogenicity (35, 52), this study is to our knowledge the first to illustrate, through leveraging the largest-ever epitope datasets, quantitative prediction of immunogenicity for each

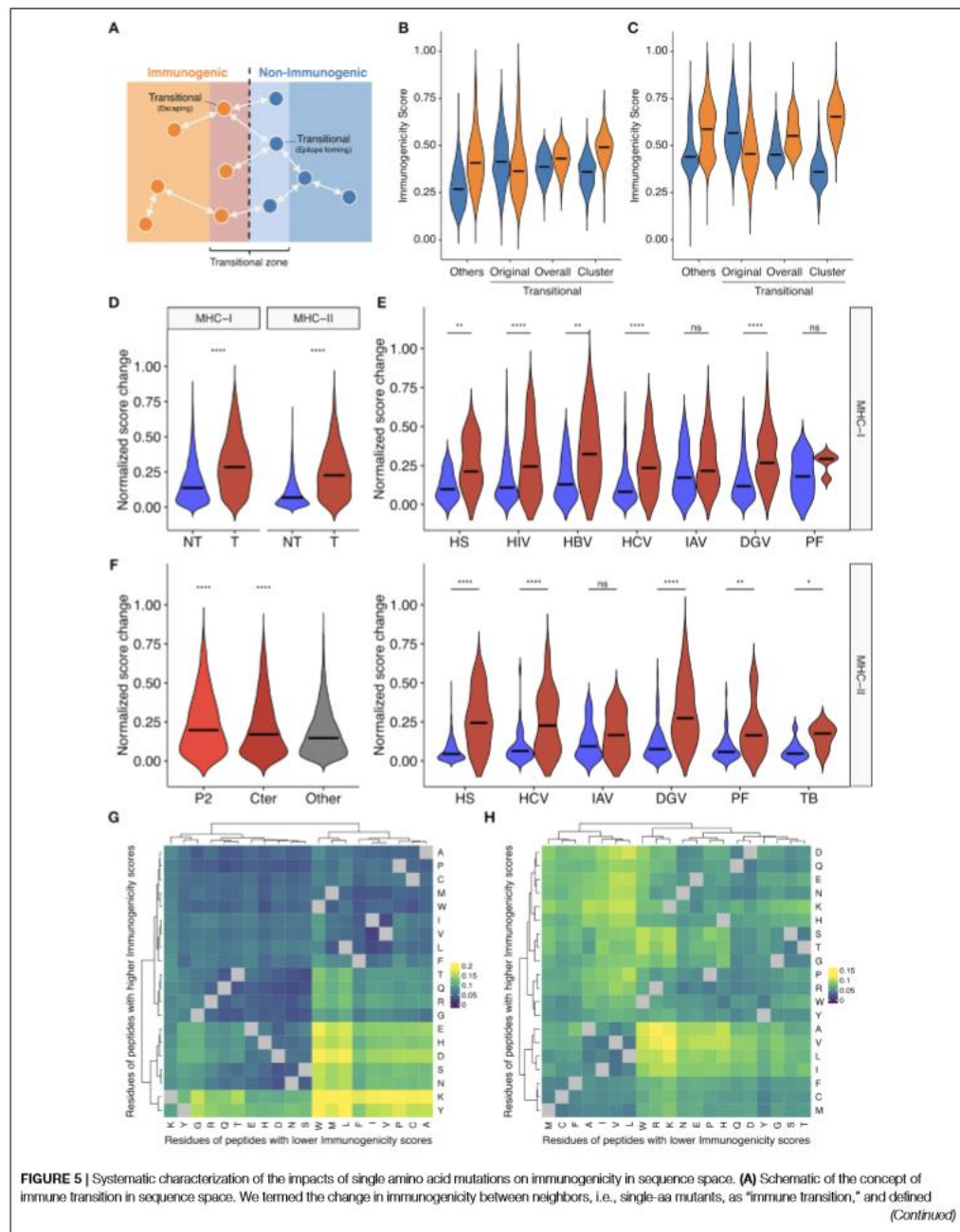
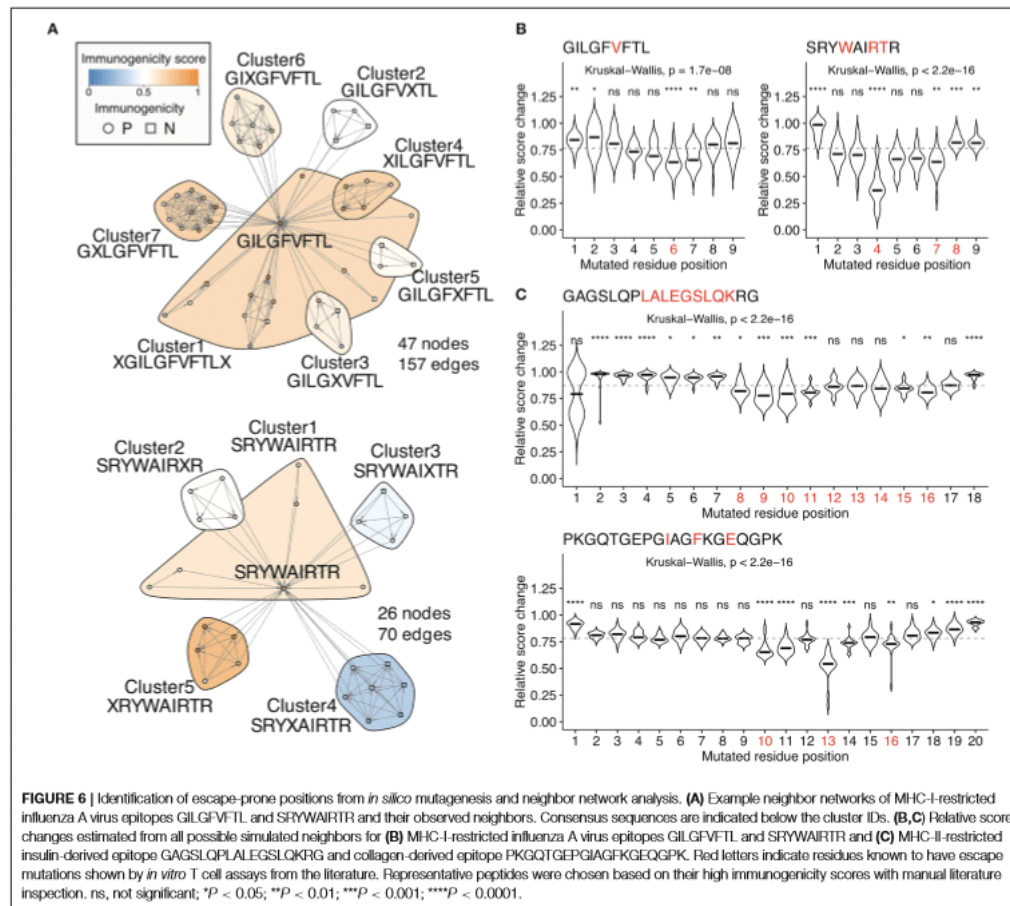


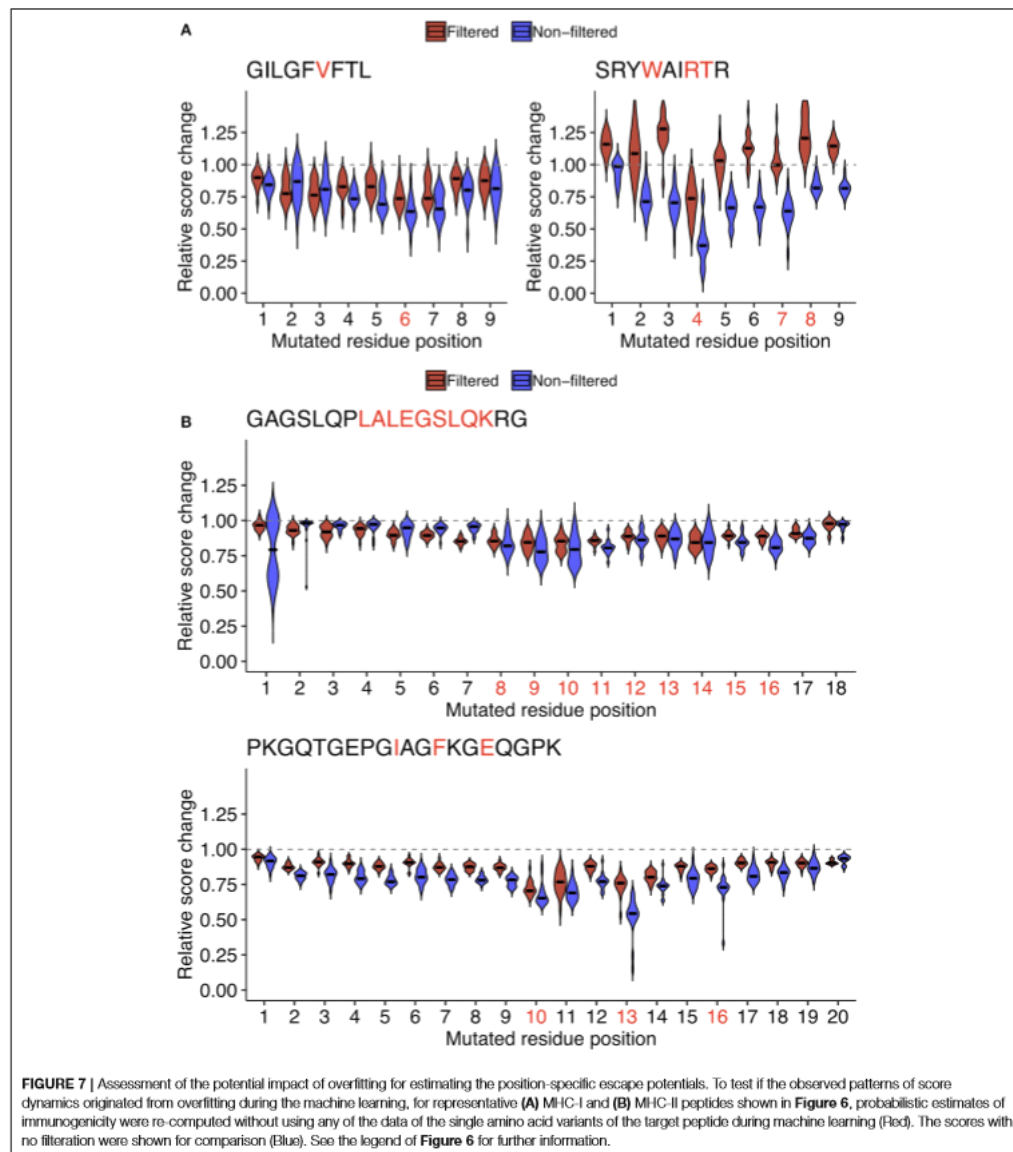
FIGURE 5 | Systematic characterization of the impacts of single amino acid mutations on immunogenicity in sequence space. **(A)** Schematic of the concept of immune transition in sequence space. We termed the change in immunogenicity between neighbors, i.e., single-aa mutants, as "immune transition," and defined (Continued)

FIGURE 5 | peptides with evidence of immune transition as “transitional.” (B,C) Immunogenicity scores among (B) MHC-I- and (C) MHC-II-restricted transitional peptides, i.e., peptides with at least one neighbor of opposite immunogenicity annotation in our dataset. We identified 1,360 and 976 transitional peptides for MHC-I and MHC-II, respectively. We expanded their neighbor networks by computing immunogenicity scores for 232,723 and 292,437 all possible single-aa mutants of transitional MHC-I- and MHC-II-restricted peptides, respectively. Original, original immunogenicity scores; Overall, mean immunogenicity scores of all neighbors. Cluster, mean immunogenicity scores of neighbors assigned to the cluster containing the parent peptide. Others, original immunogenicity scores for peptides with no evidence of immune transition (shown for comparison). Orange and blue denote immunogenic and non-immunogenic peptides, respectively. (D–F) Immunogenicity score dynamics among observed neighbor pairs. We identified 6,179 and 3,076 single-aa mutated peptide pairs for MHC-I and MHC-II, respectively. (D) Score dynamics for all possible peptide pairs, regardless of their origins. (E) Score dynamics for pairs of peptides derived from the same organism. Blue and red denote non-transitional and transitional pairs, respectively. (F) Score dynamics of MHC-I peptide pairs stratified by their mutated positions. Cter, C-terminal residue. (G,H) Heatmap clustering analysis of immunogenicity score dynamics by mutating residue pairs. We generated 2,580,890 and 3,101,092 neighbor pairs from 232,723 to 292,437 all possible single-aa mutants of 1,360 and 976 transitional MHC-I- and MHC-II-restricted peptides, respectively. Color represents median normalized score changes. Gray indicates unobserved residue pairs. In (D–F), ns, not significant; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.



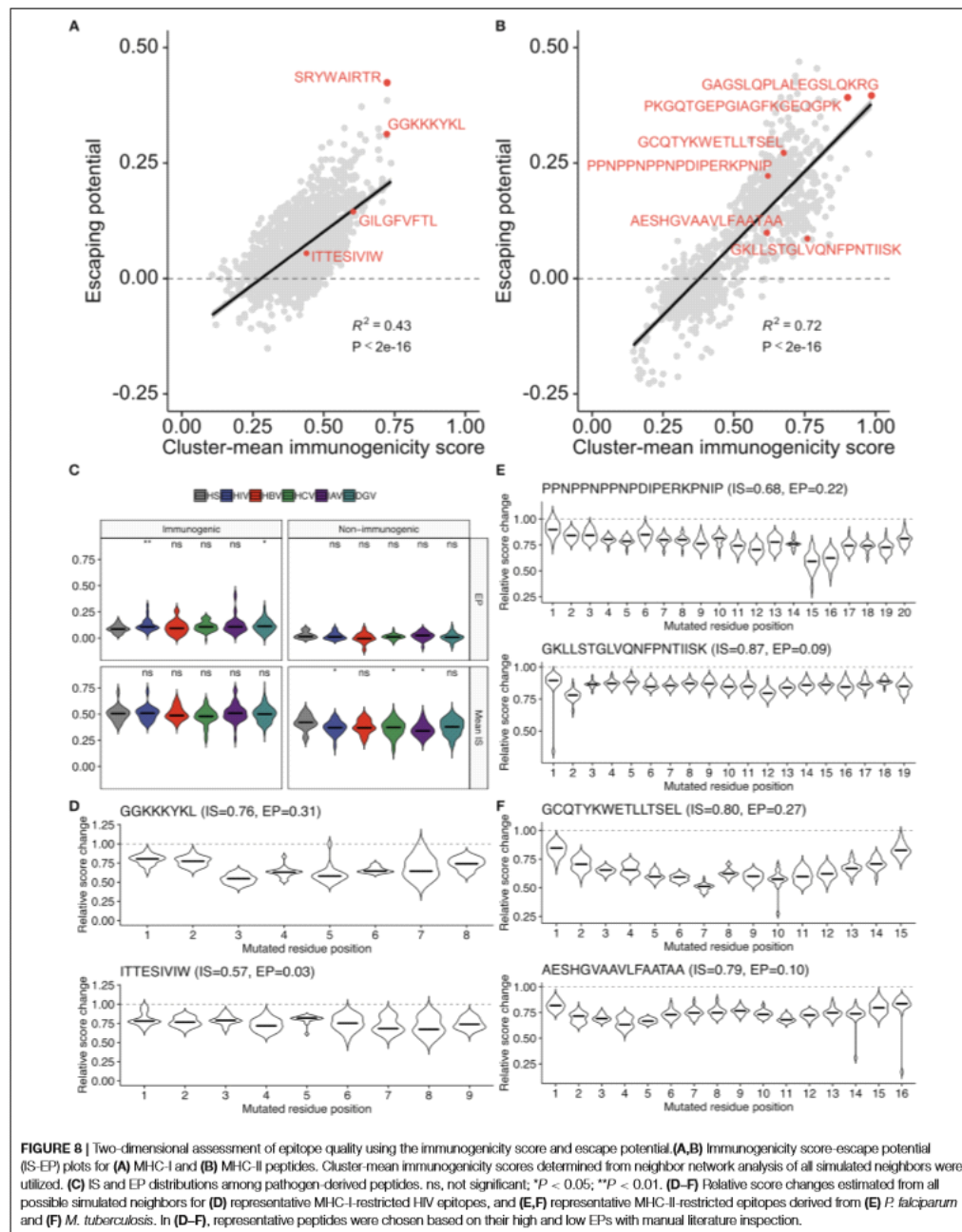
individual epitope with sufficiently high consistency with T cell-based assays. However, it should be noted that our framework is not intended for replacing other bioinformatic tools for

epitope prediction or experimental testing but rather adding an additional filter to expedite the discovery of potent epitopes. For example, our framework does not take into account the



intracellular processing of proteinous antigens, which is taken into account in other tools (53, 54). Moreover, our framework *a priori* takes peptide presentation on MHC for granted; therefore one must test whether the peptides of interest are

presented on MHC by either bioinformatic approach such as NetMHC or experimental verification (55, 56). The MHC allele frequency of the target population should also be taken into consideration if population-level immunogenicity is of interest.



Finally, experimental testing by comparing with established immunodominant epitopes is indispensable as illustrated by others in search of immunodominant Mtb-derived epitopes (57).

Analysis of contact potential profiling (CPP)-based features predictive of epitope immunogenicity is likely to provide novel clues for describing the molecular scanning process of pMHCs by the host TCR repertoire. It is of note that short (i.e., 3-aa and 4-aa) TCR fragments in combination with a restricted set of AAIIndex scales generate features contributing to both MHC-I and MHC-II immunogenicity prediction and also correlating with individual TCR-pMHC affinities because these commonly utilized parameters could potentially reflect some pan-epitope principles governing TCR-pMHC interactions regardless of MHC classes and alleles. The utilization of short fragment features is consistent with the induced fit model that only a limited complementary region of TCR is involved at least in initial engagement (31–33). Meanwhile, the best AAIIndex scale BETM990101 reflects interaction energies between amino acid residues under physiological conditions (44), and hence the inverted version of BETM990101, giving larger values for stronger interresidue contacts, is a reasonable scale for contact potential estimation. Indeed, a positive skewness in the contact potential distribution, implying the coexistence of a small and a large proportion of TCR fragments with high and low contact potentials, respectively, was associated with stronger affinities in our analysis. TCR affinity has been correlated with T cell activation (58), although there remains a debate as to whether other parameters such as avidity, on- and off-rates are also essential (59). Further studies are required to fully elucidate the biological relevance of simulated TCR-peptide contact features quintessential for immunogenicity prediction to the thermodynamic parameters of TCR-pMHC interactions and subsequent T cell activation.

Immunodominance, an immunological phenomenon in which adaptive immunity preferentially targets only a few of the antigenic peptides out of the many others, has been extensively described in multiple cases (15). For instance, only 0.03% of potential MHC-I-presented peptides derived from vaccinia virus account for >90% of the CD8⁺ T cell response in B6 mice (60). Similarly, diverse TCR repertoires specific for two immunodominant viral epitopes have been reported in human (12). Population-level immunity against various immunodominant epitopes is another line of evidence (16, 17). The fact that diverse TCRs can be utilized to achieve protective immunity against immunodominant epitopes in multiple individuals indicates the existence of some epitope-intrinsic qualities guiding “convergent evolution” of human TCR repertoire at both genomic and thymic level. To elucidate the hitherto unknown pan-epitope principles governing productive TCR-peptide contacts, we took an inductive approach using a large set of *in vitro* T cell epitope assay results which may vary depending on the specific T cell clones or TCR repertoires utilized. Our findings indicate that, unexpectedly, a limited set of TCR-peptide contact features reproducibly serve as the principal determinants of T cell reactivity. Moreover, we show that the averaged probabilistic estimates, termed immunogenicity scores, are consistently higher in epitopes with positive results

from various functional T cell reactivity assays. Interestingly, peptides physiologically presented on the thymus have lower immunogenicity scores than non-thymus counterparts, which is consistent with the biased formation of human TCR repertoire owing to negative selection (25, 45). Practically speaking, the probabilistic scoring system would be particularly useful for prioritizing peptide candidates to expedite the development of population-level immunotherapies for various infectious diseases and cancer. However, it is notable that immunogenicity scores do not distinguish immunogenic and non-immunogenic peptides from any origin. A particularly notable exception is Mtb; the Mtb epitopes with positive T cell reactivity were assigned with lower immunogenicity scores compared to those from other pathogens, like the thymically-presented self-epitopes with positive T cell reactivity. This observation implies that, from the T cells' point of view, those Mtb epitopes “look like” non-targetable MHC binders like the self-epitopes presented in the thymus. Note that the immunogenicity score is a repertoire-based probabilistic metric, whereas T cell-based assay determines the presence or absence of individual T cell clones reactive of the peptide of interest. Given the ~70,000 years of co-evolution between Mtb and humans (61), one hypothesis is that Mtb has evolutionarily adapted to human T cell immunity to mimic self-epitopes (again, not in terms of sequence-level homology but from “T cells' perspective”), and only highly distinct T cell clones can recognize those Mtb-derived epitopes. Whether these clones function as pro-inflammatory or regulatory is another layer of a question. Notably, it has been shown that T cell epitopes of Mtb are evolutionary hyperconserved and the bacteria seems to actually benefit from recognition by, rather than evasion from, human T cell immunity (62, 63). Nevertheless, we still believe that our scoring system can be useful to identify potentially immunodominant Mtb-derived epitopes, because the Mtb-derived HLA-I and HLA-II epitopes with the highest scores (KLGGVAVI and DWYSPACGKAGCQTYKWETF), derived from the 65 kDa heat shock protein and the antigen 85B, respectively, have previously been described as potentially immunodominant epitopes (64, 65). Further characterization of T cell reactivity for Mtb epitopes with high and low immunogenicity scores may unveil critical insights into the uniqueness of anti-mycobacterial T cell immunity.

One remarkable feature of our immunogenicity score system is that the score dynamics between single-aa mutants are, at least to some extent, predictive of the mutational impacts on immunogenicity. In order to drive potent and sustainable immunity against highly mutable pathogens such as HIV and HCV, the criteria of high immunogenicity may not be sufficient because of potential escape mutations. By simulating all possible single amino acid variants, our framework allows us to estimate the potential impacts of such escape mutations. It should be noted that this all-mutant strategy leads to overestimation of escape potential, because, in a real-world, some of the amino acid variations are unfeasible due to many factors such as codon usage patterns in the genome of the target organism, decreased replication fitness, loss-of-function of essential enzymes. However, since our primary interest is to obtain epitopes with relatively high immunogenicity and yet with relatively low escape potential, overestimation of

escape potential may not be a critical problem. In contrast, underestimation of escape potential could lead to failure of filtering out candidates that may lose their immunogenicity by only a single mutation. Practically, the proposed two-metrics-based approach of epitope prioritization could expedite the development of vaccines applicable to a wide range of individuals, and could also improve the predictability of the responsiveness to checkpoint blockade immunotherapies when combined with known biomarkers such as tumor mutational burden (66–68).

There are indeed several caveats to be noted in this study in addition to those discussed in the above sections. First, our current model may be suboptimal because only a small fraction of possible sequence space is experimentally covered so far, and there is considerable heterogeneity of the experimental methodologies employed. T cell-based assays of identical readouts (e.g., IFN- γ secretion) using cells obtained from multiple individuals for all candidate epitopes and their single amino acid variants would be ideal for defining a universal scale of the probability of T cell reactivity through machine learning. Second, due to the lack of sufficient annotations, we did not distinguish TCR repertoires from different T cell subsets such as CD8⁺ and CD4⁺ T cells with either pro-inflammatory or regulatory functions. Since the dynamics of TCR repertoires in these subsets are distinct, utilization of subset-specific TCR repertoire may further improve the predictability of epitope immunogenicity. Third, in a real-world setting *in vivo*, we need to take into consideration the spatiotemporal proximity of multiple epitopes simultaneously presented at the focus of inflammation. In such circumstances, the “threshold” of epitope immunogenicity itself is likely altered, and bystander activation also likely contributes to the gross immune outcome. Nevertheless, it is encouraging that the proposed framework predicted many epitopes consistently with previous experimental studies. Further validations both *in vitro* and *in vivo* (in animal models and from observational studies in humans) are warranted. Elucidation of the pan-epitope principles of epitope targeting by human T cell immunity would have significant clinical and biological implications, and we hope that our study serves as a starting point for future investigations.

MATERIALS AND METHODS

Computational Analysis

All computational analyses were conducted using R ver. 3.5.0 (<https://www.r-project.org/>) (69). The latest versions of R packages were consistently used. Compiled datasets and essential in-house functions are available as the R package *Repitope* on GitHub (<https://github.com/masato-ogishi/Repitope>). Other scripts are available upon request.

Peptide Sequence Datasets

HLA-I-restricted peptide sequences of 8-aa to 11-aa lengths with T cell assay results were collected from public databases [Immune Epitope Database (IEDB, as of May 7th, 2018) (42), the best-characterized CTL epitopes from Los Alamos National Laboratory (LANL) HIV Sequence Database (38), LANL HCV Sequence Database (36), EPIMHC (40), MHCBN (37), and

TANTIGEN (39)], and previous publications (34, 35, 41). HLA-II-restricted peptide sequences of 11-aa to 30-aa lengths with T cell assay results were collected from the IEDB database (as of May 7th, 2018). As for peptides retrieved from the IEDB database, only those with evidence of functional T cell response were included. The list of evidence included is as follows: activation, antibody help, CCL2/MCP-1 release, CCL3/MIP-1a release, CCL4/MIP-1b release, CCL5/RANTES release, CXCL10/IP-10 release, CXCL9/MIG release, cytotoxicity, decreased disease, degranulation, disease exacerbation, GM-CSF release, granulysin release, granzyme A release, granzyme B release, IFN γ release, IL-10 release, IL-12 release, IL-13 release, IL-17 release, IL-17A release, IL-1b release, IL-2 release, IL-21 release, IL-22 release, IL-23 release, IL-3 release, IL-4 release, IL-5 release, IL-6 release, IL-8 release, lymphotoxin A/TNF β release, pathogen burden after challenge, perforin release, proliferation, protection from challenge, survival from challenge, T cell-APC binding, T cell help, TGF β release, TNF release, TNF α release, tolerance, tumor burden after challenge, and type IV hypersensitivity (DTH). Peptides presented on non-human MHC molecules were discarded, whereas those presented on HLA molecules in non-human hosts (e.g., transgenic mice) were included. Peptides retrieved from LANL and TANTIGEN databases were *a priori* assumed to be immunogenic. In this manner, 21,162 and 31,693 HLA-I and HLA-II-restricted peptides were identified, respectively, of which 1,873 (8.9%) and 4,505 (14.2%) had contradicting annotations on immunogenicity. In such cases, peptides with at least one positive annotation were considered immunogenic, given that coexistence of negative assay results does not necessarily preclude the possibility of being recognized by any of the TCRs at a population-level. Eventually, 6,957 (32.9%) and 16,642 (52.5%) HLA-I and HLA-II-restricted peptides were considered immunogenic, respectively. Collected peptide sequences and annotations are summarized in **Supplementary Data Sheet 2**.

Sequences of peptides restricted on non-human MHC molecules were collected from the IEDB database (as of as of May 7th, 2018) as described above. The following species were considered primate: bonobo, chimpanzee, gorilla, marmoset, and rhesus macaque. Meanwhile, mouse and rat were considered rodent. Eventually, 411 and 8,756 MHC-I-restricted peptides, and 76 and 8,445 MHC-II-restricted peptides were identified for primates and rodents, respectively. Collected peptide sequences and annotations can be found in **Supplementary Data Sheet 2**.

TCR Sequence Datasets

TCR repertoire datasets were collected from the NCBI Sequence Read Archive (SRA) and a previous study led by Britanova et al. (21). Repertoire datasets derived from healthy donors were searched through SRA, and the following BioProjects were included: PRJNA389805, PRJNA329041, PRJNA273698, PRJNA258001, PRJNA229070, PRJNA79707, and PRJNA79435. Also, pooled sequence data from 39 healthy donors were retrieved from the paper by Britanova et al. In this manner, a total of 206 datasets, of which only five were pooled datasets, were collected. Pooled and individual datasets were derived from a total of 561 and 103 healthy donors, respectively. Fastq files were

obtained using fastq-dump script with the following options: –gzip –skip-technical –readids –read-filter pass –dumpbase –split-files –clip –accession [SRA run number]. A total of 23,066,555 CDR3 β sequences were extracted using MiXCR software (43).

Public TCR clonotypes, the ones commonly observed in multiple datasets, were identified from the pool of CDR3 β sequences mentioned above. Britanova et al. (21) reported that 10,691 clonotypes were shared between at least six of 39 (15.4%) donor-derived top 100,000 clonotype sets. Based on their findings, we decided to extract CDR3 β sequences identified in at least 22 out of 206 (10.8%) different datasets, and we subsequently identified 191,326 (0.83%) public clonotypes. A randomly sampled 10,000 clonotypes were used for TCR-peptide contact potential profiling.

TCR-Peptide Contact Potential Profiling

T cell epitopes presented on the MHC molecules must be recognized by the TCRs of CD8⁺ CTLs and CD4⁺ Th cells with sufficiently high affinity to trigger subsequent immunological cascades. Undoubtedly, the vast majority of TCR repertoire is not involved in recognition of any given peptide. Moreover, considering the substantial conformational flexibility observed upon binding of TCRs to pMHC complexes, one can assume that only a subset of residues of the peptide and the epitope-recognition domain, namely, the CDR3 β loop, serve as a “seed” of intermolecular docking during the early phase of molecular scanning (29, 32). With these in mind, identifying the best contact site between a given pair of peptide and CDR3 β sequences is conceptually similar to solving a local pairwise sequence alignment problem. One caution is that higher scores must be given to more strongly interacting residue pairs instead of more biochemically similar residue pairs as opposed to ordinary alignments. For this purpose, amino acid pairwise contact potential (AACP) scales from the AAIndex database (70) (http://www.genome.jp/aaindex/AAindex/list_of_potentials) were adopted to generate custom substitution matrices. As stronger interresidue interactions yield smaller free energy values (negative numbers), and as the Smith-Waterman local alignment algorithm attempts to maximize the alignment score of a given sequence (TCR fragment) against the whole target sequence (peptide), the optimal pairwise alignment using a custom substitution matrix derived from the sign-inverted version of a pairwise contact potential scale would in principle correspond to the best intermolecular contact. For comparative purposes, both inverted and non-inverted versions were tested. Values were rescaled to a range from zero to one for subsequent analyses. A set of non-inverted, non-rescaled AACP scales is provided as **Supplementary Data Sheet 6**. For repertoire-wide CPP analysis, the 191,326 pooled public CDR3 β sequences were randomly down-sampled to 10,000 sequences with the relative abundance ratios retained. The sequences were then fragmented by a sliding window strategy to generate a fragment library. Since the interacting orientations (either forward parallel or antiparallel) are unknown for most of the cases, reversed CDR3 β sequences were also fragmented and combined. The sizes of fragments were from 3-aa to 8-aa and from 3-aa to 11-aa

for MHC-I and MHC-II predictions, respectively. For single-TCR CPP (sCPP) analysis, every single TCR instead of public TCR repertoire was used to generate a fragment library. To perform sequence alignments, the *pairwiseAlignment* function implemented in the *Biostrings* package in *Bioconductor* (<https://www.bioconductor.org/>) (71) was utilized. This function seeks an optimal alignment which maximizes the overall alignment score defined as a sum of pairwise scores. Alignment type was set “global-local” to obtain an optimal alignment of a given set of TCR fragments against the consecutive subsequences of any given peptide. Gaps were not allowed. A set of alignment scores was summarized by calculating representative statistics. Following statistics were calculated using the functions implemented in the *psych* package: mean (Mean), standard deviation (SD), standard error of the mean (SE), median (Med), median absolute deviation (MAD), interquartile range (IQR), 10% quantile (Q10), 90% quantile (Q90), skewness (Skew), and kurtosis (Kurt). Predictive features were generated by combining the fragment length, the AACP scale, and the type of statistics. In this manner, 700 CPP features per one fragment length were generated for each of the peptides.

Peptide Descriptors

Apart from CPP features, sequence-based estimates of physicochemical properties were also calculated. Each peptide sequence was converted into a set of consecutive fragments of a defined amino acid length, and peptide descriptors were calculated against each of the fragments using functions in the *Peptide* package. Following functions were utilized: *aIndex*, *blosumIndices*, *boman*, *charge*, *crucianProperties*, *fagatVectors*, *hmoment*, *hydrophobicity*, *instatIndex*, *kideraFactors*, *mswhimScores*, *pl*, *protFP*, *vhseScales*, and *zScales*. The distributions of the values were summarized similarly to CPP features. Additionally, 20 binary features indicating whether the peptide of interest is free from a specific amino acid residue were included. The peptide length was also included as a feature.

Feature Selection

Preprocessing followed by importance-based feature selection was repeated ten times with different random seeds. The analysis workflow consisted of the following steps. First, the peptide set was randomly split in a ratio of 4:1 for the training and testing subdatasets. Second, using the training subdataset, a preprocessing function was defined; numerical features were centered and scaled; a peptide length and binary features representing existence or absence of each residue were left unaltered. The same preprocessing filter designed from the training subdataset was also applied to the testing subdataset. Third, highly correlated features were removed, with the threshold of correlation coefficient being 0.75. Fourth, features were filtered based on the importance values calculated using the *generateFilterValuesData* function with the random forest (RF) method (*randomForestSRC.rfsrc*) implemented in the *mlr* package (72). The 100 most important features were retained unless otherwise stated. In the first five repeats, the names of the peptide descriptors and the CPP features were ordered in

ascending order. In the latter five repeats, on the other hand, the names of those features were ordered in descending order. In this manner, the inherent bias of feature selection owing to the lexicographical order of the names of features was avoided. Finally, features kept in all the ten repeats were defined as the most predictive features and were utilized hereafter.

TCR-pMHC Structure Analysis

The structures of TCR-pMHC complexes with known experimentally determined affinities were collected manually from the literature using the ATLAS database (<https://zlab.umassmed.edu/atlas/web/>) (73). A total of 95 structures were retrieved, of which 82 contained non-mutated (wildtype) TCRs. Contact sites were identified using the PRODIGY server (26, 27). TCR contact footprints were defined as the distributions of the numbers of contacts at each of the peptide positions.

For the analysis of correlation with affinities, sCIPP features were computed as described in the above sections, and the single best feature was selected from a univariate analysis for each of the AAIndex scales. Multivariate regression with stepwise feature selection was conducted using the *stepAIC* function implemented in the *MASS* package. The variance inflation factors (VIFs) were computed using the *vif* function implemented in the *car* package.

Sequence permutation experiments were performed as follows. Peptide sequences were replaced with a set of random sequences of the same lengths. In contrast, because of the fragmentation strategy employed, a simple replacement would lead to incomplete disruption of hidden but essential motifs. Therefore, we instead replaced the TCR-derived fragment library with a randomly chosen set of random fragments that do not overlap with the original fragment library. Permutation experiments were iterated 1,000 times.

HLA Binding Prediction

Predicted binding strength against twelve representative HLA-I alleles (A*01:01, A*02:01, A*03:01, A*24:02, A*26:01, B*07:02, B*08:01, B*27:05, B*39:01, B*40:01, B*58:01, and B*15:01) from NetMHC 4.0 (74) and six representative HLA-II alleles (DRB1*0101, DRB3*0101, DRB4*0101, DRB5*0101, DPA1*0103-DPB1*0101, and DQA1*0101-DQB1*0201) from NetMHCIIpan 3.2 (56) for MHC-I and MHC-II immunogenicity prediction, respectively, were incorporated in the machine learning process for immunogenicity prediction because the stability of the peptide-MHC complex is a known correlate of immunogenicity (75). Default parameters were used for prediction. Thresholds of percentile ranks for strong and weak binders were set at 0.5 and 2% in MHC-I, and 2 and 10% in MHC-II, respectively.

Machine Learning

The most predictive peptide descriptors and CPP features, with or without predicted HLA bindings, were compressed into a linear coordinate system through machine learning. We utilized extremely randomized trees (ERT) algorithm implemented in the *extraTrees* package (76) because of its computational efficiency and robustness against overfitting. The model-specific parameter *mtry* was set 5. Other hyperparameters were set as defaults.

Class weights were provided to compensate for imbalanced class distributions. Probabilistic estimates were computed by conducting five times repeated 5-fold cross-validations (CVs). This strategy ensures that any peptide is subjected to five times repeated predictions by models trained from a set of peptides that do not contain the peptide in question. The averaged probability estimates were termed “immunogenicity scores.” Unless otherwise stated, we utilized immunogenicity scores from machine learning without predicted HLA binding-based features. For peptides not included in our epitope dataset, immunogenicity scores are defined as the averaged probability estimates of the 25 ERT models generated during CVs. Prediction variances were evaluated by computing coefficients of variance which are defined as standard deviations divided by means.

The IEDB Epitope Cluster Analysis Tool (<http://tools.iedb.org/cluster/>) was utilized to test the effect of sequence-level peptide homology on immunogenicity prediction. Clustering was performed with the homology threshold of 80%. When clustering filter was applied, only a single peptide was randomly chosen from each of the clusters before feature preprocessing in the machine learning workflow.

Neighbor Network

Single amino acid mutations, namely, substitutions, insertions, and deletions, have multifaceted effects on the immunogenicity of MHC-presented peptides. To systematically analyze their effects on immunogenicity, we adopted a network-style representation of the epitope data structures, termed a “neighbor network,” where pairs of peptide sequences (nodes) with just one edit distance were defined as “neighbors” and regarded as edges. Each edge was directed from the peptide with a lower immunogenicity score to the peptide with a higher score. Edge weight was defined as the ratio of the lower score to the higher score so that less immunogenically similar peptides were mapped more distant from each other in sequence space due to the smaller edge weight. Clustering was performed using a walktrap algorithm implemented in the *igraph* package (77). For each cluster, sequences were aligned using the ClustalW algorithm implemented in the *msa* package (78). A consensus sequence was generated using the method implemented in the *Biostrings* package with an ambiguity threshold of 0.5. Gaps were treated equally to ambiguities. Gaps/ambiguities were expressed as “X.”

In silico Mutagenesis

For a given “parental” peptide, the neighbor network was computationally expanded by simulating all possible single amino acid substituted mutant peptides. Insertions and deletions were not taken into consideration. These artificially introduced mutations could affect various aspects of T cell immunity, including aberrant proteolytic cleavage and altered MHC presentation, these possibilities were not rigorously examined since our primary interest is on exploring the net immunogenicity of mutated peptides if appropriately presented and scanned by TCRs. Immunogenicity scores for simulated peptides were computed by extrapolation using the classifiers trained using all experimentally annotated data. Neighbor network analysis was conducted as described. Extrapolated

immunogenicity scores of all neighbors and neighbors residing in the same cluster with the parental peptide were averaged to yield overall-mean and cluster-mean immunogenicity scores, respectively.

Immune Transition

Loss of immunogenicity due to mutation is also called escaping, evasion, and sometimes immunoeediting. In the present study, we refer to this phenomenon as escaping. In contrast, there is no appropriate terminology for the mutation-driven acquisition of immunogenicity, and thus we refer to this phenomenon as “epitope formation.” For the sake of simplicity, we do not take into consideration epitope-extrinsic mechanisms such as impaired intracellular antigen processing, somatic loss of HLA heterozygosity (79, 80), and checkpoint-mediated T cell exhaustion and apoptosis (81) in this work. Escaping and epitope formation due to mutations can be understood as two sides of the same coin in sequence space, and therefore, we propose a higher-order concept, “immune transition,” which is defined as a change in immunogenicity between any MHC-presented peptide and its single amino acid variant. Note that the synergistic effects of two or more mutations on the changes in immunogenicity are beyond the scope of the present study.

To further characterize the theoretical boundary of immunogenic and non-immunogenic peptides, we categorized peptides with evidence of immune transition found in our dataset as “transitional” and analyzed separately. Note that we cannot certainly define “non-transitional” peptides, as the lack of immune transition in our dataset may merely reflect the lack of experimental evaluation. For transitional peptides, *in silico* mutagenesis followed by neighbor network analysis was conducted. Mean immunogenicity scores of both all single amino acid mutants and those within the same cluster to the parental peptide were computed.

Immunogenicity Score Dynamics

For quantitative comparison of immunogenicity for a given pair of peptides, either the ratio of immunogenicity scores (divided by the lower score), termed “relative score change,” or the

absolute difference of immunogenicity scores divided by the lower score, termed “normalized score change,” was used as a metric. To systematically explore the position-specific mutational impacts for a given parental peptide, relative score change was computed for every pair of computationally simulated mutants. For heatmap analysis, peptide pairs were grouped based on their mutated residue combinations, and median normalized score changes were calculated for each group. Heatmaps were visualized using the *ComplexHeatmap* package (82).

Escape Potential

Escape potential of a given peptide was defined as the maximum difference of the cluster-mean immunogenicity scores between the cluster to which the target peptide belongs and the other clusters in the neighbor network constructed from all possible single amino acid variants assigned with simulated immunogenicity scores. A large escape potential means that a single mutation could cause a significant loss of predicted immunogenicity. A negative value indicates that the target peptide resides in the least immunogenic cluster, and the target epitope would not likely lose its immunogenicity further by any single mutation.

AUTHOR CONTRIBUTIONS

MO and HY conceived and discussed the concept. MO designed the entire study, performed computational analyses, and drafted the manuscript. MO and HY wrote the manuscript.

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SUPPLEMENTARY MATERIAL

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fimmu.2019.00827/full#supplementary-material>

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Conflict of Interest Statement: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Diagnostic Test Accuracy of a 2-Transcript Host RNA Signature for Discriminating Bacterial Vs Viral Infection in Febrile Children

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Diagnostic Test Accuracy of a 2-Transcript Host RNA Signature for Discriminating Bacterial vs Viral Infection in Febrile Children

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IMPORTANCE Because clinical features do not reliably distinguish bacterial from viral infection, many children worldwide receive unnecessary antibiotic treatment, while bacterial infection is missed in others.

OBJECTIVE To identify a blood RNA expression signature that distinguishes bacterial from viral infection in febrile children.

DESIGN, SETTING, AND PARTICIPANTS Febrile children presenting to participating hospitals in the United Kingdom, Spain, the Netherlands, and the United States between 2009-2013 were prospectively recruited, comprising a discovery group and validation group. Each group was classified after microbiological investigation as having definite bacterial infection, definite viral infection, or indeterminate infection. RNA expression signatures distinguishing definite bacterial from viral infection were identified in the discovery group and diagnostic performance assessed in the validation group. Additional validation was undertaken in separate studies of children with meningococcal disease (n = 24) and inflammatory diseases (n = 48) and on published gene expression datasets.

EXPOSURES A 2-transcript RNA expression signature distinguishing bacterial infection from viral infection was evaluated against clinical and microbiological diagnosis.

MAIN OUTCOMES AND MEASURES Definite bacterial and viral infection was confirmed by culture or molecular detection of the pathogens. Performance of the RNA signature was evaluated in the definite bacterial and viral group and in the indeterminate infection group.

RESULTS The discovery group of 240 children (median age, 19 months; 62% male) included 52 with definite bacterial infection, of whom 36 (69%) required intensive care, and 92 with definite viral infection, of whom 32 (35%) required intensive care. Ninety-six children had indeterminate infection. Analysis of RNA expression data identified a 38-transcript signature distinguishing bacterial from viral infection. A smaller (2-transcript) signature (*FAM89A* and *IFI44L*) was identified by removing highly correlated transcripts. When this 2-transcript signature was implemented as a disease risk score in the validation group (130 children, with 23 definite bacterial, 28 definite viral, and 79 indeterminate infections; median age, 17 months; 57% male), all 23 patients with microbiologically confirmed definite bacterial infection were classified as bacterial (sensitivity, 100% [95% CI, 85%-100%]) and 27 of 28 patients with definite viral infection were classified as viral (specificity, 96.4% [95% CI, 89.3%-100%]). When applied to additional validation datasets from patients with meningococcal and inflammatory diseases, bacterial infection was identified with a sensitivity of 91.7% (95% CI, 79.2%-100%) and 90.0% (95% CI, 70.0%-100%), respectively, and with specificity of 96.0% (95% CI, 88.0%-100%) and 95.8% (95% CI, 89.6%-100%). Of the children in the indeterminate groups, 46.3% (63/136) were classified as having bacterial infection, although 94.9% (129/136) received antibiotic treatment.

CONCLUSIONS AND RELEVANCE This study provides preliminary data regarding test accuracy of a 2-transcript host RNA signature discriminating bacterial from viral infection in febrile children. Further studies are needed in diverse groups of patients to assess accuracy and clinical utility of this test in different clinical settings.

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← Editorial page 824

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The majority of febrile children have self-resolving viral infection, but a small proportion have life-threatening bacterial infections. Although culture of bacteria from normally sterile sites remains the gold standard for confirming bacterial infection, culture results may take several days and are frequently negative when infection resides in inaccessible sites or when antibiotics have been previously administered.¹⁻³ Current practice is to admit ill-appearing febrile children to the hospital and administer parenteral antibiotics while awaiting culture results.⁴⁻⁶ Because only a minority of febrile children are ultimately proven to have bacterial infection, the process of ruling out bacterial infection results in a major burden on health care resources and in inappropriate antibiotic prescription.⁷

Molecular tests have the potential to identify bacterial and viral pathogens and improve distinction between bacterial and viral infection.⁸ Rapid molecular viral diagnostics have increased the proportion of patients shown to carry respiratory pathogens,⁹ but viruses are frequently identified in nasopharyngeal samples from healthy children.¹⁰ Thus, detection of a virus in the nasopharynx does not rule out bacterial infection and is of little help in decisions on whether to administer antibiotics.

A number of studies have suggested that specific infections can be identified by the pattern of host genes activated during the inflammatory response.¹¹⁻¹⁵ This study investigated whether bacterial infection can be distinguished from other causes of fever in children by the pattern of host genes activated or suppressed in blood in response to the infection and whether a subset of these genes could be identified as the basis for a diagnostic test.

Methods

Study Conduct and Oversight

Written informed consent was obtained from parents or guardians using locally approved research ethics committee permissions (St Mary's Research Ethics Committee (REC 09/H0712/58 and EC3263); Ethical Committee of Clinical Investigation of Galicia (CEIC ref 2010/015); UCSD Human Research Protection Program No. 140220; and Academic Medical Centre, University of Amsterdam (NL41846.018.12 and NL34230.018.10).

Discovery and Validation Groups

The overall design of the study is shown in Figure 1, Figure 2, and Figure 3.

Clinical data and samples were identified only by study number. Assignment of patients to clinical groups was made by consensus of 2 clinicians independent of those managing the patient, after review of investigation results using previously agreed-on definitions (Figure 2). Patients were recruited prospectively as part of a UK National Institute of Health Research-supported study (NIHR ID 8209), the Immunopathology of Respiratory, Inflammatory and Infectious Disease Study (IRIS), which recruited children at 3 UK hospitals; patients also were recruited in Spain (GENDRES network,

Key Points

Question Can febrile children with bacterial infection be distinguished from those with viral infection and other common causes of fever using whole-blood gene expression profiling?

Findings In this cross-sectional study that included 370 febrile children, those with bacterial infection were distinguished from those with viral infection with a sensitivity in the validation group of 100% (95% CI, 85%-100%) and specificity of 96.4% (95% CI, 89.3%-100%), using a 2-transcript signature.

Meaning This study provides preliminary data on the performance of a 2-transcript host RNA signature for discriminating bacterial from viral infection in febrile children. Further studies are needed in diverse groups of patients to assess accuracy and clinical utility of this test in different clinical settings.

Santiago de Compostela), and the United States (Rady Children's Hospital, San Diego). Inclusion criteria were fever (axillary temperature $\geq 38^{\circ}\text{C}$) and perceived illness of sufficient severity to warrant blood testing in children younger than 17 years. Patients with comorbidities likely to affect gene expression (bone marrow transplant, immunodeficiency, or immunosuppressive treatment) were excluded. Blood samples for RNA analysis were collected together with clinical blood tests at, or as close as possible to, presentation to hospital, irrespective of antibiotic use at the time of collection.

Additional Validation Groups

Additional validation groups (eMethods and eTable 1 in the Supplement) included children with meningococcal sepsis,¹⁶ inflammatory diseases (juvenile idiopathic arthritis and Henoch-Schönlein purpura), and published gene expression datasets that compared bacterial infection with viral infection^{12,15,17} or inflammatory disease.¹⁸ Healthy children were recruited from outpatient departments. Data from healthy controls were not used in identification or validation of gene expression signatures and were used only for interpretation of direction of gene regulation.

Diagnostic Process

All patients underwent routine investigations as part of clinical care, including complete blood cell count and differential, C-reactive protein level, blood chemistries, blood and urine cultures, and cerebrospinal fluid analysis where indicated. Throat swabs were cultured for bacteria, and viral diagnostics undertaken on nasopharyngeal aspirates using multiplex polymerase chain reaction for common respiratory viruses. Chest radiographs were undertaken as clinically indicated. Patients were assigned to diagnostic groups using predefined criteria (Figure 2). The definite bacterial infection group included only patients with culture-confirmed infection, and the definite viral infection group included only patients with culture, molecular, or immunofluorescent test-confirmed viral infection and no features of coexisting bacterial infection. Children in whom definitive diagnosis was not established (indeterminate infection) were categorized into probable bacterial infection, unknown bacterial or viral infection, and probable viral

Figure 1. Study Overview

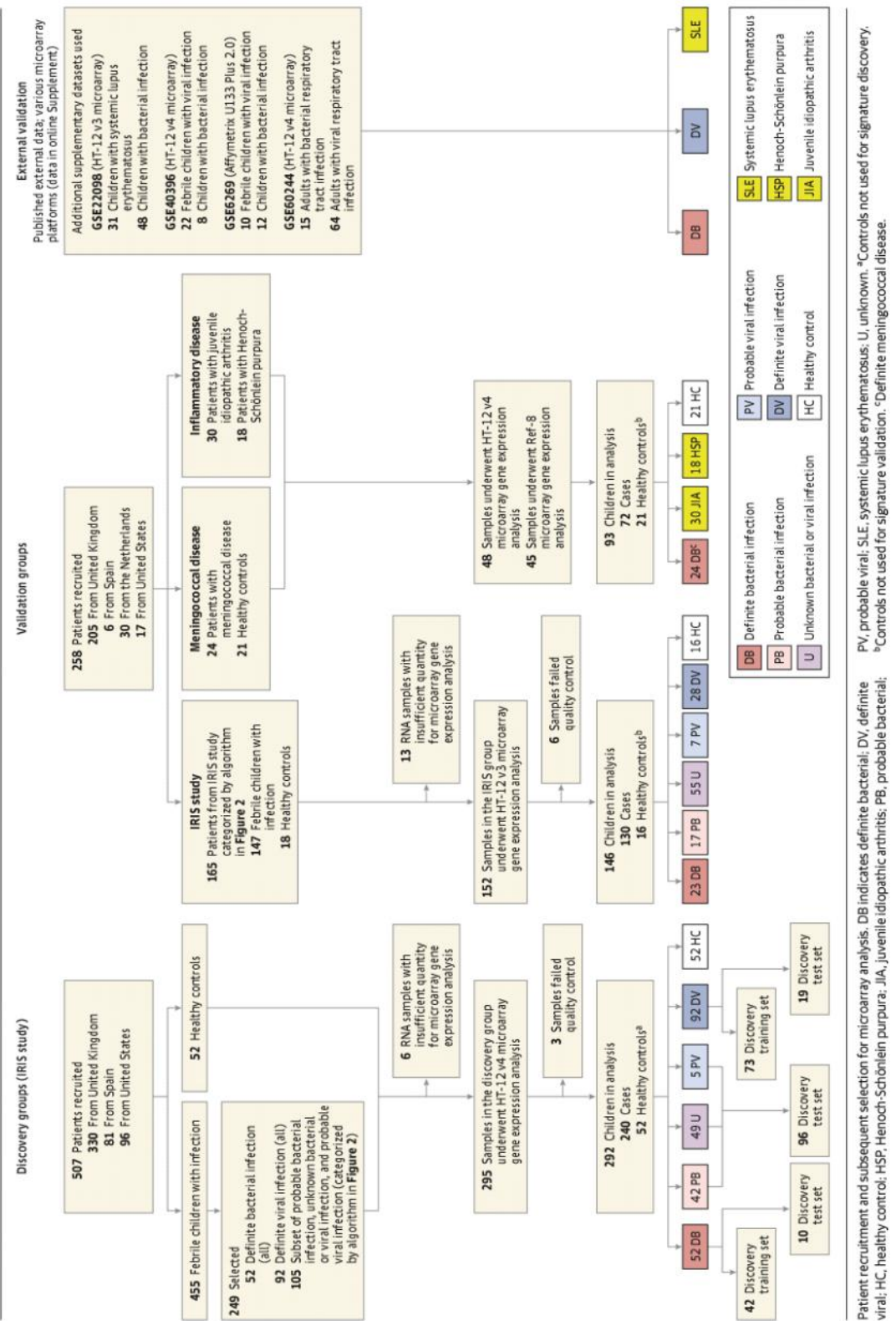
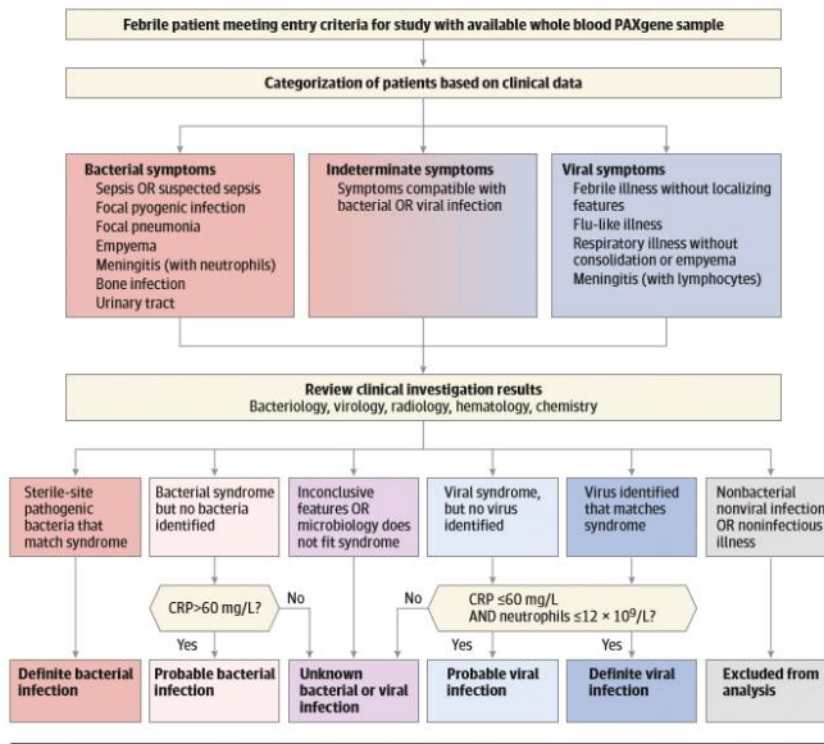


Figure 2. Classification of Patients Into Diagnostic Groups



infection groups based on level of clinical suspicion (Figure 2). Detection of virus did not prevent inclusion in the definite bacterial, probable bacterial, or unknown infection groups, because bacterial infection can occur in children co-infected with viruses.

Peripheral Blood Gene Expression by Microarray

Whole blood was collected into PAXgene blood RNA tubes (PreAnalytiX), frozen, and later extracted. Gene expression was analyzed on Illumina microarrays. Additional details of microarray method, quality control, and analysis are provided in the eAppendix (eMethods, eStatistical Methods, and eFigure 1 in the Supplement).

Statistical Analysis

Transcript Signature Discovery

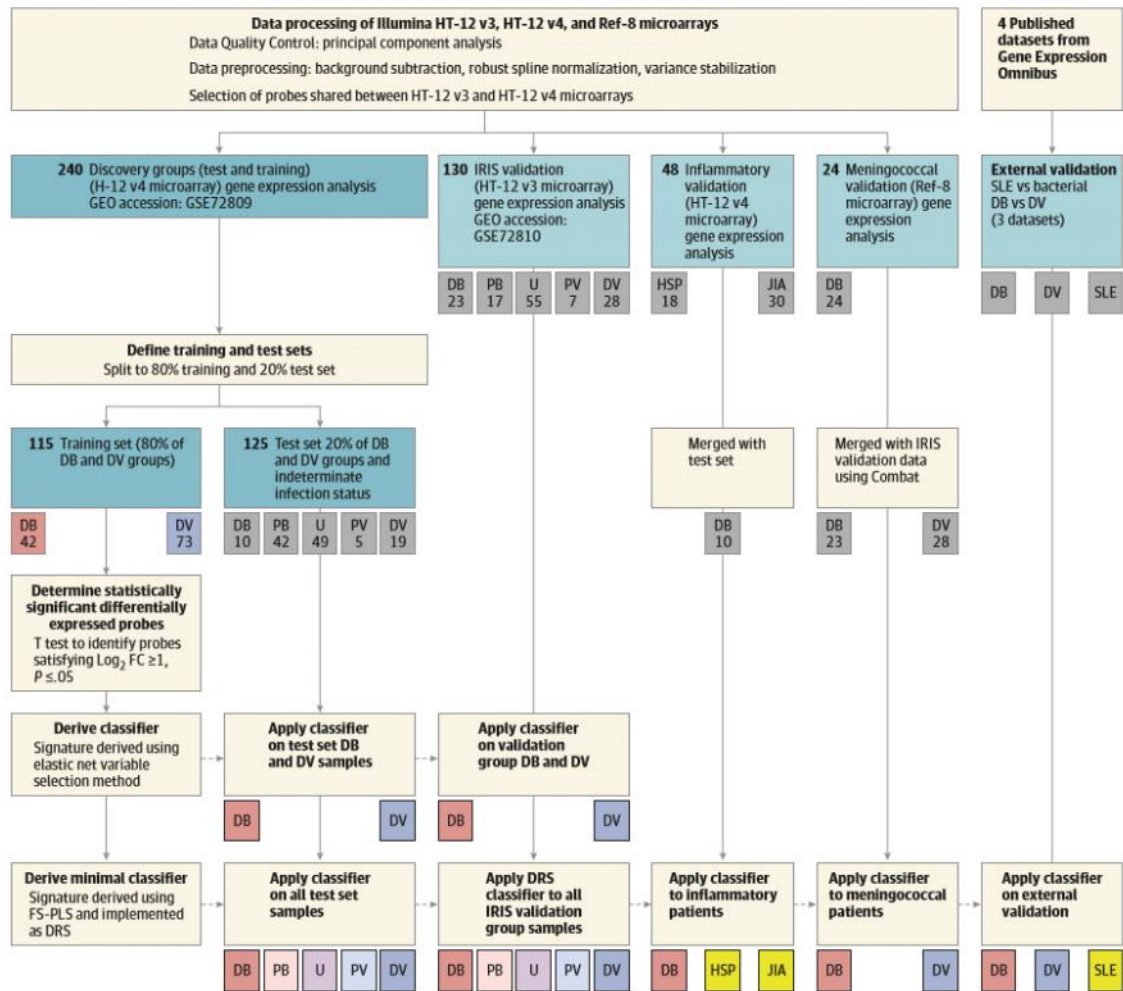
Expression data were analyzed using R version 3.1.2 (R Project for Statistical Computing). Patients with definite bacterial or viral infection in the discovery group were randomly assigned to training and test sets (80% and 20% of the patients, respectively), and significantly differentially expressed transcripts distinguishing definite bacterial infection from definite viral infection were identified in the training set (Figure 3). A linear model was fitted conditional on recruitment site, and moderated *t* statistics were calculated for each transcript. The *P* values obtained were corrected for multiple testing using the Benjamini-Hochberg false discovery rate method.¹⁹ Logistic re-

gression with variable selection was applied to the significantly differentially expressed transcripts (absolute log₂-fold change >1 and 2-sided *P* < .05) using elastic net (a variable selection algorithm that selects sparse diagnostic transcript signatures—see eMethods and eFigure 2 in the Supplement).²⁰

To further reduce the number of transcripts in the diagnostic signatures, a novel variable selection method was used that eliminates highly correlated transcripts: forward selection-partial least squares (see eAppendix in the Supplement). The disease risk score (DRS) method²¹ was applied to the resulting minimal multitranscript signature to translate it into a single value that could be assigned to each individual, to form the basis of a simple diagnostic test.^{11,21} The DRS method calculates a patient score by adding the total intensity of the up-regulated transcripts (relative to comparator group) and subtracting the total intensity of the downregulated transcripts (relative to comparator group). The signatures identified in the discovery group were externally validated on previously published validation groups,¹³ additional patient groups with meningococcal disease and inflammatory diseases, and published datasets (3 pediatric, 1 adult) (Figure 3).

To evaluate the predictive accuracy of the DRS and of models derived after variable selection analysis, point and interval metrics were calculated using the pROC package in R.²² Results obtained using elastic net and DRS models were compared with reference-standard clinically assigned diagnoses (Figure 2). The area under the receiver operating characteristic

Figure 3. Analysis Workflow



Overall study pipeline showing sample handling, derivation of test and training sets, data processing, and analysis pipeline including application of 38-transcript elastic net classifier and 2-transcript disease risk score (DRS) classifier, to the group test set, the validation group data, and published (external) validation datasets. DB indicates definite bacterial; DV, definite viral;

FC, fold change; FS-PLS, forward selection-partial least squares; HSP, Henoch-Schönlein purpura; JIA, juvenile idiopathic arthritis; PB, probable bacterial; PV, probable viral; SLE, significantly differentially expressed; SLE, systemic lupus erythematosus; U, unknown.

curve (AUC), sensitivity, and specificity were reported. Confidence intervals (95%) were calculated to measure the reliability of estimates.

Results

Two hundred forty patients (median age, 19 months; 62% male) were recruited to the discovery group between 2009-2013 in the United Kingdom (189 patients), Spain (16), and the United States (35). The definite bacterial infection group included 52 patients, of whom 36 (69%) required intensive care and 10 died. In the definite viral infection group of 92 patients, 32 (35%) re-

quired intensive care and none died (Table 1). The patients with bacterial and viral infection were subdivided into 80% (training set) and 20% (test set) (Figure 1 and Figure 3). The test set also included 96 children whose infection was not definitively diagnosed (indeterminate infection) (Figure 1 and Figure 3). The validation groups comprised 130 UK and Spanish children (median age, 17 months; 57% male) previously recruited¹³ (IRIS validation; 23 with definite bacterial infection, 28 with definite viral infection, and 79 with indeterminate infection) and 72 additional validation children—25 from the United Kingdom, 30 from the Netherlands, and 17 from the United States (24 with meningococcal infection, 30 with juvenile idiopathic arthritis, and 18 with Henoch-Schönlein

Table 1. Demographic and Clinical Characteristics of the Study Groups

Characteristics	Discovery Group			IRIS Validation Group		
	Definite Bacterial Infection (n = 52)	Definite Viral Infection (n = 92)	Indeterminate Infection ^a (n = 96)	Definite Bacterial Infection (n = 23)	Definite Viral Infection (n = 28)	Indeterminate Infection ^a (n = 79)
Age, median (IQR), mo	22 (9-46)	14 (2-39)	27 (7-71)	22 (13-52)	18 (7-48)	15 (2-44)
Male, No. (%)	22 (42)	65 (71)	62 (65)	10 (43)	17 (61)	47 (59)
White race, No./total (%) ^b	35/48 (73)	46/87 (53)	47/85 (55)	12/22 (55)	14/27 (51)	42/71 (59)
Time from symptoms to blood sampling, median (IQR), d	5 (2-8.8)	4.5 (3.0-6.0)	5 (4.8-8)	4 (2.5-8)	3.5 (2.8-5.3)	4 (3-7)
Intensive care, No. (%)	36 (69)	32 (35)	57 (59)	13 (57)	7 (23)	42 (53)
Deaths, No.	10	0	2	1	1	8
C-reactive protein, median (IQR), mg/L ^c	176 (98-275)	16 (6-27)	102 (47-176)	217 (168-285)	7 (1-20)	67 (25-128)
Blood cell differential, median (IQR), %						
Neutrophils	75 (49-85)	50 (36-63)	63 (46-79)	82 (71-88)	53 (41-69)	64 (43-82)
Lymphocytes	19 (10-36)	34 (20-44)	22 (15-42)	15 (8-23)	32 (26-48)	30 (14-42)
Monocytes	5 (3-8)	10 (4-14)	6 (2-12)	3 (0-7)	7 (5-10)	5 (2-8)
Main clinical syndrome, No.						
Bone, joint, soft tissue infection	5	0	0	1	0	0
Fever without source/sepsis	21	7	9	5	2	6
Gastroenteritis	0	0	1	0	1	2
Meningitis/encephalitis	14	3	3	5	1	1
Respiratory tract, upper + lower	10	81	83	11	23	68
Other	2	1	0	1	1	2
Virus detected, No./total (%) ^d	22/34 (65)	92/92 (100)	62/87 (71)	8/13 (62)	28/28 (100)	52/77 (68)

Abbreviations: IQR, interquartile range; IRIS, Immunopathology of Respiratory, Inflammatory and Infectious Disease Study.

SI conversion factor: To convert C-reactive protein values to nmol/L, multiply by 0.9524.

^a The indeterminate infection group in the discovery group comprised 42 probable bacterial, 49 unknown bacterial or viral, and 5 probable viral infections. The indeterminate infection group in the validation group

comprised 17 probable bacterial, 55 unknown bacterial or viral, and 7 probable viral infections.

^b Self-reported.

^c Maximum value of C-reactive protein in illness is reported.

^d Denominator denotes number of patients with viral investigations.

purpura) (Figure 1 and Figure 3). The numbers in each diagnostic category in the discovery, IRIS validation, and additional validation groups and their clinical features are shown in Table 1 and in eTable 1 in the Supplement. Details of the types of infection are shown in eTable 2 in the Supplement. Gene expression profiles of children in the discovery group clustered together on principal component analysis (eFigure 1 in the Supplement).

Identification of Minimal Transcript Signatures

Of the 8565 transcripts differentially expressed between bacterial and viral infections, 285 were identified as potential biomarkers after applying filters based on log fold change and statistical significance (see Methods). Variable selection using elastic net identified 38 transcripts (eTable 3 in the Supplement) as best discriminators of bacterial and viral infection in the discovery test set, with sensitivity of 100% (95% CI, 69%-100%) and specificity of 95% (95% CI, 84%-100%) (eTable 4 in the Supplement). In the validation group, this signature had an AUC of 98% (95% CI, 94%-100%), sensitivity of 100% (95% CI, 85%-100%), and specificity of 86% (95% CI, 71%-96%) for distinguishing bacterial from viral infection (eTable 4 and eFigures 2 and 3 in the Supplement). The putative function of the

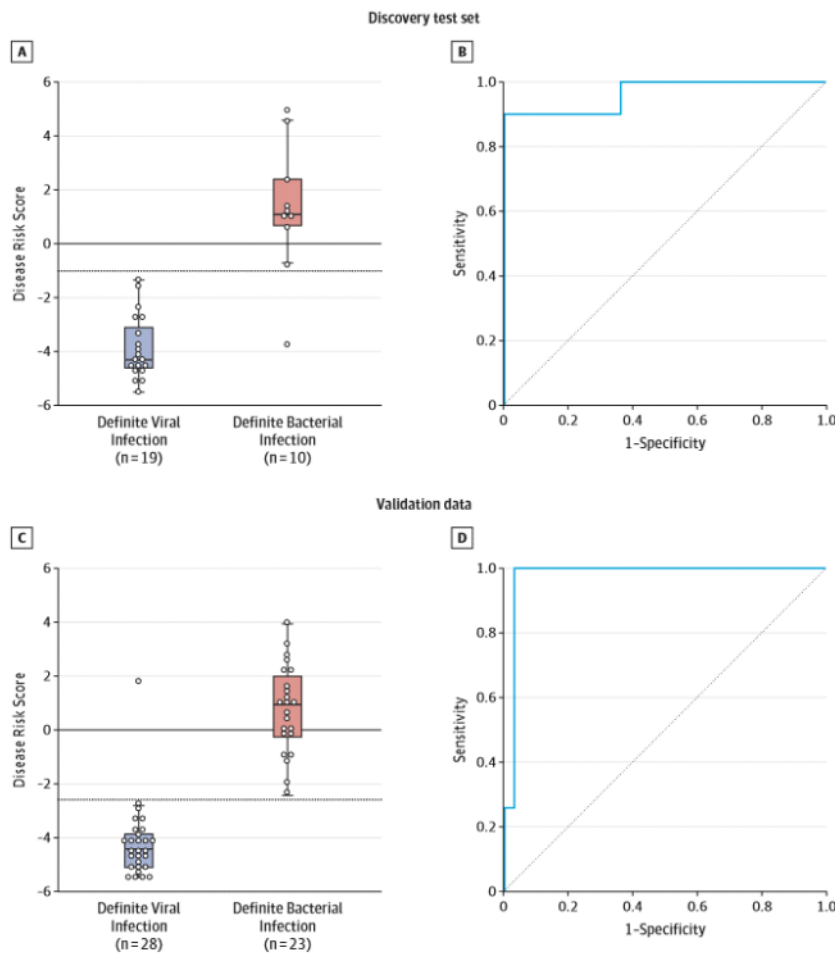
38 transcripts in our signature, as defined by gene ontology, is shown in eTable 5 in the Supplement.

After using the novel forward selection process to remove highly correlated transcripts, a 2-transcript signature was identified that distinguished bacterial from viral infections, including interferon-induced protein 44-like (*IFI44L*, RefSeq NM_006820.1), and family with sequence similarity 89, member A (*FAM89A*, RefSeq NM_198552.1). Both transcripts were included in the 38-transcript signature.

Implementation of a DRS

The expression data of both genes in the signature was combined into a single DRS for each patient, using the reported DRS method, which simplifies application of multitranscript signatures as a diagnostic test.²¹ The sensitivity of the DRS was 86% (95% CI, 74%-95%) in the discovery group training set, 90% (95% CI, 70%-100%) in the discovery group test set, and 100% (95% CI, 85%-100%) in the validation data; specificity in the validation data was 96.4% (95% CI, 89.3%-100%) (Figure 4, panels A-D; eFigure 4 and eTable 4 in the Supplement). Expression of *IFI44L* was increased in patients with viral infection and *FAM89A* was increased in patients with bacterial infection, relative to healthy children (eTable 3 in

Figure 4. DRS and ROC Curves Based on the 2-Transcript Signature Applied to Definite Bacterial and Viral Infection Groups



Classification performance and receiver operating characteristic (ROC) curve based on the 2-transcript disease risk score (DRS) (the combined *IFI44L* and *FAM89A* expression values), in the definite bacterial and viral infection groups of the discovery group test set (20% of the total discovery group), and the IRIS validation group data. Horizontal lines in boxes indicate medians; lower and upper edges of boxes, interquartile ranges; whiskers, ≤ 1 times the interquartile range. Boxes and whiskers plotted using *boxplot* in R. Sensitivity, specificity, and area under the ROC curve are reported in eTable 4 in the Supplement. Horizontal dotted line indicates the DRS threshold that separates patients predicted as bacterial (above the line) or viral (below the line), as determined by the point on the ROC curve that maximized sensitivity and specificity.

the Supplement). The summary of diagnostic test accuracy, including STARD flow diagrams, is shown in eFigure 5 in the Supplement.

For additional validation, the 2-transcript signature was applied to patients with meningococcal disease (eFigure 6 in the Supplement) and inflammatory diseases (juvenile idiopathic arthritis and Henoch-Schönlein purpura). Bacterial infection was identified with a sensitivity of 91.7% (95% CI, 79.2%-100%) for patients with meningococcal disease and 90.0% (95% CI, 70.0%-100%) for those with inflammatory diseases and with a specificity of 96.0% (95% CI, 88.0%-100%) and 95.8% (95% CI, 89.6%-100%), respectively. When applied to 4 published datasets for children and adults with bacterial or viral infection and inflammatory disease (pediatric systemic lupus erythematosus),^{12,15,17,18} the 2-transcript signature distinguished bacterial infection from viral infection and inflammatory disease in all these datasets, with AUCs ranging from 89.2% to 96.6% (eTable 6 and eFigures 7 and 8 in the Supplement).

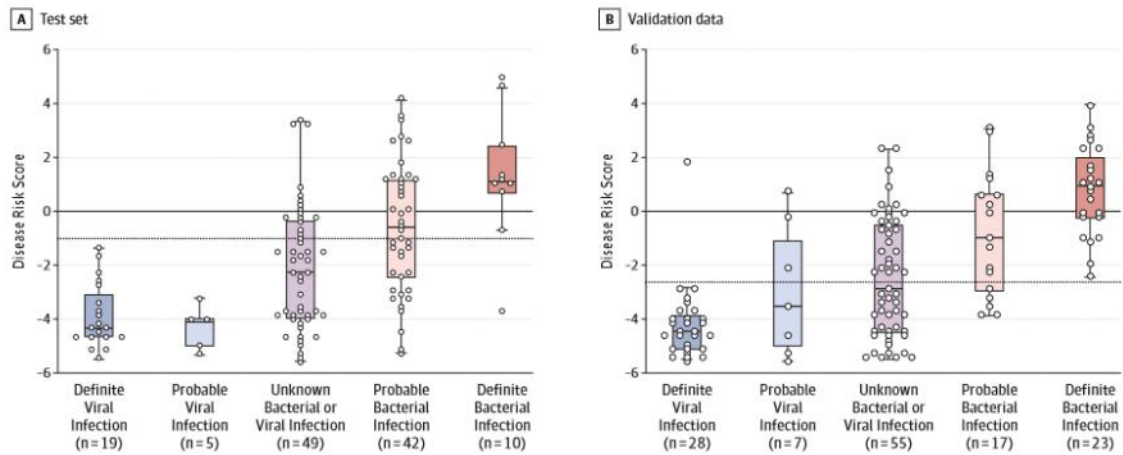
Effect of Viral and Bacterial Co-infection

The effect of viral co-infection on the signatures was investigated (Table 1). In the definite bacterial infection group, 30 of 47 patients tested (64%) had a virus isolated from nasopharyngeal samples. There was no significant difference in DRS between those with and without viral co-infection.

DRS in Patients With Indeterminate Infection Status

The classification performance of the DRS was investigated in patients with indeterminate viral or bacterial infection status. Patients were separated into those with clinical features strongly suggestive of bacterial infection (probable bacterial infection group), those with features consistent with either bacterial or viral infection (unknown infection group), and those with clinical features and results suggestive of viral infection (probable viral infection group) (Figure 2). The probable bacterial and unknown infection groups included patients with DRS values that indicated

Figure 5. Performance of the 2-Transcript DRS Signature in Indeterminate Infection Group



Classification performance of the 2-transcript disease risk score (DRS) (the combined *IFI44L* and *FAM89A* expression values) in the indeterminate groups of probable bacterial, probable viral, and unknown infection of the discovery test and IRIS validation sets. Horizontal lines in boxes indicate medians; lower and upper edges of boxes, interquartile ranges; whiskers, ≤ 1 times the interquartile range. Boxes and whiskers plotted using *boxplot* in R. Horizontal

dotted line indicates the DRS threshold (threshold_{test data set} = -1.03; threshold_{validation data set} = -2.63) that separates patients predicted as bacterial (above the line) or viral (below the line), as determined by the point in the receiver operating characteristic curve that maximized sensitivity and specificity. For the discovery group test set, the training threshold was used.

Table 2. Comparison of Disease Risk Score Prediction With Antibiotic Treatment^a

	No. (%)				
	Definite Bacterial Infection	Probable Bacterial Infection	Unknown Bacterial or Viral Infection	Probable Viral Infection	Definite Viral Infection
Patients with information on antibiotic use ^b	28 (84.8)	49 (83.1)	77 (74.0)	10 (83.3)	39 (83.0)
Patients receiving antibiotics	28 (100)	49 (100)	73 (94.8)	7 (70)	33 (84.6)
Patients receiving antibiotics and suggested by DRS to have bacterial infection	27 (96.4)	32 (65.3)	29 (39.7)	2 (28.6)	1 (3.0)

Abbreviation: DRS, disease risk score.

^a Comparison of the proportion of patients in the combined discovery group test set and validation group receiving antibiotics, and the proportion of predicted bacterial, as predicted by the DRS (the combined *IFI44L* and

FAM89A expression values).

^b The denominator is 33 definite bacterial, 59 probable bacterial, 104 unknown bacterial or viral, 12 probable viral, and 47 definite viral infections.

viral infection, despite having clinical features that justified initiation of antibiotics by the clinical team. The median DRS showed a gradient of assignment that followed the degree of certainty in the clinical diagnosis, although many of the indeterminate infection group DRS values overlapped with those of the definite bacterial and definite viral infection groups (Figure 5A, Figure 4B).

DRS assignment as viral or bacterial was compared with clinical variables in the indeterminate infection group (eTable 7 in the Supplement). Measurement of C-reactive protein (CRP) levels is widely used to aid distinction of bacterial from viral infection and was included in the categorization of definite viral, probable bacterial, and probable viral infection in this study; patients categorized as having bacterial infection by DRS had higher CRP levels than those categorized as having viral infection (median, 101 [interquartile range {IQR}, 48-192] mg/L vs 71 [IQR, 27-120] mg/L; $P = .02$ [to convert values to nmol/L, multiply by 0.9524]). They also had increased incidence of shock ($P = .006$), requirement for ventilator support ($P = .048$),

and intensive care admission ($P = .07$). There was a nonsignificant increase in white cell and neutrophil counts in patients assigned by DRS as having bacterial or viral infection (median, 14.1 [IQR, 8.3-19.4] and 11.1 [IQR, 7.3-16.0] for white cells [$P = .08$]; 8.7 [IQR, 5.0-13.8] and 6.8 [IQR, 3.5-11.4] for neutrophils [$P = .11$]).

Antibiotic Use

The number of children treated with antibiotics was compared with DRS prediction of bacterial or viral infection. There were high rates of antibiotic use in all groups, including greater than 90% in the unknown infection group. The high rate of antibiotic use in the indeterminate infection group contrasted with the low numbers predicted to have bacterial infection by both DRS and clinical assignment (Table 2).

Illness Severity and Duration

The study recruited a high proportion of seriously ill patients needing intensive care, thus raising concern that selection bias

might have influenced performance of the signature. To exclude bias based on severity or duration of illness, performance of the DRS was evaluated after stratification of patients into those with milder illness or severe illness requiring intensive care and by duration of reported illness before presentation. The DRS distinguished bacterial from viral infection in both severe and milder groups (eFigure 9 in the Supplement) and irrespective of day of illness (eFigure 10 in the Supplement).

Discussion

This study identified a host whole blood RNA transcriptomic signature that distinguished bacterial from viral infection with 2 gene transcripts. The signature also distinguished bacterial infection from childhood inflammatory diseases, systemic lupus erythematosus, juvenile idiopathic arthritis, and Henoch-Schönlein purpura and discriminated bacterial from viral infection in published adult studies.^{12,15,17,18} The results extend previous studies suggesting that bacterial and viral infections have different signatures.^{12,13,17,23}

The transcripts identified in the 38-transcript elastic net signature comprise a combination of transcripts up-regulated by viruses or by bacteria. The 2 transcripts *IFI44L* and *FAM89A* in the smaller signature show reciprocal expression in viral and bacterial infection. *IFI44L* has been reported to be up-regulated in antiviral responses mediated by type I interferons,²⁴ and *FAM89A* was reported as elevated in children with septic shock.²⁵

An obstacle in the development of improved tests to distinguish bacterial from viral infection is the lack of a reference standard. Some studies include patients with clinically diagnosed bacterial infection who have features of bacterial infection but whose cultures remain negative. Negative cultures may reflect prior antibiotic use, low numbers of bacteria, or inaccessible sites of infection. If patients with indeterminate status are included in biomarker discovery, there is a risk that the identified biomarker will not be specific for true infection. This study adopted the rigorous approach of identifying the signature in culture-confirmed cases and using the signature to explore likely proportions of true infection in the indeterminate infection group.

The proportion of children predicted by DRS signature to have bacterial infection follows the level of clinical suspicion (greater in the probable bacterial infection group and less in the probable viral infection group), thus supporting the hypothesis that the signatures may be an indication of the true proportion of bacterial infection in each group. Furthermore, a higher proportion of patients in the indeterminate infection group, assigned as bacterial by the signature (probable infection and unknown infection groups), had clinical features normally associated with severe bacterial infection, including increased need for intensive care, higher neutrophil counts, and higher CRP levels, suggesting that the signature may be providing additional clues to the presence of bacterial infection.

The decision to initiate antibiotics in febrile children is largely driven by concern about missing bacterial infection. A test that correctly distinguishes children with bacterial infection from those with viral infections would reduce inappropriate antibiotic prescription and investigation. The DRS suggests that many children who were prescribed antibiotics did not have a bacterial illness. If the score reflects the true likelihood of bacterial infection, its implementation could reduce unnecessary investigation, hospitalization, and treatment with antibiotics. Confirmation that the DRS provides an accurate estimate of bacterial infection in the large group of patients with negative cultures, for whom there is no reference standard, can only be achieved in prospective clinical trials. Careful consideration will be needed to design an ethically acceptable and safe trial in which observation without antibiotic administration is undertaken for febrile children suggested by DRS to be at low risk of bacterial infection.

In comparison with the high frequency of common viral infections in febrile children presenting to health care, inflammatory and vasculitic illness are very rare.²⁶⁻²⁹ However, children presenting with inflammatory or vasculitic conditions commonly undergo extensive investigation to exclude bacterial infection and treatment with antibiotics before the correct diagnosis is made. Although children with inflammatory conditions were not included in the discovery process, the 2-transcript signature was able to distinguish bacterial infection from systemic lupus erythematosus, juvenile idiopathic arthritis, and Henoch-Schönlein purpura. Additional studies including a wider range of inflammatory diseases are needed to assess use of the signature for excluding bacterial infection in inflammatory diseases.

This study has a number of important limitations. The cross-sectional design aimed to recruit equal numbers of children with bacterial and viral infections. The numbers of children recruited thus did not reflect the usual frequency of bacterial infection in febrile children presenting to health care facilities. Further studies of a test based on the 2-transcript signature in unselected febrile children will be needed to provide information on positive and negative predictive performance of the test.

A second limitation is that validation of the signatures was undertaken in groups that included a high proportion of patients requiring intensive care, and with a relatively narrow spectrum of pathogens, which may not reflect the spectrum of infection in other settings. The signature performed well, both in patients with less severe infection and those admitted to intensive care, and performance was not influenced by duration of illness. However, further studies will be needed to evaluate the DRS signature in less severely ill patients with a wider range of infections or in settings such as emergency departments or outpatient offices. Another limitation is the use in validation of published datasets and data obtained using different microarray platforms. Although batch effects were minimized computationally, additional studies are needed in which gene expression is measured on identical platforms.

A major challenge in using transcriptomic signatures for diagnosis is the translation of multitranscript signatures into

clinical tests suitable for use in hospital laboratories or at the bedside. The DRS signature, distinguishing viral from bacterial infections with only 2 transcripts, has potential to be translated into a clinically applicable test using current technology such as polymerase chain reaction.³⁰ Furthermore, new methods for rapid detection of nucleic acids, including nanoparticles and electrical impedance, have potential for low-cost, rapid analysis of multitranscript signatures.

Conclusions

This study provides preliminary data regarding test accuracy of a 2-transcript host RNA signature discriminating bacterial from viral infection in febrile children. Further studies are needed in diverse groups of patients to assess accuracy and clinical utility of this test in different clinical settings.

ARTICLE INFORMATION

Correction: This article was corrected online on February 7, 2017, for errors in the text, Figure 1, and the online Supplement.

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Author Contributions: Drs Herberg and Levin had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Herberg, Kaforou, and Wright contributed equally to this work. *Concept and design:* Herberg, Kaforou, Wright, Levin.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Herberg, Kaforou, Wright, Levin.

Critical revision of the manuscript for important intellectual content: All authors.

Statistical analysis: Herberg, Kaforou, Wright, Eleftherohorinou, Hoggart, Carter, Janes, Barendregt, Levin.

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Additional Information: People linked to the IRIS consortium through membership of the GENDRES consortium (<http://www.gendres.org>) are listed in the eAppendix in the Supplement.

Reproducible Research Statement: The data discussed in this publication have been deposited in NCBI's Gene Expression Omnibus (Edgar R, Domrachev M, Lash AE, et al. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Res.* 2002;30(1):207-210) and are accessible through GEO Series accession number GSE72829 (<http://www.ncbi.nlm.nih.gov/geo/>).

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Vaccine Development

Monday, June 21, 2021 7:02 PM

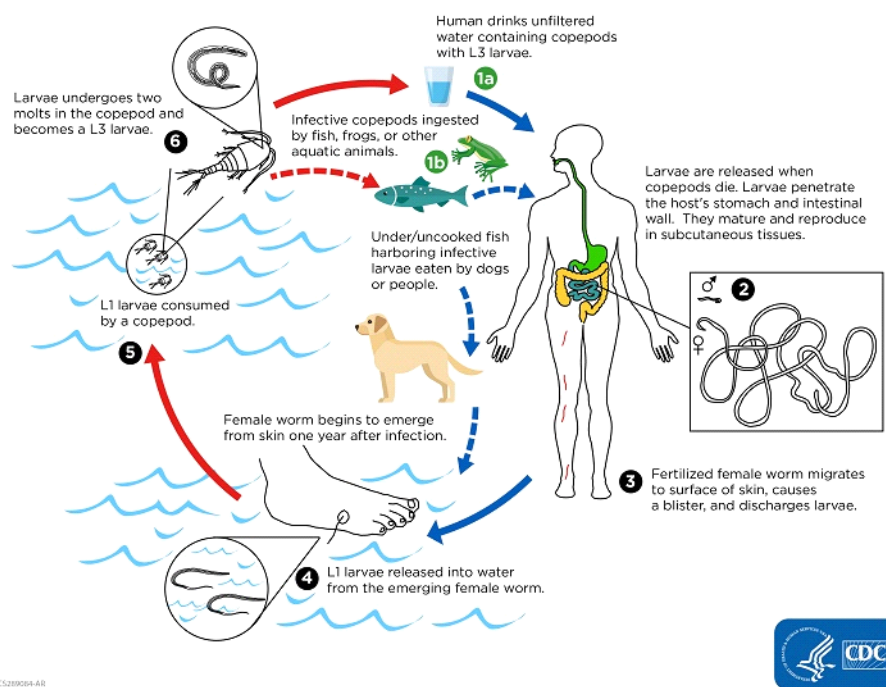
- 06/21/21

Monday, June 21, 2021 7:03 PM

Target Identification

Websites:

1. <https://www.cdc.gov/parasites/guineaworm/biology.html>
2. <https://www.uniprot.org/proteomes/?query=+Dracunculus+medinensis&sort=score>
3. http://weizhong-lab.ucsd.edu/cdhit_suite/cgi-bin/index.cgi?cmd=cd-hit



Dracunculiasis (guinea worm disease) is caused by the nematode (roundworm) *Dracunculus medinensis*.

Humans become infected by drinking unfiltered water containing copepods (small crustaceans) which are infected with larvae of *D. medinensis*.

①

Following ingestion, the copepods die and release the larvae, which penetrate the host stomach and intestinal wall and enter the abdominal cavity and retroperitoneal space.

②

After maturation into adults and copulation, the male worms die and the females (length: 70 to 120 cm) migrate in the subcutaneous tissues towards the skin surface.

③

3

Approximately one year after infection, the female worm induces a blister on the skin, generally on the distal lower extremity, which ruptures. When this lesion comes into contact with water, a contact that the patient seeks to relieve the local discomfort, the female worm emerges and releases larvae.

4

The larvae are ingested by a copepod

5

and after two weeks (and two molts) have developed into infective larvae

6

Ingestion of the copepods closes the cycle.

1

Procedure:

1. Obtain fasta file for *Dracunculus medinensis*. There are 2, so download both from website #2. (Image 1)

<https://www.uniprot.org/proteomes/?query=+Dracunculus+medinensis&sort=score>



Uniprot file

2. Remove duplicates, by uploading the file onto website #3 (Image 2)

http://weizhong-lab.ucsd.edu/cdhit_suite/cgi-bin/index.cgi?cmd=cd-hit

3. Download file, paste in a word document, and upload to website #4 (BLAST) (Image 3)

<https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins>

a. Needs to be done one by one (Protein by Protein)

```
>tr|A0A158Q4M0|A0A158Q4M0_DROME_GDP-D-mannose dehydratase OS=Dracunculus medinensis OX=318479 GN=DME_LOCUS8993 PE=3 SV=1
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TARLHLYSNFVTHYGDQNFLEHYGDMTDSCLIKLISQIQPTEVYHLAAQSHVKSFDL
VEYTAEDVAVGTLRLDLSHACGLTNKIRFYQASTSELYGMQVEVPQKETTPFYPRSPYA
VAKLYSVIVINREAYNMFACNGLFNHESPRGCTFVTRKITRAVAKIALGQDSDIEL
QNLNSRRDNGHAREYVEAMRILQHDVDDPFIATGKNYAVRDFVQYAFNEIGETIVWEG
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SDLOMKANPLA
>tr|A0A0N4UP02|A0A0N4UP02_DROME Protein disulfide-isomerase OS=Dracunculus medinensis OX=318479 GN=DME_LOCUS10720 PE=3 SV=1
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LEFHWQAAELDGKVKLGALDATVHVNMASRYGKGFPTIKYFAPGSSASDAEDYNGGR
SASDIQWALEKAAENRPAPFVLEGLIAQTVFDDACKERQLCIAVLDPILDQSKCRNDY
IQVLDKLSKFKKLNKGNVVEATKQPELEAFGMGGFGYPAMVAVNYRKMFSMLKGSF
GKTDIHEFLRDLISYGGQTPAPVKGAEFFKIRKIQFWDGKGEMPMVEDIDLSDVDLDGDK
TEI
>tr|A0A158Q4T2|A0A158Q4T2_DROME Phosphatidylinositol-3-phosphate phosphatase OS=Dracunculus medinensis OX=318479 GN=DME_LOCUS700 PE=3
SV=1
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RQGIADKHLKLNINDPVDYRYCDYPELYCTPSSATQILIGSKFRSRARLPVLTYYW
RRPGRFCRCAGPLTGFTARCVEDEGLANLIATNPSYQSCLIDTRFLVNAMNKKWGG
KGYDIKNYPNRRFFPDIENTHVNRSLSKLLDAGQRTKITIDYKQLESGLRHIRS
LLDQAFPLADLVSRGISAVHSCSDGNDRTSQTVSLAQLLLDPFFRTIRGFVLIKDWLG
PCHVDDRCCHIGALNEAAKEVSPITPQFLDCVYQIMRLQPRAFEFNERYLIDIEHAY
SCQVTFELGCKDCKRDLNLAKRTLSLWSYIDHDDYINPLYEFQANKFSYLANVDLHP
SAIVWTAMYNRFDTGLPRESTDTVTIATLEHIGVLEAQMASLQAVTFAPSLPMKMI
SGVMSDSGHSSTVSSGAQPTPLQSLPELCMLLIFLHGIIDNTSFPALQWQSLRADEC
SSKNCIAEFATRVRRFRCYRCCKIHCRCCTVTSQDRDRICYSCAAVNAAT
```

GDP-D-mannose dehydratase

```

SDLOLMKANPLA
>tr|A0A0N4UP02|A0A0N4UP02_DROME Protein disulfide-isomerase OS=Dracunculus medinensis OX=318479 GN=DME_LOCUS10720 PE=3 SV=1
VVGCVAFVNDLDVIELTDSNPNKSVIGSNDVWVVEFYAPWCGHCQSFAQEYKAGKIL
KGFVKGAVDATKYSSLQRFSLQGFPPTVKIFGSNNKQPIDFNGARTQAALSDASISEIR
KVIISRLGVKTDSSGSGGTGGGANKVIELTDSNFDLVLNSKDIWLVEFFAPWCGHCCKN
LEFWRQAAALDGGKVLGALDATVHNVMSRYGKGFPTIKYFAPGSSASDAEDYNGGR
SASDIQWALEKAAENRPAPEVLGTAQTVFDDACKELQICIIAVLPDILDCQSKCRNDY
IQKLDLSEKFKNNLGNWVVEATKQPELEAFGMGGFGYPAMVAVNYRKMKFSMLKGSF
GRTIIEFRLDLSYGRGQTAPVKGAEFFKIRKIQPDGDKGEMFVMDIDLSDVDLGDHK
TEL
>tr|A0A158Q4T2|A0A158Q4T2_DROME Phosphatidylinositol-3-phosphate phosphatase OS=Dracunculus medinensis OX=318479 GN=DME_LOCUS700 PE=3
SV=1
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KGYDIKNYPNMRFBFFDIENIHVMRSNLNLLDACQRTKITDYYKQLESSGMLRHRS
LLDCAFPLADLVSRGISAVVHCSGWDRTSQTVSLAQLLLDPFFRTIRGFOVLTEDWL
FGHMDDRCGHIGALNEEAKEVSPITQFLDCVQIMRLQPAFEFNERYLIDIEHAY
SCQYTFLGNCKDKRDLNLAKRTLSLWSYIDHDDYINPLYEQANKFSYLANVDLHP
SAIIVWTAMYNRFDTGLPRESTDTVTIATLEHIGVLEAQMASLQAVTFLAPSFLPMKI
SGVMSDGHSTVSSGAQPTTFLQSLPELCMLIFLHGIIDNTSFFALQWQSLRADEC
SSKNCAEFAFTRVRRFHCYRCGKIHCRRKCVTTSQDRICYSCAAVNAT

```

Image 1. Screenshot of the code generated by Uniprot.

The screenshot shows the NCBI BLAST web interface. In the 'Enter Query Sequence' section, a protein sequence is pasted into the text box. The 'Choose Search Set' section shows 'Reference proteins (refseq_protein)' selected for the database and 'human (taxid:9606)' selected for the organism. The 'Program Selection' section shows 'Blastp (protein-protein BLAST)' selected as the algorithm. The 'BLAST' button is circled in red.

Image 2. BLAST (Basic Local Alignment Search Tool) is a program used to generate alignment between a nucleotide or protein sequence, referred to as 'query' and nucleotide or protein sequences within a database, referred to as 'subject' sequences.

Chrome File Edit View History Bookmarks Profiles Tab Window Help 48.1% 93 °C 1995 rpm 39% Mon 7:47 PM f4l

blast.ncbi.nlm.nih.gov/Blast.cgi

NIH U.S. National Library of Medicine
National Center for Biotechnology Information

BLAST® » **blastp suite** » RID-D2G9WAZW016 Home Recent Results Saved Strategies Help

[Formatting options]

Job Title: trjA0A158Q4M0|A0A158Q4M0_DRAVE GDP-D-mannose...

Request ID	D2G9WAZW016
Status	Searching
Submitted at	Mon Jun 21 21:47:40 2021
Current time	Mon Jun 21 21:47:53 2021
Time since submission	00:00:13

This page will be automatically updated in 7 seconds

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National Center for Biotechnology Information
8600 Rockville Pike

Popular	Resources	Actions
PubMed	Literature	Submit
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Bookshelf	Genomes	Learn

Blasting Protein sequences help us find similarities with human subject sequences in our *Dracunculus medinensis* query of proteins. For example: GDP-D-Mannose Dehydratase has 12 sequences match with at least 67% percent identity.

NCBI Blast:trjA0A158Q4M0|A0A158Q4M0_DRAVE GDP-D-mannose... RCSB PDB - 7DMU: Structure

blast.ncbi.nlm.nih.gov/Blast.cgi

< Edit Search Save Search Search Summary How to read this report? BLAST Help Videos Back to Traditional Results Page

Your search is limited to records that include: human (taxid:9606)

Job Title: trjA0A158Q4M0|A0A158Q4M0_DRAVE GDP-D-mannose...
 RID: D2G9WAZW016 Search expires on 06-23 09:47 am Download All
 Program: BLASTP Citation
 Database: refseq_protein See details
 Query ID: lcl|Query_12068
 Description: trjA0A158Q4M0|A0A158Q4M0_DRAVE GDP-D-mannose...
 Molecule type: amino acid
 Query Length: 372
 Other reports: Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism only top 20 will appear ☐ exclude
 Type common name, binomial, taxid or group name
 + Add organism

Percent Identity to E value to Query Coverage to
 Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

☒ select all 12 sequences selected

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
☒ GDP-mannose 4.6 dehydratase isoform 1 (Homo sapiens)	Homo sapiens	528	528	95%	0.0	67.42%	372	NP_001491.1
☒ GDP-mannose 4.6 dehydratase isoform X1 (Homo sapiens)	Homo sapiens	509	509	91%	0.0	67.65%	342	XP_011512802.1
☒ GDP-mannose 4.6 dehydratase isoform 2 (Homo sapiens)	Homo sapiens	509	509	91%	0.0	67.85%	342	NP_001240775.1
☒ GDP-mannose 4.6 dehydratase isoform X2 (Homo sapiens)	Homo sapiens	480	480	84%	2e-170	69.33%	334	XP_011512804.1
☒ GDP-mannose 4.6 dehydratase isoform X3 (Homo sapiens)	Homo sapiens	388	388	66%	6e-135	70.28%	290	XP_006715129.1
☒ GDP-mannose 4.6 dehydratase isoform X4 (Homo sapiens)	Homo sapiens	385	385	71%	6e-134	65.41%	285	XP_016666241.1
☒ GDP-mannose 4.6 dehydratase isoform X5 (Homo sapiens)	Homo sapiens	329	329	53%	1e-112	73.87%	222	XP_011512809.1

7dmu.pdb Feedback Show All

- 06/22/21

Monday, June 21, 2021 8:36 PM

<http://www.ddg-pharmfac.net/vaxijen/VaxiJen/VaxiJen.html>

```
>tr|A0A158Q4M0|A0A158Q4M0|DRAKE-GDP-D-mannose dehydratase OS=Dracunculus  
medinensis OX=318479 GN=DME LOCUS8993 PE=3 SV=1
```

```
MQGFESCLYQTIMNYTEAFRRRKIALITGISQDQGSYLAELLSSKGYKVHGIIRSSSFN  
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PEYTAEDVAVGTLRLLDSIHACGLTNKIRFYQASTSELYGMVQEVPOKETTPFYRSPYA  
VAKLYSYWIVINYRHAYNMFACNGILFNHESPRRGETFVTRKITRAVAKIALGQODSIEL  
GNLNSRRDWGHAREYVEAMWRILQHDVPDDFVIATGKNYAVRDFVQYAFNEIGETIVWEG  
EGINEIGKEQRTGKIRVKVNPKYRPTVDLLIGDSTKAKLKLGWAKVQLQRTIVKEMVA  
SDLQLMKANPLA
```

Overall Prediction for the Protective Antigen = **0.3189** (Probable **NON-ANTIGEN**).

The screenshot shows the VaxiJen v2.0 web interface. The browser address bar displays the URL <http://www.ddg-pharmfac.net/vaxijen/VaxiJen/VaxiJen.html>. The page title is "VaxiJen v2.0". Below the title, there is a red banner with the text "VaxiJen: Prediction of Protective Antigens and Subunit Vaccines." and "For jobs containing >100 proteins please contact ldimitrov@pharmfac.mu-sofia.bg." The main form area contains a text input field for the protein sequence, which has been populated with the sequence: MQGFESCLYQTIMNYTEAFRRRKIALITGISQDQGSYLAELLSSKGYKVHGIIRSSSFN TARIEHLY\$NPVTHYGDQNFKLHYGDMTDSSCLIKLISQIQPTEVYHLAAQSHVKVSFDL PEYTAEDVAVGTLRLLDSIHACGLTNKIRFYQASTSELYGMVQEVPOKETTPFYRSPYA VAKLYSYWIVINYRHAYNMFACNGILFNHESPRRGETFVTRKITRAVAKIALGQODSIEL GNLNSRRDWGHAREYVEAMWRILQHDVPDDFVIATGKNYAVRDFVQYAFNEIGETIVWEG EGINEIGKEQRTGKIRVKVNPKYRPTVDLLIGDSTKAKLKLGWAKVQLQRTIVKEMVA SDLQLMKANPLA. Below the sequence input, there is a section for selecting a target organism, with "Virus" selected. There is also a threshold input field set to 0.5. At the bottom of the form, there are checkboxes for "ACC Output", "Sequence Output" (checked), and "Summary Mode". The "Submit" button is visible. The footer of the page includes links to "VaxiJen Help Page", "Other Prediction Services", "Citation", "Bacterial Immunogens dataset", and "Other EJIIR Bioinformatics Web-Sites: AntiJen, EpiJen, MHCPre, LipPre, BPROMP".

Choose Service | BioCode x Table Browser x Outbound - Google Sheets x Screening Antigenicity Of P... x VaxiJen - predicting prote... x

Not Secure ddg-pharmfac.net/vaxijen/scripts/VaxiJen_scripts/VaxiJen3.pl

VaxiJen v2.0

VaxiJen RESULTS

Model selected: parasite

Threshold for this model: 0.5

Your Sequence:

```

MQGFESCLYQTIMNYTEAFRRRKIALITGIS
GQDGSYLAELLLSGYKVHGIIRSSSFNTA
RIEHLYSNPVTHYGDQNFKLHYGDMTSSCL
IKLISIQIPTVYHLAAQSHVKVFDLPEYT
AEVDAVGTLRLLDSIHACGLTNKIRFYQAST
SELYGMVQEVQKETTPFYPRSPYAVAKLYS
YWIVINYREAYNMFACNGILFNHESPRRGET
FVTRKITRAVAKIALGQDSEIELGNLSRRD
WGHAREYVEAMWRILQHDVPDDFVIATGKNY
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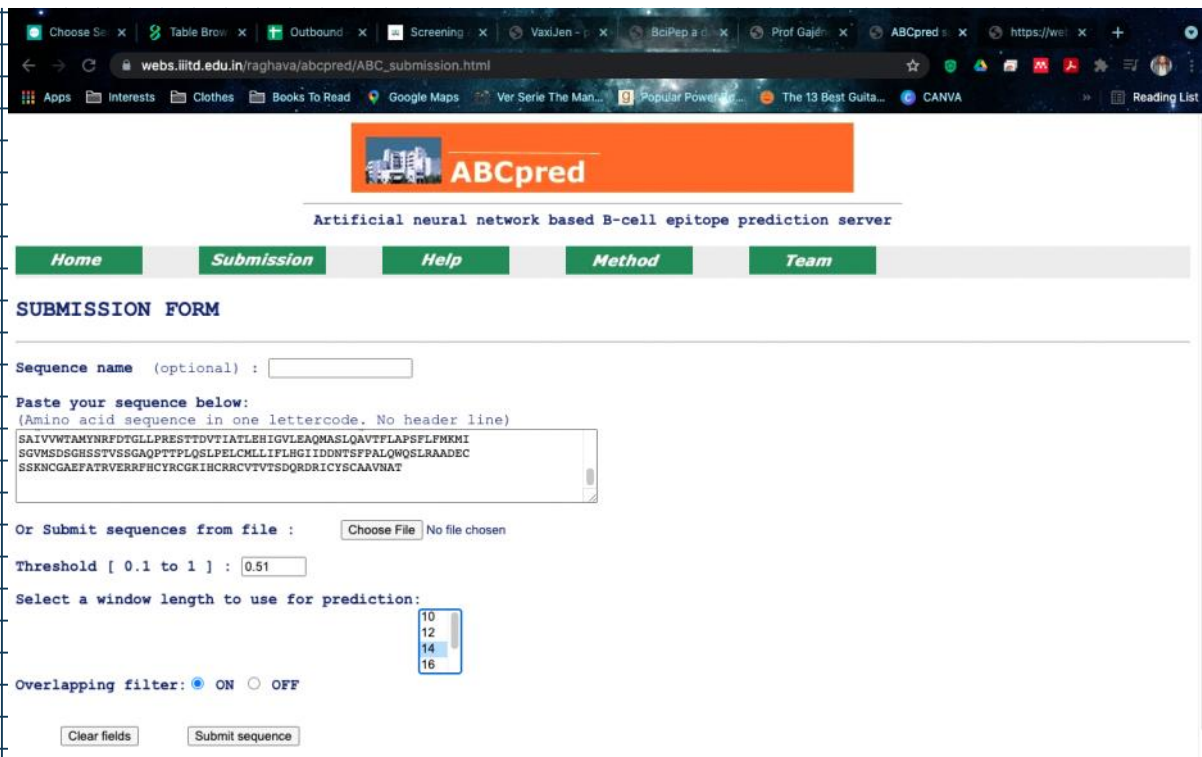
```

Overall Prediction for the Protective Antigen = 0.3189 (Probable **NON-ANTIGEN**).

>tr|A0A158Q4T2_A0A158Q4T2_DROME_Phosphatidylinositol-3-phosphate phosphatase
OS=Dracunculus medinensis OX=318479 GN=DME_LOCUS700 PE=3 SV=1

QLIDKFGSESNIIVGTIHTTSHFIFRADNSSKEIWWANALIGNIEKGNLSAIGTYLTIRC
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RQGIGIADKHLWKINDFNVDYRYCDTYPEYLCTPSS\$ATKQIIGSCKFRSRARLPVITYW
HRPNFGRFCRCAQPLTGFTARCVEDQELMNLIAKTNPSYQSICLIDTRPLVNAMNMKMQG
KGYEDIKNYPNMRFFHFDIENIHVMRSSLNKLDDACQRTKTITDYYKQLESSGWLRIHS
LLDCSAFLADLVSRGISAVVHCSDGWDRTSQTVSLAQLLDFFFTIRGFOVLEKDWLG
FCHKFDDRCGHICALNEEAKEVSPIFTQFLDCVYQIMRIQPRAFEFNERYLIDIEHAY
SQQYGTFLGNCDKDRKDLNIAKRTLSLWSYIDDHDDYINPIYEFQANKESYLANVDLHP
SAIVVWTAMYNRFDTGILLPRESTTDVTIATLEHIGVLEAQMASLOAVTFLAP\$FLFMKMI
SGVMSDSGHSSTVSSGAQPTTFLQSLPELCMLLI FLHGIIDDNTSFPALQWQSLRAADEC
SSKNCGAEFATRVERRFHCYRCGKIHCRCVTVTSDQDRICYS CAAVNAT

Overall Prediction for the Protective Antigen = 0.5253 (Probable **ANTIGEN**).



The screenshot shows the ABCpred website interface. At the top, there's a navigation bar with links: Home, Submission, Help, Method, and Team. Below this is the 'SUBMISSION FORM' section. It includes a 'Sequence name (optional)' text box, a 'Paste your sequence below' section with a text area containing an amino acid sequence, and a 'Choose File' button for submitting sequences from a file. There are also input fields for 'Threshold [0.1 to 1]' (set to 0.51) and a 'Select a window length to use for prediction' dropdown menu (set to 14). An 'Overlapping filter' section has radio buttons for 'ON' (selected) and 'OFF'. At the bottom of the form are 'Clear fields' and 'Submit sequence' buttons.

[RAGHAVA Web Services - B Cell Epitope prediction server](#)



Predicted
B-cell epit...

[Sheets Document W/ Antigenicity & Immunogenicity](#)

[IFDB Analysis Resource - Class I Immunogenicity](#)

Rank	Sequence	Start position	Score
1	RCGHIGALNEEAAK	368	0.91
2	SLAQLLLDPFFRTI	334	0.89
3	TITDYYKQLESSGW	281	0.88
4	EKNGRKRCWDRFNW	102	0.87
5	ATKQLIGSCKFRS	157	0.86
5	WKINDFNVDYRYCD	132	0.86
6	KCGRPVNIIDVFAF	85	0.85
6	NFLIAKDKDCQDLY	67	0.85
6	DEQLMNLIAKTNPS	205	0.85
6	FARQGIGIADKHLW	119	0.85
6	CWDRFNWQIEFARQ	109	0.85
7	LDACQRTKTITDYY	273	0.84
8	SDSGHSSTVSSGAQ	545	0.83
8	SDGWDRTSQTVSLA	323	0.83
9	IDVFAFENREKNR	93	0.82
9	VTIATLEHIGVLEA	506	0.82
9	EKGNLSAIGTYLTI	45	0.82
9	SNIVGTIHITTSHF	10	0.82
10	VVWTAMYNRFDTG	484	0.81
10	PLTGFTARCEDEQ	194	0.81
11	VLEKDWLGFCHKF	352	0.80
12	QANKFSYLANVDLH	466	0.79
12	SAVVHCSDGWDRTS	317	0.79
13	LAKRTLSLWSYIDD	440	0.78
13	IFRADNSSKEIWVA	24	0.78
13	QKGKGYEDIKNYPNM	239	0.78
14	YQIMRLQPRAFEFN	395	0.77
14	RFHFFDIENIHVMR	253	0.77
14	NPSYQSLCLIDTRP	216	0.77
14	NVDYRYCDTYPEYL	138	0.77
15	KDRKDLNLAKRSL	433	0.76
15	LIGSCKFRSRARLP	162	0.76
16	STVSSGAQPTPLQ	551	0.75
16	QMASLQAVTFLAPS	520	0.75
17	APSFLEFMKMSGVM	531	0.74
17	NYPNMRFHFFDIEN	248	0.74
17	KFRSRARLPVLTW	167	0.74
18	HIGVLEAQMASLQA	513	0.73
18	VANALIGNIEKGNL	36	0.73
18	NIHVMRSSLNKLDD	261	0.73
19	YLCTPSSATKQILI	150	0.72
20	RSLLDCSAFLADLV	299	0.71
20	HRPNFGRFCRCAQP	181	0.71
21	KMISGVMSDSGHSS	538	0.70
21	FLADLVSRGISAVV	307	0.70
22	WSYIDDDHDDYINP	448	0.69
22	FEFNERYLIDIEH	405	0.69
22	FRTIRGFQVLEKD	344	0.69
23	YNRFDTGLLPREST	490	0.68
23	HKFDDRCGHIGALN	363	0.68
24	LHPSAIVVWTAMYN	478	0.67
24	RFCRCAQPLTGFTA	187	0.67
25	DLYETLLKCGRPVN	78	0.66
25	GALNEEAAKEVSPI	373	0.66

26	QAVTFLAPSLFMK	525	0.63
26	MMNKMQGKGYEDIK	234	0.63
26	ARLPVLTYWHRPNF	172	0.63
27	AKEVSPFTQFLDC	380	0.62
27	RTSQTVSLAQLLLD	328	0.62
28	NSSKEIWVANALIG	29	0.58
29	HHDDYINPLYEFQA	454	0.57
30	GAQPTTPLQSLPEL	556	0.56
30	TIHITTSHFIFRAD	15	0.56
31	IDTRPLVNAMMNKM	225	0.53
32	HEHAYSCQYGTFLG	416	0.51

Choc x Table x Outb x Scre x VaxiJ x BciPe x Phos x Prof G x ABCp x https x Class x Phos x +

Not Secure | tools.iedb.org/immunogenicity/

Apps Interests Clothes Books To Read Google Maps Ver Serie The Man... Popular Power... The 13 Best Guita... CANVA Reading List

IEDB Analysis Resource

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Class I Immunogenicity

Specify sequence(s) *

Enter peptide sequence(s)

RCGHIGALNEEAAR

Or select file containing sequence(s) Choose File No file chosen

Choose which positions to mask

☒ Default (1st, 2nd, and C-terminus amino acids)

☐ Custom User Defined

Specify which positions to mask

(Comma separated numbers)

Peptide lengths must be equal when using custom masking.

Submit Reset

*The tool was only validated for 9-mer peptides. However, predictions can be made for peptides of any length.

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Choc x Table x Outb x Scre x VaxiJ x BciPe x Phos x Prof G x ABCp x https x Immun x +

Not Secure | tools.iedb.org/immunogenicity/result/

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Class I Immunogenicity

Masking: default
Masked variables: [1, 2, 'cterm']

Peptide	Length	Score
RCGHIGALNEEAAR	14	0.47478

Download result

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Supported by a contract from the [National Institute of Allergy and Infectious Diseases](#), a component of the National Institutes of Health in the Department of Health and Human Services.

The screenshot shows a web browser window with the URL ddg-pharmfac.net/vaxijen/scripts/VaxiJen_scripts/VaxiJen3.pl. The page title is "VaxiJen v2.0". Below the title, there is a red header bar with the text "VaxiJen RESULTS". The main content area displays the following information:

- Model selected:** parasite
- Threshold for this model:** 0.5
- Your Sequence:** RCGHIGALNEEAAK
- Overall Prediction for the Protective Antigen = 1.0658 (Probable ANTIGEN).**

3 proteins of choice from genome:

1. Phosphatidylinositol-3-phosphate phosphatase (OS - Dracunculus Medinensis)

Fundamentals for T & B cell epitope prediction

Thursday, June 24, 2021 5:25 PM



da1dc063b
517139b8...

Review Article

Fundamentals and Methods for T- and B-Cell Epitope Prediction

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Adaptive immunity is mediated by T- and B-cells, which are immune cells capable of developing pathogen-specific memory that confers immunological protection. Memory and effector functions of B- and T-cells are predicated on the recognition through specialized receptors of specific targets (antigens) in pathogens. More specifically, B- and T-cells recognize portions within their cognate antigens known as epitopes. There is great interest in identifying epitopes in antigens for a number of practical reasons, including understanding disease etiology, immune monitoring, developing diagnosis assays, and designing epitope-based vaccines. Epitope identification is costly and time-consuming as it requires experimental screening of large arrays of potential epitope candidates. Fortunately, researchers have developed *in silico* prediction methods that dramatically reduce the burden associated with epitope mapping by decreasing the list of potential epitope candidates for experimental testing. Here, we analyze aspects of antigen recognition by T- and B-cells that are relevant for epitope prediction. Subsequently, we provide a systematic and inclusive review of the most relevant B- and T-cell epitope prediction methods and tools, paying particular attention to their foundations.

1. Introduction

The immune system is typically divided into two categories, innate and adaptive. Innate immunity involves nonspecific defense mechanisms that act immediately or within hours after a microbe appearance in the body. All multicellular beings exhibit some kind of innate immunity. In contrast, adaptive immunity is only present in vertebrates and it is highly specific. In fact, the adaptive immune system is able to recognize and destroy invading pathogens individually. Moreover, the adaptive immune system remembers the pathogens that fights, acquiring a pathogen-specific long-lasting protective memory that enables stronger attacks each time the pathogen is reencountered [1]. Nonetheless, innate and adaptive immune mechanisms work together and adaptive immunity elicitation is contingent on prior activation of innate immune responses [1].

Adaptive immunity is articulated by lymphocytes, more specifically by B- and T-cells, which are responsible for the

humoral and cell-mediated immunity. B- and T-cells do not recognize pathogens as a whole, but molecular components known as antigens. These antigens are recognized by specific receptors present in the cell surface of B- and T-cells. Antigen recognition by these receptors is required to activate B- and T-cells but not enough, as second activation signals stemming from the activation of the innate immune system are also needed. The specificity of the recognition is determined by genetic recombination events that occur during lymphocyte development, which lead to generating millions of different variants of lymphocytes in terms of the antigen-recognizing receptors [1]. Antigen recognition by B- and T-cells differ greatly.

B-cells recognize solvent-exposed antigens through antigen receptors, named as B-cell receptors (BCR), consisting of membrane-bound immunoglobulins, as shown in Figure 1. Upon activation, B-cells differentiate and secrete soluble forms of the immunoglobulins, also known as antibodies, which mediate humoral adaptive immunity.

How?

Only in vertebrates, then: hemopoietic & lymph

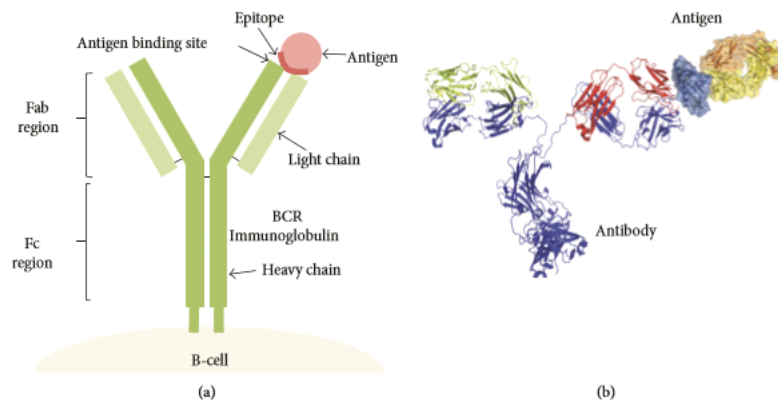


FIGURE 1: B-cell epitope recognition. B-cell epitopes are solvent-exposed portions of the antigen that bind to secreted and cell-bound immunoglobulins. (a) B-cell receptors encompass cell-bound immunoglobulins, consisting of two heavy chains and two light chains. The different chains and regions are annotated. (b) Molecular representation of the interaction between an antibody and the antigen. Antibodies are secreted immunoglobulins of known specificity.

Antibodies released by B-cells can have different functions that are triggered upon binding their cognate antigens. These functions include neutralizing toxins and pathogens and labeling them for destruction [1].

A B-cell epitope is the antigen portion binding to the immunoglobulin or antibody. These epitopes recognized by B-cells may constitute any exposed solvent region in the antigen and can be of different chemical nature. However, most antigens are proteins and those are the subjects for epitope prediction methods.

On the other hand, T-cells present on their surface a specific receptor known as T-cell receptor (TCR) that enables the recognition of antigens when they are displayed on the surface of antigen-presenting cells (APCs) bound to major histocompatibility complex (MHC) molecules. T-cell epitopes are presented by class I (MHC I) and II (MHC II) MHC molecules that are recognized by two distinct subsets of T-cells, CD8 and CD4 T-cells, respectively (Figure 2). Subsequently, there are CD8 and CD4 T-cell epitopes. CD8 T-cells become cytotoxic T lymphocytes (CTL) following T CD8 epitope recognition. Meanwhile, primed CD4 T-cells become helper (Th) or regulatory (Treg) T-cells [1]. Th cells amplify the immune response, and there are three main subclasses: Th1 (cell-mediated immunity against intracellular pathogens), Th2 (antibody-mediated immunity), and Th17 (inflammatory response and defense against extracellular bacteria) [2].

Identifying epitopes in antigens is of great interest for a number of practical reasons, including understanding disease etiology, immune monitoring, developing diagnosis assays, and designing epitope-based vaccines. B-cell epitopes can be identified by different methods including solving the 3D structure of antigen-antibody complexes, peptide library screening of antibody binding or performing functional assays in which the antigen is mutated and the interaction antibody-antigen is evaluated [3, 4]. On the other hand,

experimental determination of T-cell epitopes is carried out using MHC multimers and lymphoproliferation or ELISPOT assays, among others [5, 6]. Traditional epitope identification has depended entirely upon experimental techniques, being costly and time-consuming. Thereby, scientists have developed and implemented epitope prediction methods that facilitate epitope identification and decrease the experimental load associated with it. Here, we will first analyze aspects of antigen recognition by T- and B-cells that are relevant for a better understanding of the topic of epitope prediction. Subsequently, we will provide a systematic and inclusive review of the most important prediction methods and tools, paying particular attention to their foundations and potentials. We will also discuss epitope prediction limitations and ways to overcome them. We will start with T-cell epitopes.

2. T-Cell Epitope Prediction

T-cell epitope prediction aims to identify the shortest peptides within an antigen that are able to stimulate either CD4 or CD8 T-cells [7]. This capacity to stimulate T-cells is called immunogenicity, and it is confirmed in assays requiring synthetic peptides derived from antigens [5, 6]. There are many distinct peptides within antigens and T-cell prediction methods aim to identify those that are immunogenic. T-cell epitope immunogenicity is contingent on three basic steps: (i) antigen processing, (ii) peptide binding to MHC molecules, and (iii) recognition by a cognate TCR. Of these three events, MHC-peptide binding is the most selective one at determining T-cell epitopes [8, 9]. Therefore, prediction of peptide-MHC binding is the main basis to anticipate T-cell epitopes and we will review it next.

2.1. Prediction of Peptide-MHC Binding. MHC I and MHC II molecules have similar 3D-structures with bound peptides sitting in a groove delineated by two α -helices overlying a

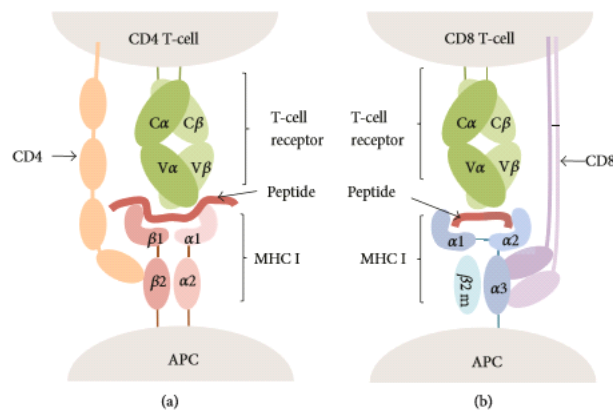


FIGURE 2: T-cell epitope recognition. T-cell epitopes are peptides derived from antigens and recognized by the T-cell receptor (TCR) when bound to MHC molecules displayed on the cell surface of APCs. (a) CD4 T-cells express the CD4 coreceptor, which binds to MHC II, and recognize peptides presented by MHC II molecules. (b) CD8 T-cells express the CD8 coreceptor, which binds to MHC I, and recognize peptides presented by MHC I molecules.

floor comprised of eight antiparallel β -strands. However, there are also key differences between MHC I and II binding grooves that we must highlight for they condition peptide-binding predictions (Figure 3). The peptide-binding cleft of MHC I molecules is closed as it is made by a single α chain. As a result, MHC I molecules can only bind short peptides ranging from 9 to 11 amino acids, whose N- and C-terminal ends remain pinned to conserved residues of the MHC I molecule through a network of hydrogen bonds [10, 11]. The MHC I peptide-binding groove also contains deep binding pockets with tight physicochemical preferences that facilitate binding predictions. There is a complication however. Peptides that have different sizes and bind to the same MHC I molecule often use alternative binding pockets [12]. Therefore, methods predicting peptide-MHC I binding require a fixed peptide length. However, since most MHC I peptide ligands have 9 residues, it is generally preferable to predict peptides with that size. In contrast, the peptide-binding groove of MHC II molecules is open, allowing the N- and C-terminal ends of a peptide to extend beyond the binding groove [10, 11]. As a result, MHC II-bound peptides vary widely in length (9–22 residues), although only a core of nine residues (peptide-binding core) sits into the MHC II binding groove. Therefore, peptide-MHC II binding prediction methods often target to identify these peptide-binding cores. MHC II molecule binding pockets are also shallower and less demanding than those of MHC I molecules. As a consequence, peptide-binding prediction to MHC II molecules is less accurate than that of MHC I molecules.

Given the relevance of the problem, there are numerous methods to predict peptide-MHC binding. The most relevant with free online use are collected on Table 1. They can be divided in two main categories: data-driven and structure-based methods. Structure-based approaches generally rely

on modeling the peptide-MHC structure followed by evaluation of the interaction through methods such as molecular dynamic simulations [8, 13]. Structure-based methods have the great advantage of not needing experimental data. However, they are seldom used as they are computationally intensive and exhibit lower predictive performance than data-driven methods [14].

Data-driven methods for peptide-MHC binding prediction are based on peptide sequences that are known to bind to MHC molecules. These peptide sequences are generally available in specialized epitope databases such as IEDB [15], EPIMHC [16], Antigen [17, 18]. Both MHC I and II binding peptides contain frequently occurring amino acids at particular peptide positions, known as anchor residues. Thereby, prediction of peptide-MHC binding was first approached using sequence motif (SM) reflecting amino acid preferences of MHC molecules at anchor positions [19]. However, it was soon shown that nonanchor residues also contribute to the capacity of a peptide to bind to a given MHC molecule [20, 21]. Subsequently, researchers developed motif matrices (MM), which could evaluate the contribution of each and all peptide positions to the binding with the MHC molecule [22–25]. The most sophisticated form of motif matrices consists of profiles [24–26] that are similar to those used for detecting sequence homology [27]. We would like to remark that motif matrices are often mistaken with quantitative affinity matrices (QAMs) since both produce peptide scores. However, MMs are derived without taking in consideration values of binding affinities and, therefore, resulting peptide scores are not suited to address binding affinity. In contrast, QAMs are trained on peptides and corresponding binding affinities, and aim to predict binding affinity. The first method based on QAMs was developed by Parker et al. [28] (Table 1). Subsequently, various approaches were developed to obtain QAMs from peptide

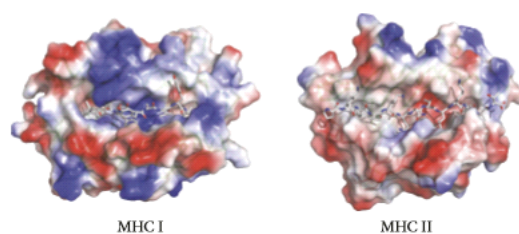


FIGURE 3: MHC molecule binding groove. The figure depicts the molecular surface as seen by the TCR of representative MHC I and II molecules. Note how the binding groove of the MHC I molecule is closed but that of MHC II is open. As a result, MHC I molecules bind short peptides (8–11 amino acids), while MHC II molecules bind longer peptides (9–22 amino acids). The figure was prepared from PDB files 1QRN (MHC I) and 1FYT (MHC II) using PyMol.

TABLE 1: Selected T-cell epitope prediction tools available online for free public use.

Tool	URL	Method ¹	MHC class	A	S	T	P	Ref.
EpiDOCK	http://epidock.ddg-pharmfac.net	SB	II	—	—	—	—	[86]
MotifScan	https://www.hiv.lanl.gov/content/immunology/motif_scan/motif_scan	SM	I and II	—	X	—	—	—
Rankpep	http://imed.med.ucm.es/Tools/rankpep.html	MM	I and II	—	—	—	X	[26]
SYFPEITHI	http://www.syfpeithi.de/	MM	I and II	—	—	—	—	[23]
MAPPP	http://www.mpiib-berlin.mpg.de/MAPPP/	MM	I	—	X	—	X	[87]
PREDIVAC	http://predivac.biosci.uq.edu.au/	MM	II	—	—	—	—	[88]
PEPVAC	http://imed.med.ucm.es/PEPVAC/	MM	I	—	X	—	X	[63]
EPISOPT	http://bio.med.ucm.es/episopt.html	MM	I	—	X	—	—	[64]
Vaxign	http://www.violinet.org/vaxign/	MM	I and II	—	—	—	—	[89]
MHCPred	http://www.ddg-pharmfac.net/mhcpred/MHCPred/	QSAR	I and II	X	—	—	—	[34]
EpiTOP	http://www.pharmfac.net/EpiTOP	QSAR	II	X	—	—	—	[90]
BIMAS	http://bimas.cit.nih.gov/molbio/hla_bind/	QAM	I	X	—	—	—	[28]
TEPITOPE	http://datamining-iip.fudan.edu.cn/service/TEPITOPEpan/TEPITOPEpan.html	QAM	II	X	—	—	—	[32]
Propred	http://www.imtech.res.in/raghava/propred/	QAM	II	X	X	—	—	[91]
Propred-1	http://www.imtech.res.in/raghava/propred1/	QAM	I	X	X	—	X	[92]
EpiJen	http://www.ddg-pharmfac.net/epijen/EpiJen/EpiJen.htm	QAM	I	X	—	X	X	[82]
IEDB-MHCI	http://tools.immuneepitope.org/mhci/	Combined	I	X	—	—	—	[93]
IEDB-MHCII	http://tools.immuneepitope.org/mhcii/	Combined	II	X	—	—	—	[93]
IL4pred	http://webs.iitd.edu.in/raghava/il4pred/index.php	SVM	II	—	—	—	—	[67]
MULTIPRED2	http://cvc.dfci.harvard.edu/multipred2/index.php	ANN	I and II	—	X	—	—	[62]
MHC2PRED	http://www.imtech.res.in/raghava/mhc2pred/index.html	SVM	II	—	—	—	—	[38]
NetMHC	http://www.cbs.dtu.dk/services/NetMHC/	ANN	I	X	—	—	—	[49]
NetMHCII	http://www.cbs.dtu.dk/services/NetMHCII/	ANN	II	X	—	—	—	[30]
NetMHCpan	http://www.cbs.dtu.dk/services/NetMHCpan/	ANN	I	X	—	—	—	[54]
NetMHCIIpan	http://www.cbs.dtu.dk/services/NetMHCIIpan/	ANN	II	X	—	—	—	[55]
nHLApred	http://www.imtech.res.in/raghava/nhlapred/	ANN	I	—	—	—	X	[94]
SVMHC	http://abi.inf.uni-tuebingen.de/Services/SVMHC/	SVM	I and II	—	—	—	—	[95]
SVRMHC	http://us.accurascience.com/SVRMHCdb/	SVM	I and II	X	—	—	—	[46]
NetCTL	http://www.cbs.dtu.dk/services/NetCTL/	ANN	I	X	X	X	X	[83]
WAPP	https://abi.inf.uni-tuebingen.de/Services/WAPP/index_html	SVM	I	—	—	X	X	[37]

¹Method used for prediction of peptide-MHC binding. Keys for methods: SM: sequence motif; SB: structure-based; MM: motif matrix; QAM: quantitative affinity matrix; SVM: support vector machine; ANN: artificial neural network; QSAR: quantitative structure-activity relationship model; combined: tool uses different methods including ANN and QAM, selecting the more appropriate method for each distinct MHC molecule. The table also indicates whether the tools predict quantitative binding affinity (A), supertypes (S), TAP binding (T), and proteasomal cleavage (P); marked with an X in the affirmative case.

affinity data and predict peptide binding to MHC I and II molecules [29–32].

QAMs and motif matrices assume an independent contribution of peptide side chains to the binding. This assumption is well supported by experimental data but there is also evidence that neighboring peptide residues interfere with others [33]. To account for those interferences, researchers introduced quantitative structure-activity relationship (QSAR) additive models wherein the binding affinity of peptides to MHC is computed as the sum of amino acid contributions at each position plus the contribution of adjacent side chain interactions [34]. However, machine learning (ML) is the most popular and robust approach introduced to deal with the nonlinearity of peptide-MHC binding data [8]. Researchers have used ML for two distinct problems: the discrimination of MHC binders from nonbinders and the prediction of binding affinity of peptides to MHC molecules.

For developing discrimination models, ML algorithms are trained on data sets consisting of peptides that either bind or do not bind to MHC molecules. Relevant examples of ML-based discrimination models are those based on artificial neural networks (ANNs) [35, 36], support vector machines (SVMs) [37–39], decision trees (DTs) [40, 41], and Hidden Markov models (HMMs), which can also cope with nonlinear data and have been used to discriminate peptides binding to MHC molecules. However, unlike other ML algorithms, they have to be trained only on positive data. Three types of HMMs have been used to predict MHC-peptide binding: fully connected HMMs [42], structure-optimized HMMs [43], and profile HMMs [43, 44]. Of these, only fully connected HMMs (fcHMMs) and structure-optimized HMMs (soHMMs) can recognize different patterns in the peptide binders. In fact, profile HMMs that are derived from sets of ungapped alignments (the case for peptides binding to MHC) are nearly identical to profile matrices [45] (Table 1).

With regard to predicting binding affinity, ML algorithms are trained on datasets consisting of peptides with known affinity to MHC molecules. Both SVMs and ANNs have been used for such purpose. SVMs were first applied to predict peptide-binding affinity to MHC I molecules [46] and later to MHC II molecules [47] (Table 1). Likewise, ANNs were also applied first to the prediction of peptide binding to MHC I [48, 49] and later to MHC II molecules [50] (Table 1). Benchmarking of peptide-MHC binding prediction methods appears to indicate that those based on ANNs are superior to those based on QAMs and MMs. However, the differences between the distinct methods are marginal and vary for different MHC molecules [51]. Moreover, it has been shown that the performance of peptide-MHC predictions is improved by combining several methods and providing consensus predictions [52].

A major complication for predicting T-cell epitopes through peptide-MHC binding models is MHC polymorphism. In humans, MHC molecules are known as human leukocyte antigens (HLAs), and there are hundreds of allelic variants of class I (HLA I) and class II (HLA II) molecules. These HLA allelic variants bind distinct sets of peptides [53] and require specific models for predicting peptide-

MHC binding. However, peptide-binding data is only available for a minority of HLA molecules. To overcome this limitation, some researchers have developed pan-MHC-specific methods by training ANNs on input data combining MHC residues that contact the peptide with peptide-binding affinity that are capable of predicting peptide-binding affinities to uncharacterized HLA alleles [54, 55].

HLA polymorphism also hampers the development of worldwide covering T-cell epitope-based vaccines as HLA variants are expressed at vastly variable frequencies in different ethnic groups [56]. Interestingly, different HLA molecules can also bind similar sets of peptides [57, 58] and researchers have devised methods to cluster them in groups, known as HLA supertypes, consisting of HLA alleles with similar peptide-binding specificities [59–61]. The HLA-A2, HLA-A3, and HLA-B7 are relevant examples of supertypes; 88% of the population expresses at least an allele included in these supertypes [25, 57, 58]. Identification of promiscuous peptide-binding to HLA supertypes enables the development of T-cell epitope vaccines with high-population coverage using a limited number of peptides. Currently, several web-based methods allow the prediction of promiscuous peptide-binding to HLA supertypes for epitope vaccine design including MULTIPRED [62] and PEPVAC [63] (Table 1). A method to identify promiscuous peptide-binding beyond HLA supertypes was developed and implemented by Molero-Abraham et al. [64] with the name of EPI-SOPT. EPISOPT predicts HLA I presentation profiles of individual peptides regardless of supertypes and identifies epitope combinations providing a wider population protection coverage.

Prediction of peptide binding to MHC II molecules readily discriminate CD4 T-cell epitopes, but cannot tell their ability to activate the response of specific CD4 T-cell subsets (e.g., Th1, Th2, and Treg). However, there is evidence that some CD4 T-cell epitopes appear to stimulate specific subsets of Th cells [65, 66]. Distinguishing the ability of MHC II-restricted epitopes to elicit distinct responses is clearly relevant for epitope vaccine development and has prompted researchers' attention. A relevant example is the work by Dhanda et al. [67] who generated classifiers capable of predicting potential peptide inducers of interleukin 4 (IL-4) secretion, typical of Th2 cells, by training SVM models on experimentally validated IL4 inducing and noninducing MHC class II binders (Table 1).

2.2. Prediction of Antigen Processing and Integration with Peptide-MHC Binding Prediction. Antigen processing shapes the peptide repertoire available for MHC binding and is a limiting step determining T-cell epitope immunogenicity [68]. Subsequently, computational modeling of the antigen processing pathway provides a mean to enhance T-cell epitope predictions. Antigen presentation by MHC I and II molecules proceed by two different pathways. MHC II molecules present peptide antigens derived from endocytosed antigens that are degraded and loaded onto the MHC II molecule in endosomal compartments [69]. Class II antigen degradation is poorly understood, and there is lack of good prediction algorithms yet [70]. In contrast, MHC I molecules

present peptides derived mainly from antigens degraded in the cytosol. The resulting peptide antigens are then transported to the endoplasmic reticulum by TAP where they are loaded onto nascent MHC I molecules [69] (Figure 4). Prior to loading, peptides often undergo trimming by ERAAP N-terminal amino peptidases [71].

Proteasomal cleavage and peptide-binding to TAP have been studied in detail and there are computational methods that predict both processes. Proteasomal cleavage prediction models have been derived from peptide fragments generated *in vitro* by human constitutive proteasomes [72, 73] and from sets of MHC I-restricted ligands mapped onto their source proteins [74–76]. On the other hand, TAP binding prediction methods have been developed by training different algorithms on peptides of known affinity to TAP [77–80]. Combination of proteasomal cleavage and peptide-binding to TAP with peptide-MHC binding predictions increases T-cell epitope predictive rate in comparison to just peptide-binding to MHC I [37, 77, 81–83]. Subsequently, researchers have developed resources to predict CD8 T-cell epitopes through multistep approaches integrating proteasomal cleavage, TAP transport, and peptide-binding to MHC molecules [26, 37, 82–85] (Table 1).

3. Prediction of B-Cell Epitopes

B-cell epitope prediction aims to facilitate B-cell epitope identification with the practical purpose of replacing the antigen for antibody production or for carrying structure-function studies. Any solvent-exposed region in the antigen can be subject of recognition by antibodies. Nonetheless, B-cell epitopes can be divided in two main groups: linear and conformational (Figure 5). Linear B-cell epitopes consist of sequential residues, peptides, whereas conformational B-cell epitopes consist of patches of solvent-exposed atoms from residues that are not necessarily sequential (Figure 5). Therefore, linear and conformational B-cell epitopes are also known as continuous and discontinuous B-cell epitopes, respectively. Antibodies recognizing linear B-cell epitopes can recognize denatured antigens, while denaturing the antigen results in loss of recognition for conformational B-cell epitopes. Most B-cell epitopes (approximately a 90%) are conformational and, in fact, only a minority of native antigens contains linear B-cell epitopes [3]. We will review both, prediction of linear and conformational B-cell epitopes.

3.1. Prediction of Linear B-Cell Epitopes. Linear B-cell epitopes consist of peptides which can readily be used to replace antigens for immunizations and antibody production. Therefore, despite being a minority, prediction of linear B-cell epitopes have received major attention. Linear B-cell epitopes are predicted from the primary sequence of antigens using sequence-based methods. Early computational methods for the prediction of B-cell epitopes were based on simple amino acid propensity scales depicting physicochemical features of B-cell epitopes. For example, Hopp and Wood applied residue hydrophilicity calculations for B-cell epitope prediction [96, 97] on the assumption that hydrophilic regions are predominantly located on the protein surface and are potentially

antigenic. We know now, however, that protein surfaces contain roughly the same number of hydrophilic and hydrophobic residues [98]. Other amino acid propensity scales introduced for B-cell epitope prediction are based on flexibility [99], surface accessibility [100], and β -turn propensity [101]. Current available bioinformatics tools to predict linear B-cell epitopes using propensity scales include PREDITOP [102] and PEOPLE [103] (Table 2). PREDITOP [102] uses a multiparametric algorithm based on hydrophilicity, accessibility, flexibility, and secondary structure properties of the amino acids. PEOPLE [103] uses the same parameters and in addition includes the assessment of β -turns. A related method to predict B-cell epitopes was introduced by Kolasakar and Tongaonkar [104], consisting on a simple antigenicity scale derived from physicochemical properties and frequencies of amino acids in experimentally determined B-cell epitopes. This index is perhaps the most popular antigenic scale for B-cell epitope prediction, and it is actually implemented by GCG [105] and EMBOSS [106] packages. Comparative evaluations of propensity scales carried out in a dataset of 85 linear B-cell epitopes showed that most propensity scales predicted between 50 and 70% of B-cell epitopes, with the β -turn scale reaching the best values [101, 107]. It has also been shown that combining the different scales does not appear to improve predictions [102, 108]. Moreover, Blythe and Flower [109] demonstrated that single-scale amino acid propensity scales are not reliable to predict epitope location.

The poor performance of amino acid scales for the prediction of linear B-cell epitopes prompted the introduction of machine learning- (ML-) based methods (Table 2). These methods are developed by training ML algorithms to distinguish experimental B-cell epitopes from non-B-cell epitopes. Prior to training, B-cell epitopes are translated into feature vectors capturing selected properties, such as those given by different propensity scales. Relevant examples of B-cell epitope prediction methods based on ML include BepiPred [110], ABCpred [111], LBtope [112], BCPREDS [113], and SVMtrip [114]. Datasets, training features, and algorithms used for developing these methods differ. BepiPred is based on random forests trained on B-cell epitopes obtained from 3D-structures of antigen-antibody complexes [110]. Both BCPREDS [113] and SVMtrip [114] are based on support vector machines (SVM) but while BCPREDS was trained using various string kernels that eliminate the need for representing the sequence into length-fixed feature vectors, SVMtrip was trained on length-fixed tripeptide composition vectors. ABCpred and LBtope methods consist on artificial neural networks (ANNs) trained on similar positive data, B-cell epitopes, but differ on negative data, non-B-cell epitopes. Negative data used for training ABCpred consisted on random peptides while negative data used for LBtope was based on experimentally validated non-B-cell epitopes from IEDB [15]. In general, B-cell epitope prediction methods employing ML-algorithm are reported to outperform those based on amino acid propensity scales. Nevertheless, some authors have reported that ML algorithms show little improvement over single-scale-based methods [115].

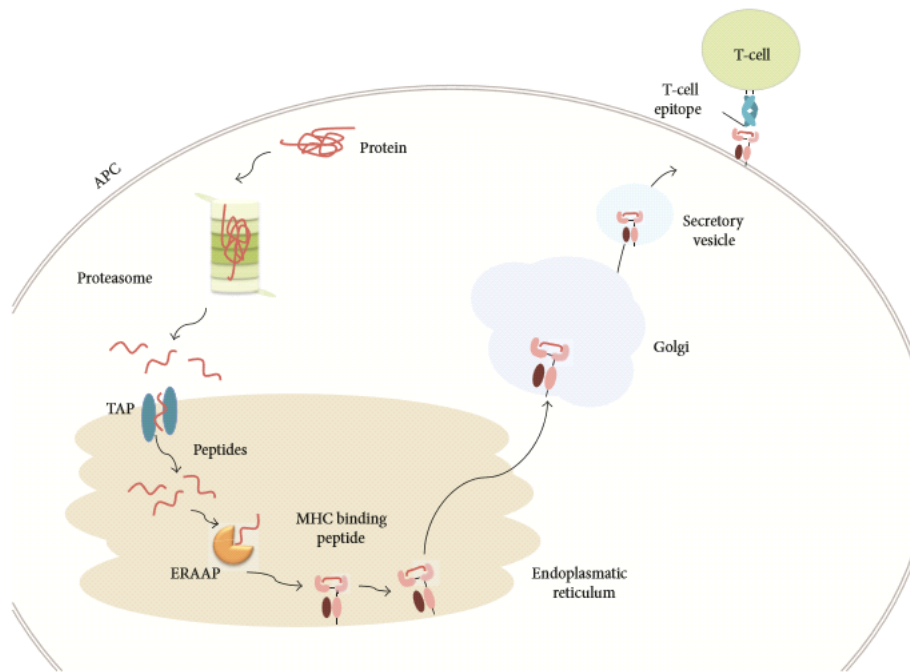


FIGURE 4: Class I antigen processing. The figure depicts the major steps involved in antigen presentation by MHC I molecules. Proteins are degraded by the proteasome and peptide fragments transported to the endoplasmic reticulum (ER) by TAP where they are loaded onto nascent MHC I molecules. TAP transports peptides ranging from 8 to 16 amino acids. Long peptides cannot bind MHC I molecules but often become suitable for binding after N-terminal trimming by ERAAP.

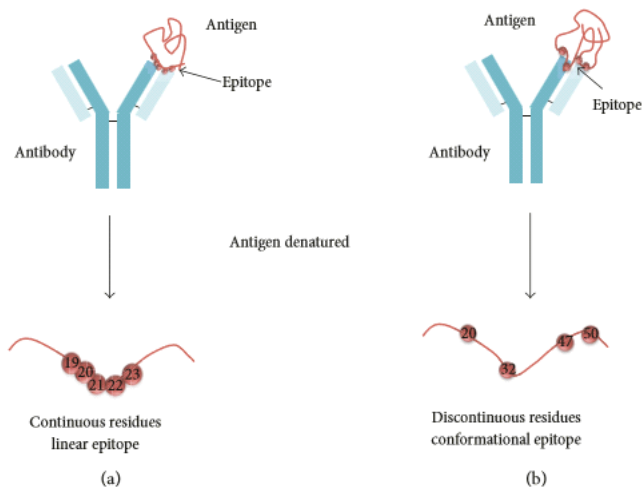


FIGURE 5: Linear and conformational B-cell epitopes. Linear B-cell epitopes (a) are composed of sequential/continuous residues, while conformational B-cell epitopes (b) contain scattered/discontinuous residues along the sequence.

TABLE 2: Selected B-cell epitope prediction methods available for free online use.

Tool	Method	Server (URL)	Ref.
<i>Linear B cell epitope</i>			
PEOPLE	Propensity scale method	http://www.iedb.org/	[103]
BepiPred	ML (DT)	http://www.cbs.dtu.dk/services/BepiPred/	[110]
ABCPred	ML (ANN)	http://www.imtech.res.in/raghava/abcpred/	[111]
LBtope	ML (ANN)	http://www.imtech.res.in/raghava/lbtope/	[112]
BCPREDS	ML (SVM)	http://ailab.ist.psu.edu/bcpred/	[113]
SVMtrip	ML (SVM)	http://sysbio.unl.edu/SVMTriP/prediction.php	[114]
<i>Conformational B-cell epitope</i>			
CEP	Structure-based method (solvent accessibility)	http://bioinfo.ernet.in/cep.htm	[118]
DiscoTope	Structure-based method (surface accessibility and propensity amino acid score)	http://tools.iedb.org/discotope/	[119]
ElliPro	Structure-based method (geometrical properties)	http://tools.iedb.org/ellipro/	[121]
PEPITO	Structure-based method (physicochemical properties and geometrical structure)	http://pepito.proteomics.ics.uci.edu/	[122]
SEPPA	Structure-based method (physicochemical properties and geometrical structure)	http://lifecenter.sgst.cn/seppa/	[123]
EPITOPIA	Structure-based method (ML-naïve Bayes)	http://epitopia.tau.ac.il/	[125]
EPSVR	Structure-based method (ML-SVR)	http://sysbio.unl.edu/EPSVR/	[126]
EPIPREP	Structure-based method (ASEP, Docking)	http://opig.stats.ox.ac.uk/webapps/sabdab-sabpred/EpiPred.php	[129]
PEASE	Structure-based method (ASEP, ML)	http://www.ofranlab.org/PEASE	[130]
MIMOX	Mimotope	http://immunet.cn/mimox/helps.html	[131]
PEPITOPE	Mimotope	http://pepitope.tau.ac.il/	[132]
EpiSearch	Mimotope	http://curie.utmb.edu/episeach.html	[133]
MIMOPRO	Mimotope	http://informatics.nenu.edu.cn/MimoPro	[134]
CBTOPE	Sequence based (SVM)	http://www.imtech.res.in/raghava/cbtope/submit.php	[136]

Antibodies elicited in the course of an immune response are generally of a given isotype that determines their biological function. A recent advance in B-cell epitope prediction is the development of a method by Gupta et al. [116] that allows the identification of B-cell epitopes capable of inducing specific class of antibodies. This method is based on SMVs trained on a dataset that includes linear B-cell epitopes known to induce IgG, IgE, and IgA antibodies.

3.2. Prediction of Conformational B-Cell Epitopes. Most B-cell epitopes are conformational and yet, prediction of conformational B-cell epitopes has lagged behind that of linear B-cell epitopes. There are two main practical reasons for that. First of all, prediction of conformational B-cell epitopes generally requires the knowledge of protein three-dimensional (3D) structure and this information is only available for a fraction of proteins [117]. Secondly, isolating conformational B-cell epitopes from their protein context for selective antibody production is a difficult task that requires suitable scaffolds for epitope grafting. Thereby, prediction of conformational B-cell prediction is currently of little relevance for epitope vaccine design and antibody-based technologies. Nonetheless, prediction of conformational B-cell epitopes is interesting for carrying structure-function studies involving antibody-antigen interactions.

There are several available methods to predict conformational B-cell epitopes (Table 2). The first to be introduced was CEP [118], which relied almost entirely on predicting patches of solvent-exposed residues. It was followed by DiscoTope [119], which, in addition to solvent accessibility, considered amino acid statistics and spatial information to predict conformational B-cell epitopes. An independent evaluation of these two methods using a benchmark dataset of 59 conformational epitopes revealed that they did not exceed a 40% of precision and a 46% of recall [120]. Subsequently, more methods were developed, like ElliPro [121] that aims to identify protruding regions in antigen surfaces and PEPITO [122] and SEPPA [123] that combine single physicochemical properties of amino acids and geometrical structure properties. The reported area under the curve (AUC) of these methods is around 0.7, which is indicative of a poor discrimination capacity yet better than random. Though, in an independent evaluation, SEPPA reached an AUC of 0.62 while all the mentioned methods had an AUC around 0.5 [124]. ML has also been applied to predict conformational B-cell epitopes in 3D-structures. Relevant examples include EPITOPIA [125] and EPSVR [126] which are based on naïve Bayes classifiers and support vector regressions, respectively, trained on feature vectors combining different scores. The reported AUC of these two methods is around 0.6.

The above methods for conformational B-cell epitope prediction identify generic antigenic regions regardless of antibodies, which are ignored [127]. However, there are also methods for antibody-specific epitope prediction. This approach was pioneered by Soga et al. [128] who defined an antibody-specific epitope propensity (ASEP) index after analyzing the interfaces of antigen-antibody 3D-structures. Using this index, they developed a novel method for predicting epitope residues in individual antibodies that worked by narrowing down candidate epitope residues predicted by conventional methods. More recently, Krawczyk et al. [129] developed EpiPred, a method that uses a docking-like approach to match up antibody and antigen structures, thus identifying epitope regions on the antigen. A similar approach is used by PEASE [130], adding that this method utilizes the sequence of the antibody and the 3D-structure of the antigen. Briefly, for each pair of antibody sequence and antigen structure, PEASE uses a machine learning model trained on properties from 120 antibody-antigen complexes to identify pair combination of residues from complementarity-determining regions (CDRs) of the antibody and the antigen that are likely to interact.

Another approach to identify conformational B-cell epitopes in a protein with a known 3D-structure is through mimotope-based methods. Mimotopes are peptides selected from randomized peptide libraries for their ability to bind to an antibody raised against a native antigen. Mimotope-based methods require to input antibody affinity-selected peptides and the 3D-structure of the selected antigen. Examples of bioinformatics tools for conformational B-cell epitope prediction using mimotopes include MIMOX [131], PEPITOPE [132], EPISEARCH [133], MIMOPRO [134], and PEPMAPPER [135] (Table 2).

As remarked before, methods for conformational B-cell epitope prediction generally require the 3D-structure of the antigen. Exceptionally, however, Ansari and Raghava [136] developed a method (CBTOPE) for the identification of conformational B-cell epitope from the primary sequence of the antigen. CBTOPE is based on SVM and trained on physico-chemical and sequence-derived features of conformational B-cell epitopes. CBTOPE reported accuracy was 86.6% in crossvalidation experiments.

4. Concluding Remarks

Currently, T-cell epitope prediction is more advanced and reliable than that of B-cell prediction. However, while it is possible to confirm experimentally the predicted binding to MHC molecules of most peptides predicted, only ~10% of those are shown to be immunogenic (able to elicit a T-cell response) [68]. Such a low T-cell epitope discovery rate is due to the fact that we do not have adequate models for predicting antigen processing yet [68]. The economic toll of low T-cell epitope discovery rate can be overcome, at least in part, by prioritizing protein antigens for epitope prediction [137–139]. For T-cell epitope vaccine development, researchers can also resort to experimentally known T-cell epitopes, available in epitope databases, selecting through immunoinformatics those that provide maximum

population protection coverage [64, 140, 141]. In any case, T-cell epitope prediction remains an integral part of T-cell epitope mapping approaches. In contrast, B-cell epitope prediction utility is currently much more limited. There are several reasons to that. First of all, prediction of B-cell epitopes is still unreliable for both linear and conformational B-cell epitopes. Secondly, linear B-cell epitopes do usually elicit antibodies that do not crossreact with native antigens. Third, the great majority of B-cell epitopes are conformational and yet predicting conformational epitopes have few applications, as they cannot be isolated from their protein context. Under this scenario, the key is not only to improve current methods for B-cell epitope prediction but also to develop novel approaches and platforms for epitope grafting onto suitable scaffolds capable of replacing the native antigen.

To conclude, we wish to make two final remarks that are relevant for epitope vaccine design. First of all, it is that epitope prediction methods can provide potential epitopes from any given protein query but not all the antigens are equally relevant for vaccine development. Therefore, researchers have also developed tools to identify vaccine candidate antigens [142, 143], those likely to induce protective immunity, which can then be targeted for epitope prediction and epitope vaccine design. Second, it should be borne in mind that epitope peptides exhibit little immunogenicity and need to be used in combination with adjuvants, which increase immunogenicity by inducing strong innate immune responses that enable adaptive immunity [144–146]. Consequently, the discovery of new adjuvants is particularly relevant for epitope-based vaccines [146] and to that end, Nagpal et al. [147] developed a pioneered method that can predict the immunomodulatory activity of RNA sequences.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Authors' Contributions

Jose L. Sanchez-Trincado and Marta Gomez-Perosanz contributed equally to this work.

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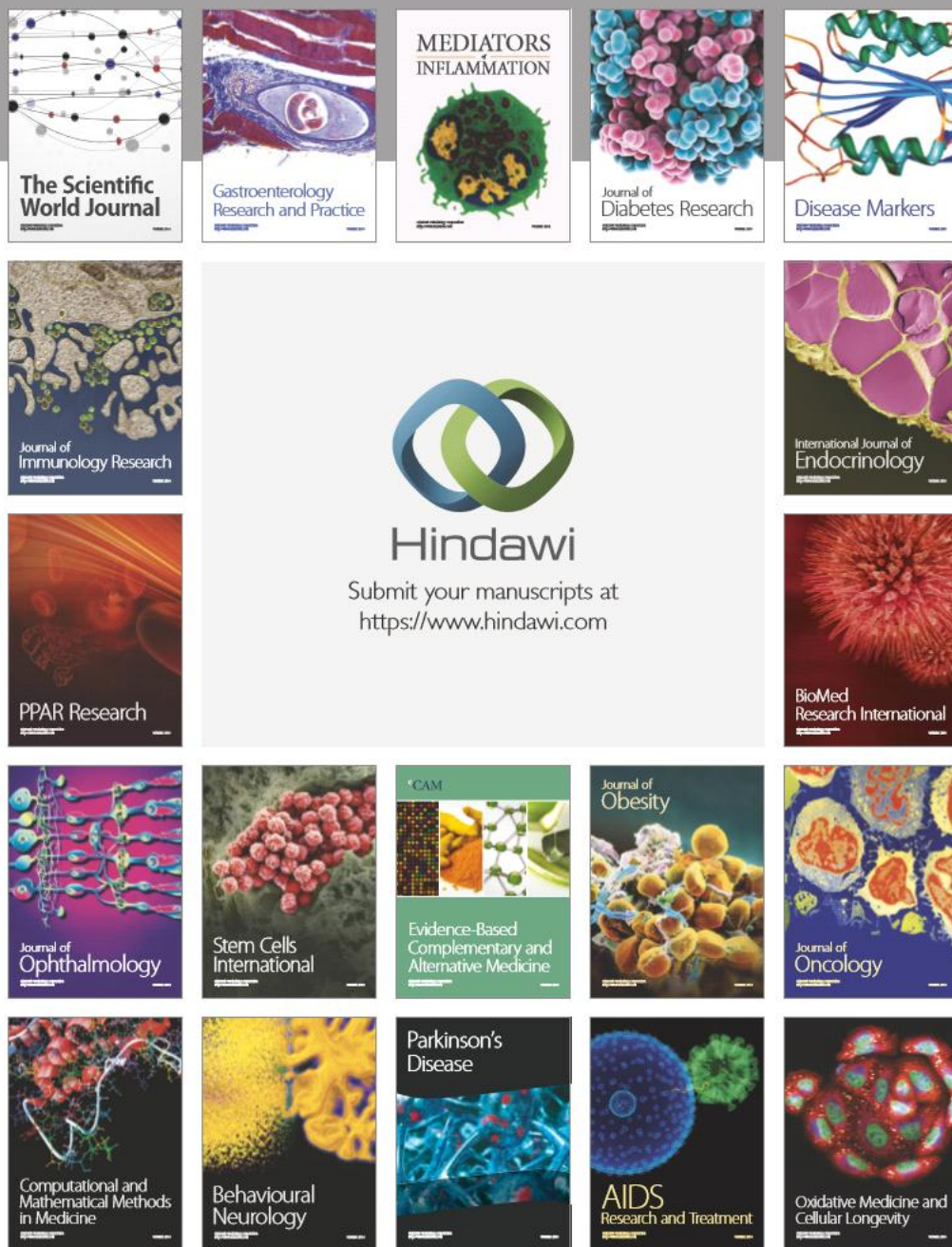
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Sunday, July 11, 2021 10:41 PM

The screenshot displays the IEDB MHC Binding Prediction tool. The browser window shows the URL 'tools.iedb.org/mhci/'. The tool's interface includes a top navigation bar with a 'MHC Binding' tab. Below this, a message states 'Space_separated format detected.'. The main form is organized into several sections:

- Choose a Prediction Method:** A dropdown menu is set to 'Consensus'. A link 'Help on prediction method selections' is provided.
- Specify what to make binding predictions for:** A dropdown menu is set to 'human'.
- MHC source species:** A table lists various MHC alleles and their lengths. The table has two columns: 'Allele' and 'Length'. The alleles listed are HLA-A*01:01, HLA-A*02:01, HLA-A*02:06, HLA-A*03:01, HLA-A*11:01, HLA-A*23:01, HLA-A*26:01, HLA-A*68:01, HLA-B*14:02, HLA-B*07:02, HLA-E*01:01, and HLA-A*32:01. The lengths are all 14. There are expand/collapse icons next to each row.
- Specify Output:** A dropdown menu is set to 'Percentile Rank'.
- Output format:** A dropdown menu is set to 'XHTML table'.
- Email address (optional):** An empty text input field.

At the bottom right, there are 'Submit' and 'Reset' buttons. The BioCode logo is visible in the bottom right corner of the browser window.

tools.eab.org/mhc/result_in_text

MBIC Ending Prediction Results

Method used: consensus

allele	seq_num	start	end	length	peptide	consensus_percentile_rank	ann_ic50	ann_rank	simm_ic50	simm_rank	comblib_sidney2008_score	comblib_sidney2008_rank
HLA-A*01:01	136	149	14	SAAKYSDSAISVY	0.02	5.39	0.02	-	-	-	-	-
HLA-A*01:01	137	150	14	AAKYSDSAISVY	0.03	13.42	0.03	-	-	-	-	-
HLA-A*01:01	139	152	14	KYSDSAISVYLPV	0.04	14.79	0.04	-	-	-	-	-
HLA-A*01:01	6	19	14	KQENYFILFWYF	0.05	11.59	0.05	-	-	-	-	-
HLA-A*01:01	138	151	14	AAKYSDSAISVYLP	0.05	20.97	0.05	-	-	-	-	-
HLA-A*01:01	255	268	14	QCFTENRTKQLQY	0.06	25.8	0.06	-	-	-	-	-
HLA-A*01:01	308	321	14	YSDSAISVYLNWNI	0.06	19.14	0.06	-	-	-	-	-
HLA-A*01:01	135	148	14	LSAAKYSDSAISVY	0.07	29.62	0.07	-	-	-	-	-
HLA-A*01:01	140	153	14	YSDSAISVYLPVT	0.07	30.96	0.07	-	-	-	-	-
HLA-A*11:01	577	590	14	FMSISFNLDLHRLFK	0.07	17.04	0.07	-	-	-	-	-
HLA-A*23:01	7	20	14	QKFNIFILFWYFR	0.11	33.63	0.11	-	-	-	-	-
HLA-A*01:01	222	235	14	VLFPFGNLDTRFVL	0.12	5766.53	0.12	-	-	-	-	-
HLA-A*03:01	488	501	14	DLVLVHVFPCSLFK	0.12	33.69	0.12	-	-	-	-	-
HLA-A*23:01	8	21	14	KENYFILFWYFR	0.13	41.79	0.13	-	-	-	-	-
HLA-A*23:01	309	322	14	DQKCYFLHNIWNFI	0.13	41.6	0.13	-	-	-	-	-
HLA-A*23:01	9	22	14	ENYFILFWYFR	0.14	43.22	0.14	-	-	-	-	-
HLA-A*23:01	10	23	14	NYFILFWYFR	0.14	45.35	0.14	-	-	-	-	-
HLA-A*23:01	45	58	14	NIYFLNCPFGCARF	0.14	44.11	0.14	-	-	-	-	-
HLA-A*23:01	447	460	14	GFDLILMSCGYVF	0.15	49.69	0.15	-	-	-	-	-
HLA-A*11:01	143	156	14	SAISVYLPVTSTK	0.16	29.61	0.16	-	-	-	-	-
HLA-A*01:01	256	269	14	QCFTENRTKQLQY	0.16	95.59	0.16	-	-	-	-	-
HLA-A*23:01	310	323	14	QKCYFLHNIWNFI	0.17	58.67	0.17	-	-	-	-	-
HLA-A*23:01	312	325	14	CYFLHNIWNFI	0.17	58.68	0.17	-	-	-	-	-
HLA-A*01:01	481	494	14	AYENGNADLLHLY	0.17	98.19	0.17	-	-	-	-	-
HLA-A*23:01	311	324	14	KCYFLHNIWNFI	0.18	63.86	0.18	-	-	-	-	-
HLA-A*02:01	162	175	14	IGDYLQEDNVALV	0.2	22.34	0.2	-	-	-	-	-
HLA-A*23:01	11	24	14	YFLVHNYRILTY	0.22	75.19	0.22	-	-	-	-	-
HLA-A*01:01	113	126	14	ATRPAGDQSGTGY	0.22	152.28	0.22	-	-	-	-	-
HLA-A*11:01	261	274	14	NTKQYGVHPTK	0.22	39.39	0.22	-	-	-	-	-
HLA-A*01:01	488	501	14	DLVLVHVFPCSLFK	0.22	39.32	0.22	-	-	-	-	-
HLA-A*01:01	489	502	14	LLVHVFPCSLFK	0.22	56.70	0.22	-	-	-	-	-
HLA-A*23:01	514	527	14	LYVHVFPCSLFK	0.22	72.83	0.22	-	-	-	-	-
HLA-A*01:01	352	365	14	RYLVHVFHAGGSD	0.23	7823.30	0.23	-	-	-	-	-
HLA-A*07:02	372	385	14	FPRLGLTSSAVP	0.23	64.96	0.23	-	-	-	-	-
HLA-A*01:01	297	270	14	FPRLGLTSSAVP	0.24	165.02	0.24	-	-	-	-	-
HLA-A*07:02	371	384	14	FPRLGLTSSAVP	0.24	66.37	0.24	-	-	-	-	-
HLA-A*11:01	79	92	14	YSDSAISVYLNWNI	0.27	46.18	0.27	-	-	-	-	-
HLA-A*01:01	351	364	14	RYLVHVFHAGGSD	0.27	6560.50	0.27	-	-	-	-	-
HLA-A*03:01	418	431	14	VLFGADNGHGLTK	0.27	69.96	0.27	-	-	-	-	-
HLA-A*23:01	504	517	14	NILNFFENPLVYF	0.27	95.07	0.27	-	-	-	-	-
HLA-A*23:01	46	59	14	YLNCPFGCARF	0.28	97.91	0.28	-	-	-	-	-
HLA-A*23:01	313	326	14	YFLHNIWNFI	0.28	99.73	0.28	-	-	-	-	-

AutoSave

CT: epitopes of lactamase B domain protein - Comp... Search

MAN TALHA SARFRAZ

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allele seq_num start end length peptide consensus_percentile_rank ann_ic50 ann_rank simm_ic50

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HLA-A*01:01 1 137 150 14 AAKYSDSAISVY 0.03 13.42 0.03 - - -

HLA-A*01:01 1 139 152 14 KYSDSAISVYLPV 0.04 14.79 0.04 - - -

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HLA-A*01:01 1 138 151 14 AAKYSDSAISVYLP 0.05 20.97 0.05 - - -

HLA-A*01:01 1 255 268 14 QCFTENRTKQLQY 0.06 25.8 0.06 - - -

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HLA-A*11:01 1 577 590 14 FMSISFNLDLHRLFK 0.07 17.04 0.07 - - -

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HLA-A*01:01 1 222 235 14 VLFPFGNLDTRFVL 0.12 5766.53 0.12 - - -

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HLA-A*11:01 1 143 156 14 SAISVYLPVTSTK 0.16 29.61 0.16 - - -

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HLA-A*23:01 1 310 323 14 QKCYFLHNIWNFI 0.17 58.67 0.17 - - -

1336 words English (Pakistan)

BioCode

7:08 PM Sun Jul 11 47%

VaxiJen - predicting prote... ToxinPred Bioinformatics tool for aler... tools.iedb.org/imh... Immunogenicity Result

Not secure | ddg-pharmfac.net/vaxijen/scripts/VaxiJen3.pl

VaxiJen v2.0

VaxiJen RESULTS

Model selected: parasite

Threshold for this model: 0.5

Your Sequence:
SAARYSDSAISVYY

Overall Prediction for the Protective Antigen = **0.5922** (Probable **ANTIGEN**).

BioCode

VaxiJen - predicting prote... Bioinformatics tool for aler... tools.iedb.org/imh... ToxinPred Immunogenicity Result

Not secure | tools.iedb.org/immunogenicity/result/

IEDB Analysis Resource

Home Help Example Reference Download Contact

Class I Immunogenicity

Masking: default
Masked variables: [1, 2, 'cterm']

Peptide	Length	Score
SAARYSDSAISVYY	14	0.5922

Download result

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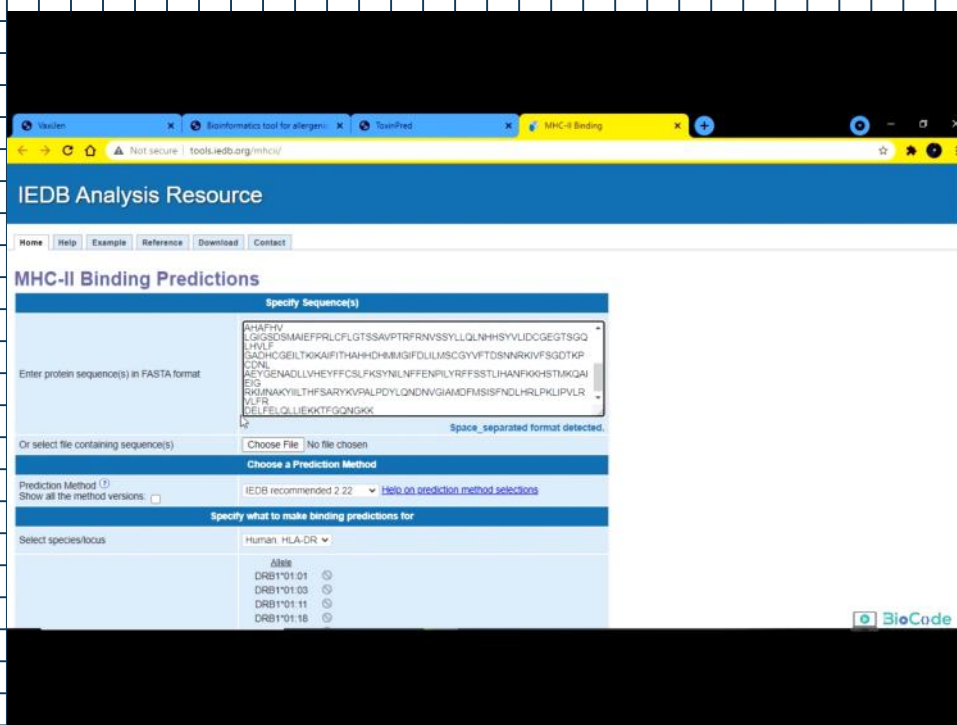
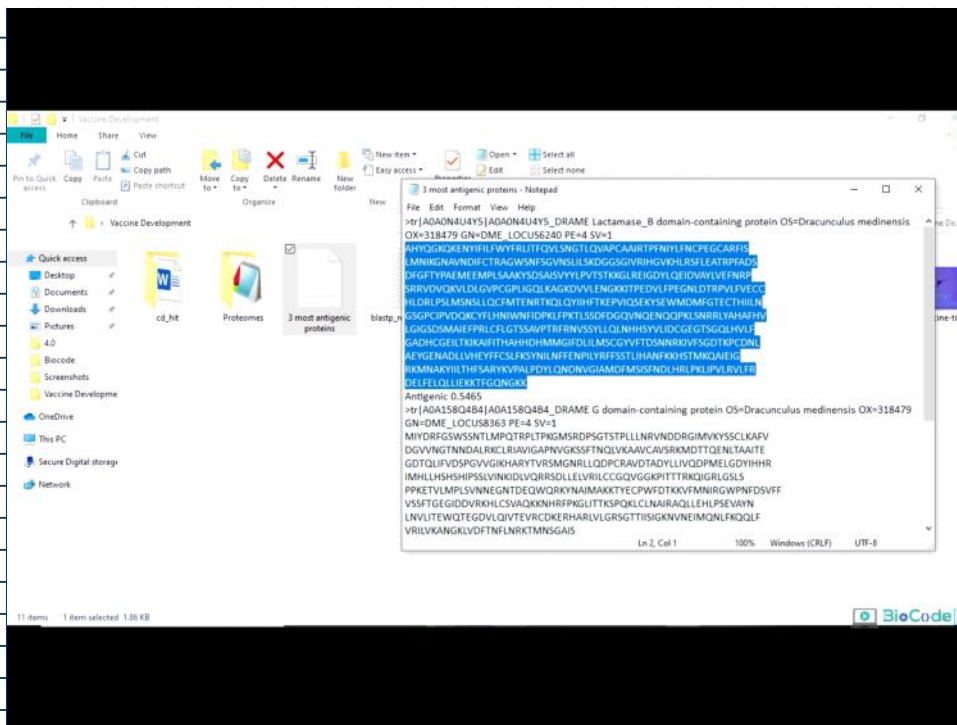
BioCode

CTL Epitopes	Protein	Position	Antigenicity	Immunogenicity	Alleles	Binding Score
KQKENYIFILFW	Lactamase_B domain-containing protein	6-17	1.45	0.44	HLA-B*44:03	-201.52
QKENYIFILFWY	Lactamase_B domain-containing protein	7-18	1.72	0.61	HLA-B*44:03	-199.33
KENYIFILFWYF	Lactamase_B domain-containing protein	9-19	2.01	0.65	HLA-B*44:03	-197.59
VRCDKERHARLV	G domain-containing protein	381-392	2.40	0.04	HLA-C*06:02	-188.95
KLVDFTNFLNRK	G domain-containing protein	430-441	0.76	0.30	HLA-A*11:01	-196.06
FPKILWTVILHL	Ferrochelatase	63-74	1.32	0.48	HLA-B*51:01	-183.28
RVIPAFYDNYLY	Ferrochelatase	184-195	1.00	0.18	HLA-A*11:01	-224.00

The screenshot shows the IEDB Analysis Resource website, specifically the MHC-II Binding Predictions page. The browser's address bar shows the URL `tools.iedb.org/mhcii/`. The page has a blue header with the text "IEDB Analysis Resource". Below the header is a navigation bar with links: Home, Help, Example, Reference, Download, and Contact. The main heading is "MHC-II Binding Predictions". The interface is divided into several sections:

- Specify Sequence(s)**: A section for entering protein sequences. It includes a text input field labeled "Enter protein sequence(s) in FASTA format" and a button labeled "Choose File" with the text "No file chosen" next to it.
- Choose a Prediction Method**: A section for selecting a prediction method. It shows "Prediction Method (1)" as "IEDB recommended 2.22" with a dropdown arrow and a link to "Help on prediction method selections". Below this is a checkbox labeled "Show all the method versions".
- Specify what to make binding predictions for**: A section for selecting species/locus. It shows "Select species/locus" as "Human_HLA-DR" with a dropdown arrow. Below this is a list of alleles: DRB1*01:01, DRB1*01:03, DRB1*01:11, DRB1*01:18, and DRB1*01:15, each with a radio button next to it.

The BioCode logo is visible in the bottom right corner.

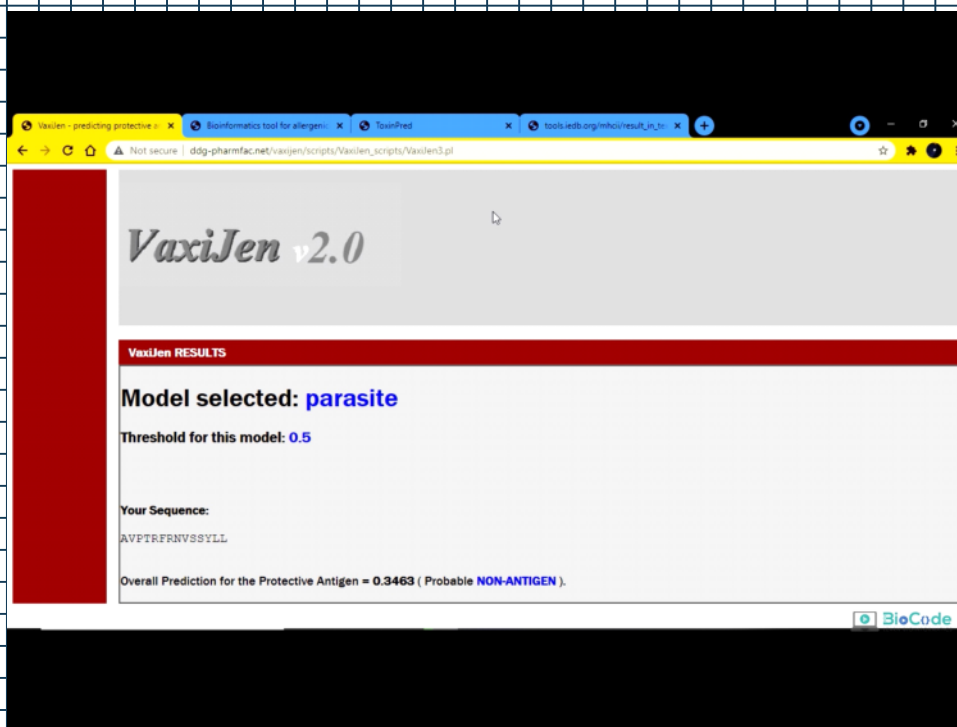


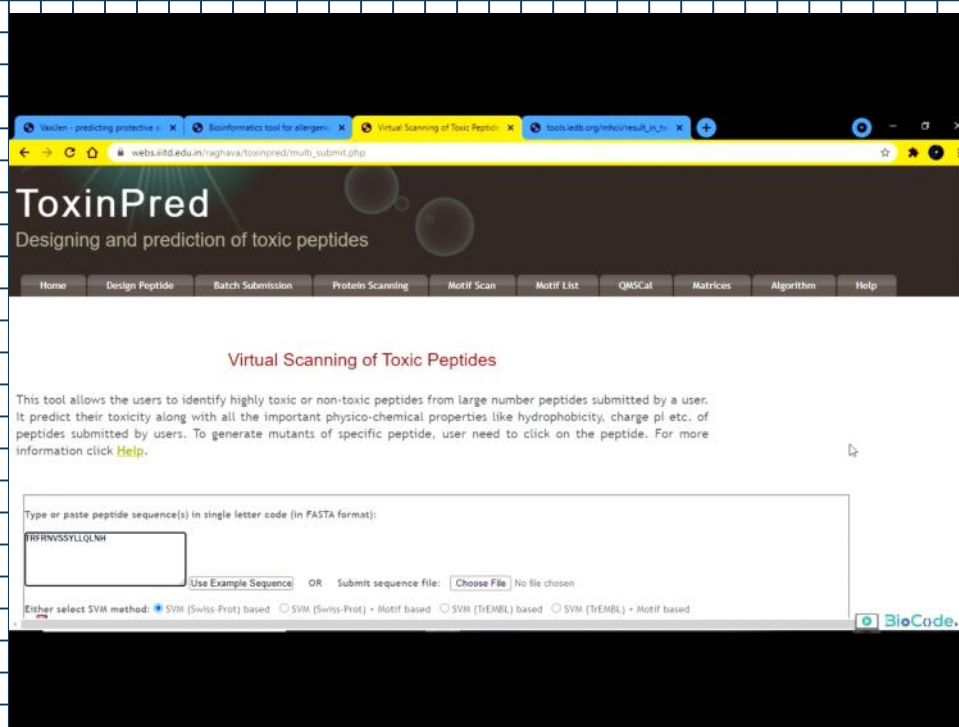
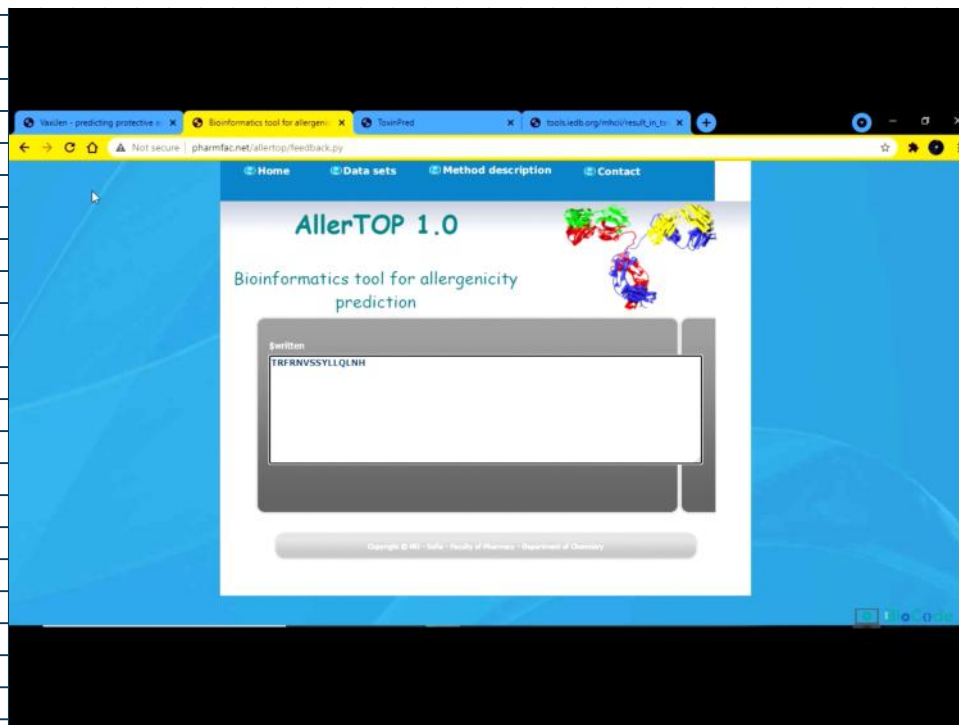
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 Supported by a contract from the National Institute of Allergy and Infectious Diseases, a component of the National Institutes of Health in the Department of Health and Human Services

 BioCode

MHC-II Binding Prediction Results

Waiting for tools.edb.org...





toxigen - predicting protective x Bioinformatics tool for allergen x Virtual Scanning of Toxin Peptide x tools.webd.org/info/result_n/...

webs.iitd.edu.in/raghava/toxinpred/multi_submit.php

peptides submitted by users. To generate mutants or specific peptide, user need to click on the peptide. For more information click [Help](#).

Type or paste peptide sequence(s) in single letter code (in FASTA format):

TRFRNVSSYLLQLNH

[Use Example Sequence](#) OR Submit sequence file: [Choose File](#) No file chosen

Either select SVM method: ☒ SVM (Swiss-Prot) based ☐ SVM (Swiss-Prot) + Motif based ☐ SVM (TrEMBL) based ☐ SVM (TrEMBL) + Motif based

OR

Select Quantitative Matrix (QM) method: ☐ Monopeptide(Swiss-Prot) ☐ Monopeptide (TrEMBL) ☐ Dipeptide (Swiss-Prot) ☐ Dipeptide (TrEMBL)

Choose E-value cut-off for motif-based method: 10

Choose SVM threshold: 0.0

Physicochemical Properties to Be Displayed:

☒ Hydrophobicity ☐ Steric hindrance ☐ Side bulk ☒ Hydrophobicity ☐ Amphipathicity ☒ Hydrophilicity ☐ Net Hydrogen

☒ Charge ☐ pI ☒ Molecular weight ☐ All

[Clear All](#) [Run Analysis](#)

DATASETS | TEAM | CONTACT US | RAGHAVA | CSIR-IMTECH | CSIR

BioCode

toxigen - predicting protective x Bioinformatics tool for allergen x https://webs.iitd.edu.in/raghava x tools.webd.org/info/result_n/...

webs.iitd.edu.in/raghava/toxinpred/multi_submit.php?name=1025

ToxinPred

Designing and prediction of toxic peptides

[Home](#) [Design Peptide](#) [Batch Submission](#) [Protein Scanning](#) [Motif Scan](#) [Motif List](#) [QMScal](#) [Matrices](#) [Algorithm](#) [Help](#)

Peptide ID	Peptide Sequence	SVM Score	Prediction	Hydrophobicity	Hydrophobicity	Hydrophobicity	Charge	Mol wt
TRFRNVSSYLLQLNH		-1.12	Non-Toxin	-0.26	-0.53	-0.36	2.50	1848.32

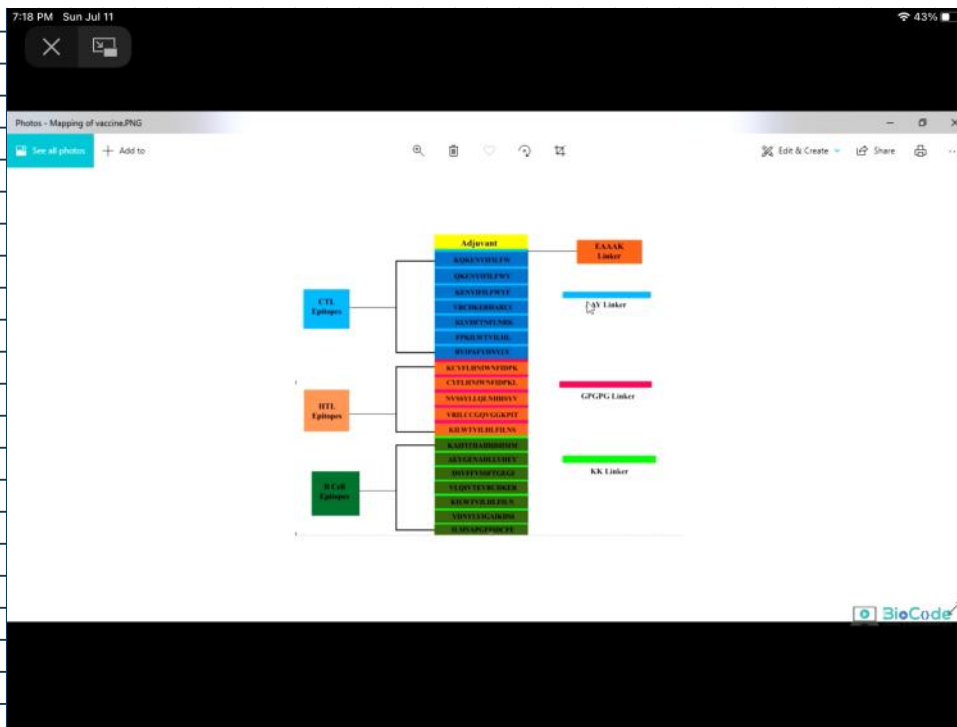
1/1 50

Your job id is 1025

[Go Back](#)

tools.webd.org/info/result_n/...

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Prediction and Assessment of pV... UniProt

uniportal.org

UniProtKB

UniRef

UniParc

Proteomes

Supporting data

News

BioCode

The mission of UniProt is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of protein sequence and functional information.

UniProtKB: UniProt Knowledgebase. Swiss-Prot (564,638). Manually annotated and reviewed. Records with information extracted from literature and curator-evaluated computational analysis.

UniRef: Sequence clusters.

UniParc: Sequence archive.

Proteomes: Proteome sets.

Supporting data: Literature citations, Cross-ref. databases, Taxonomy, Diseases, Subcellular locations, Keywords.

News: New UniProt portal for the latest SARS-CoV-2 coronavirus protein entries and receptors, updated independent of the general UniProt release cycle. View SARS-CoV-2 Proteins and Receptors.

Forthcoming changes: Planned changes for UniProt.

UniProt release 2021_02: With a little help from my friend | SwissBioPics subcellular location visualization | Change of evidence codes for combinatorial evidence.

UniProt release 2021_01: (Almost) all about that CBASS | Cross-references to VEuPathDB | Changes to humsavat.txt and related keywords | Reference proteomes downlo...

Waiting for www.scopex-analytics.com...

UniProtKB 2021_02 results

UniProtKB consists of two sections:

- Reviewed (Swiss-Prot) - Manually annotated**
Records with information extracted from literature and curator-evaluated computational analysis.
- Unreviewed (TrEMBL) - Computationally analyzed**
Records that await full manual annotation.

The UniProt Knowledgebase (UniProtKB) is the central hub for the collection of functional information on proteins, with accurate, consistent and rich annotation. In addition to capturing the core data mandatory for each UniProtKB entry (mainly, the amino acid sequence, protein name or description, taxonomic data and citation information), as much annotation information as possible is added.

Filter by: [BLAST](#) [Align](#) [Download](#) [Add to basket](#) [Columns](#)

Quote terms: "cholera enterotoxin subunit"

Entry	Entry name	Protein names	Gene names	Organism	Length
☐ P01556	CHTB_VIBCH	Cholera enterotoxin subunit B	ctxB toxB, VC_1456	Vibrio cholerae serotype O1 (strain ATCC 39315 / El Tor Inaba N16961)	124
☐ P01555	CHTA_VIBCH	Cholera enterotoxin subunit A	ctxA toxA, VC_1457	Vibrio cholerae serotype O1 (strain ATCC 39315 / El Tor Inaba N16961)	258
☐ P06917	ELAB_ECOLY	Heat-labile	eltA toxA	Escherichia coli	

Popular organisms: Human (2), Bovine (1), Mouse (1)

7:20 PM Sun Jul 11

UniProtKB 2021_02 results

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Filter by: [BLAST](#) [Align](#) [Download](#) [Add to basket](#) [Columns](#)

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☐ P06917	ELAB_ECOLY	Heat-labile	eltA toxA	Escherichia coli	

Popular organisms: Human (2), Bovine (1), Mouse (1)

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Sunday, July 11, 2021 10:45 PM

The image displays two screenshots of the UniProtKB website, showing the entry for P01556 (CHTB_VIBCH).

Top Screenshot: UniProtKB - P01556 (CHTB_VIBCH)

The page shows the entry for **Cholera enterotoxin subunit B** (ctxB) from *Vibrio cholerae* serotype O1 (strain ATCC 39315 / El Tor Inaba N16961). The entry is reviewed, with an annotation score of 5 stars and experimental evidence at the protein level.

Function:

The B subunit pentameric ring directs the A subunit to its target by binding to the GM1 gangliosides present on the surface of the intestinal epithelial cells. It can bind five GM1 gangliosides. It has no toxic activity by itself.

GO - Molecular function:

- galactose binding (Source: CAFA)
- host cell surface binding (Source: TIGR)
- toxin activity (Source: UniProtKB-KW)

GO - Biological process:

Complete GO annotation on QuickGO ...

Bottom Screenshot: UniProtKB - P01556 (CHTB_VIBCH) - Sequence

The page shows the sequence status as complete. The displayed sequence is further processed into a mature form.

Sequence:

P01556-1 (UniParc) [FASTA](#) [Add to basket](#)

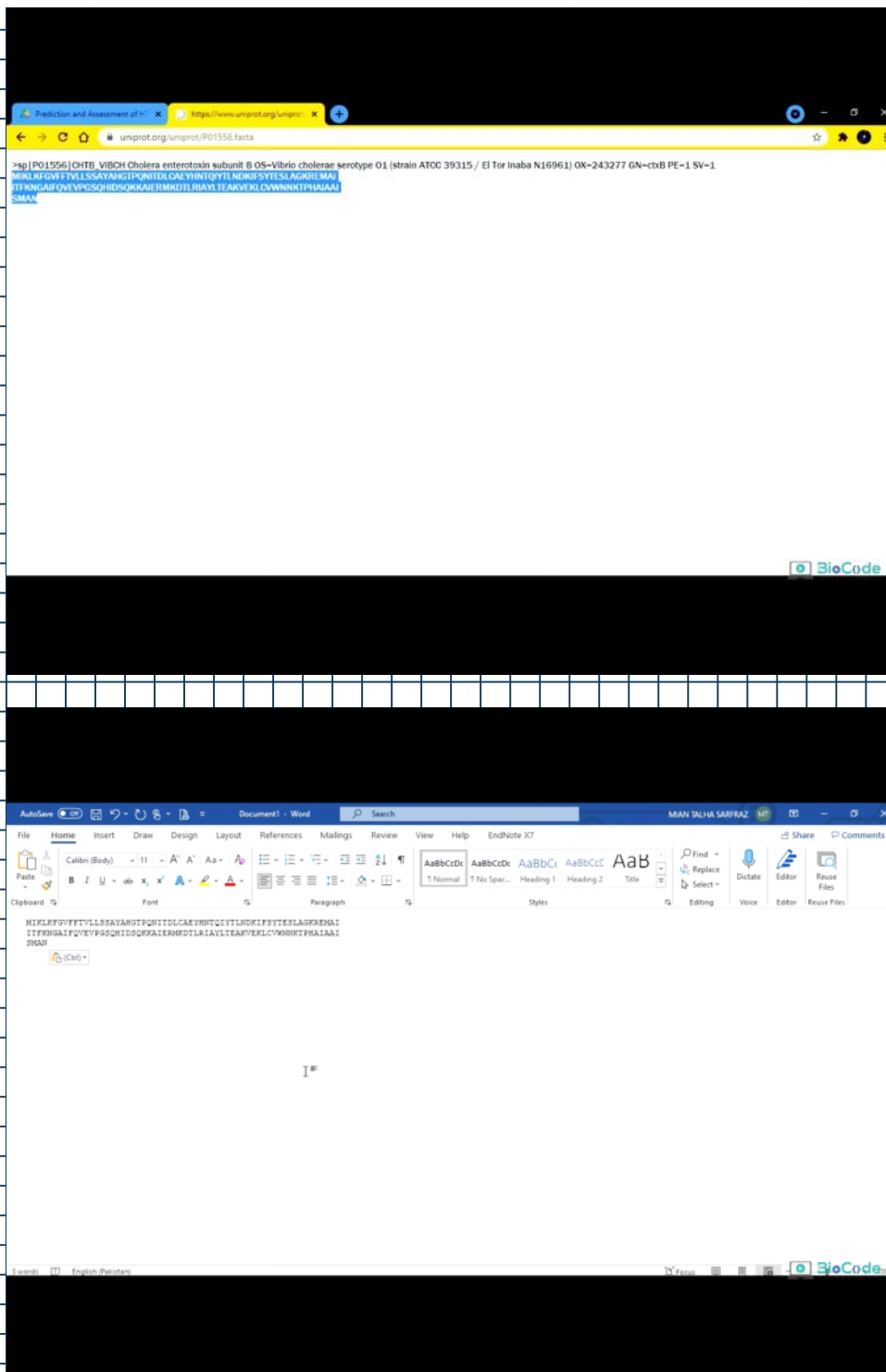
Length: 124
Mass (Da): 13,957
Last modified: August 13, 1987 - v1
Checksum: 9AA393E3EABE3EBF

Sequence:

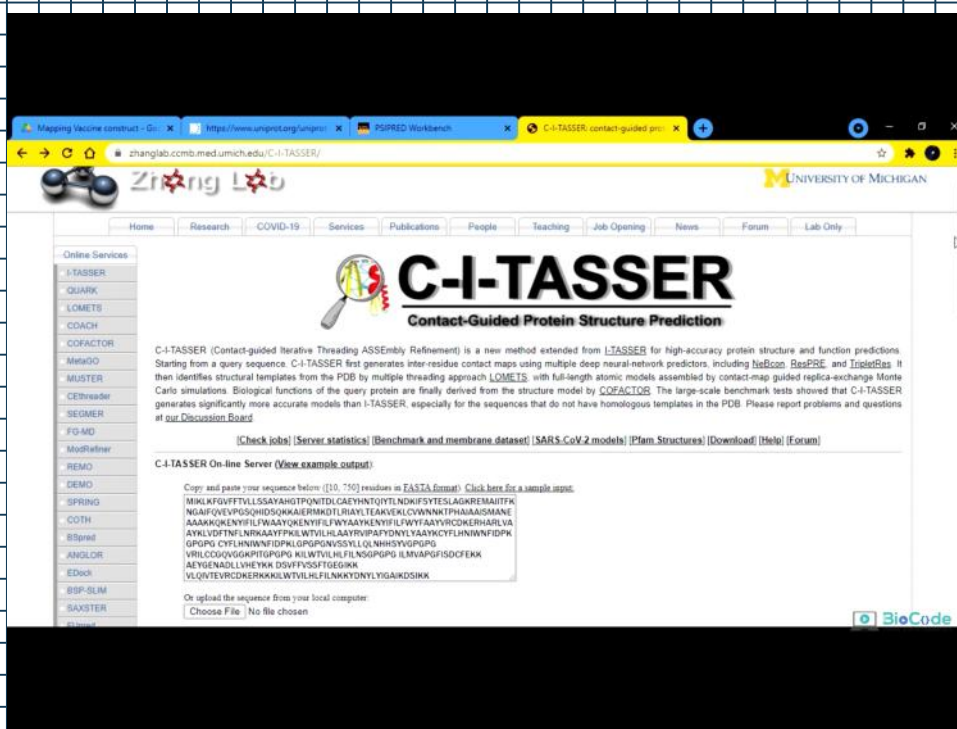
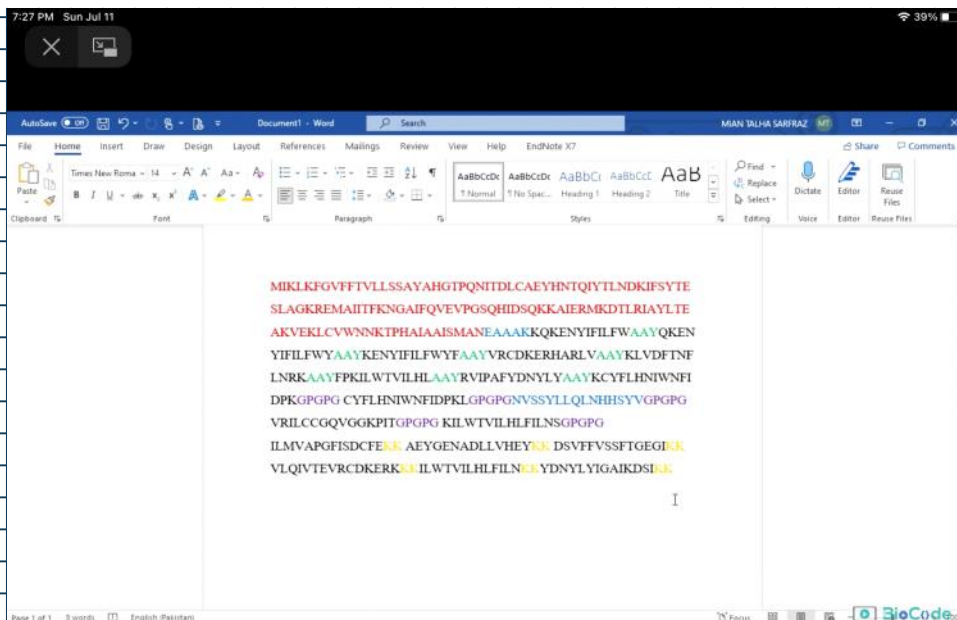
```
10 20 30 40 50
MIKLKFGVFF TVLLSSAYAH GTPQMTDLG AEYHNTQYT LNDKIPSYTE
60 70 80 90 100
SLAGKREMAI ITRKGAIFQ VEPGSGHID SQKKAIERMK DTLRIAYLTE
110 120
AKVERLCVWN NKTPHAUAI SMAN
```

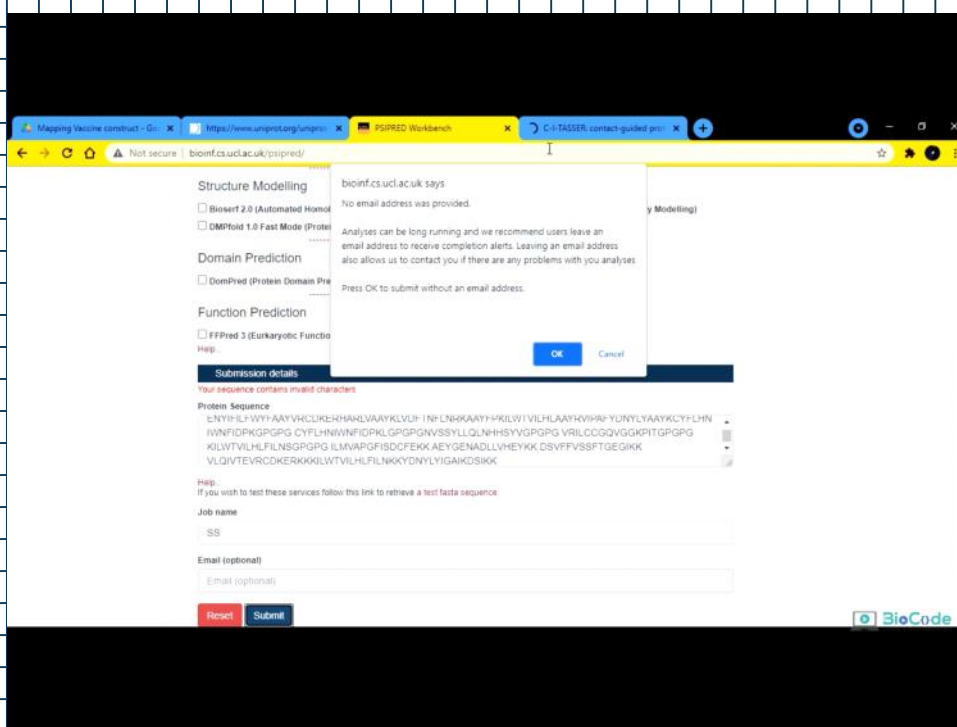
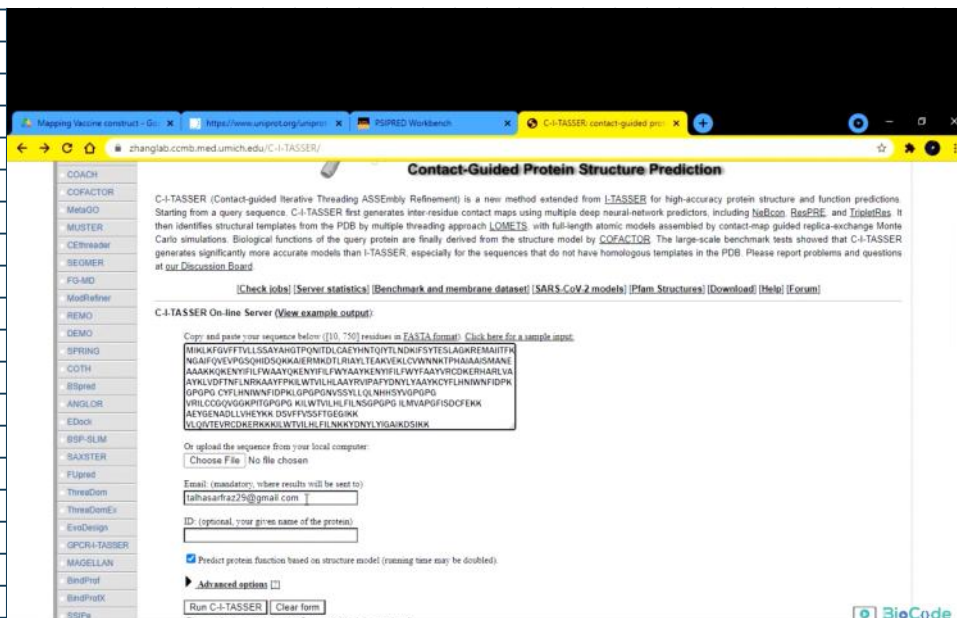
Experimental Info:

Feature key	Position(s)	Description	Actions	Graphical view	Length
Sequence conflict ¹	23 Y → S	In CAA24996 (PubMed:6646234). Curated			1
Sequence conflict ¹	39 Y → H	AA sequence (PubMed:903363). Curated			1
Sequence conflict ¹	39 Y → H	AA sequence (PubMed:903362). Curated			1
Sequence conflict ¹	43 D → N	AA sequence (PubMed:903363). Curated			1
Sequence conflict ¹	43 D → N	AA sequence (PubMed:903362). Curated			1
Sequence conflict ¹	68 I → T	AA sequence (PubMed:903363). Curated			1
Sequence conflict ¹	68 I → T	AA sequence (PubMed:903362). Curated			1



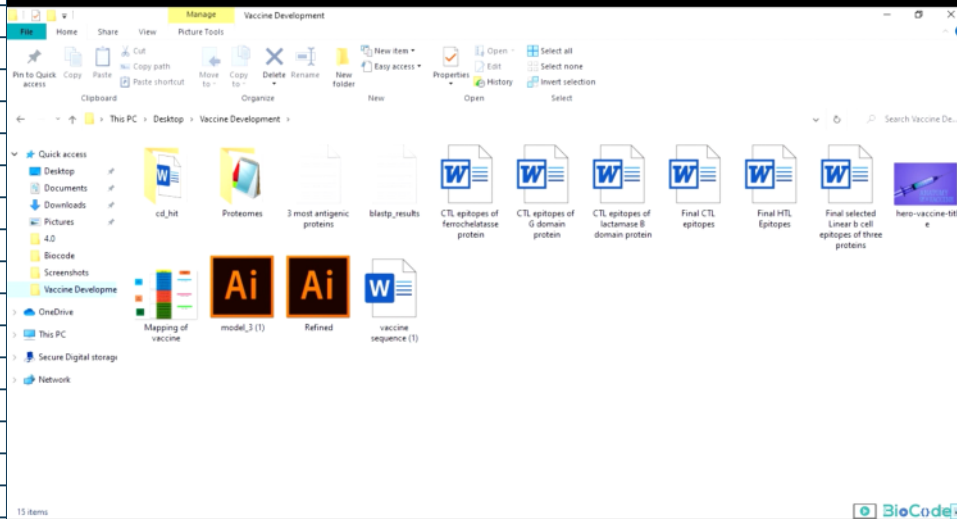


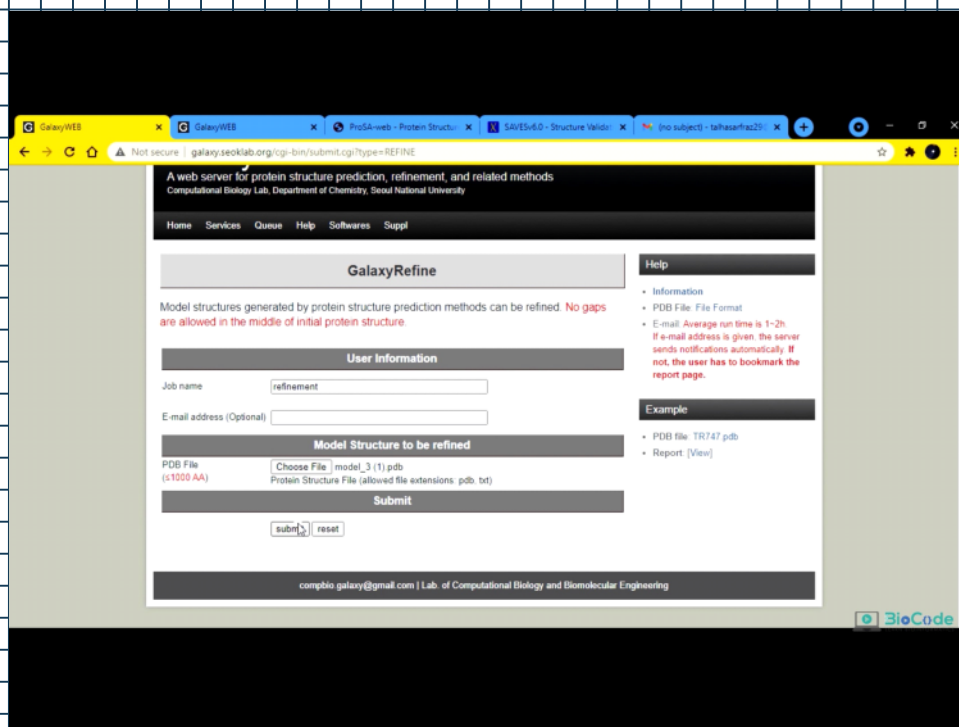
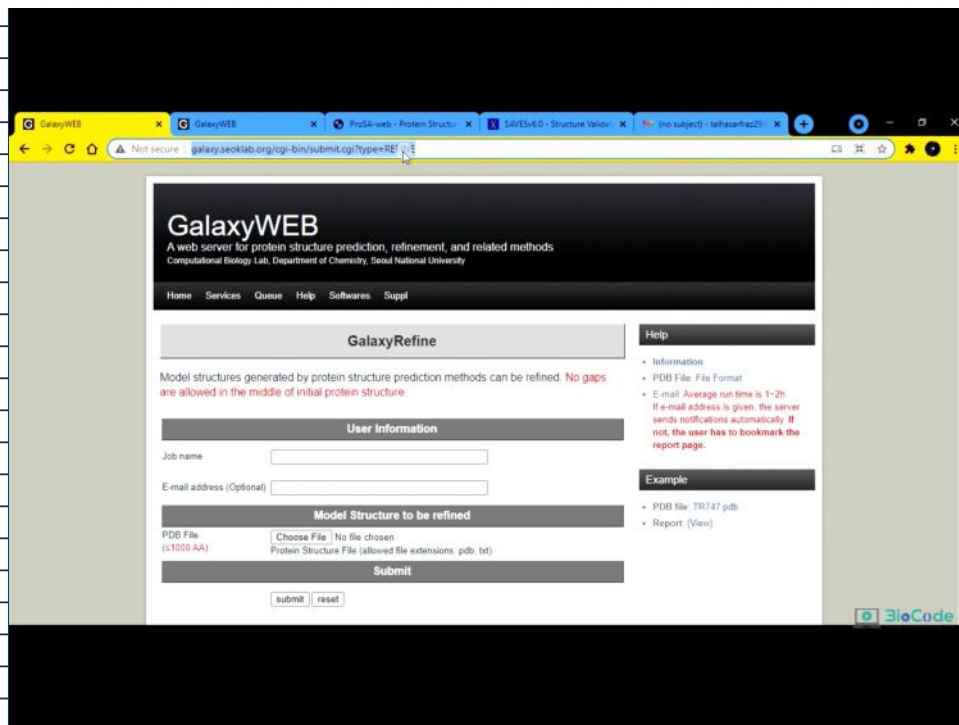




Tertiary Structure Refinement And Validation

www.BioCode.Ltd





GalaxyWEB

galaxy.secklab.org/cgi-bin/report_REFINE.cgi?key=d6021feada7d385a828cd0fb676baf5



View in PV [Model 1] [Model 2] [Model 3] [Model 4] [Model 5] [All Models]
Download [Model 1] [Model 2] [Model 3] [Model 4] [Model 5] [All Models]

Structure information

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	1.895	12.5	0.0	95.8
MODEL 1	0.9847	0.310	1.684	12.9	2.2	99.0
MODEL 2	0.9821	0.322	2.018	17.2	2.2	97.9
MODEL 3	0.9847	0.316	1.694	13.5	1.1	97.9
MODEL 4	0.9858	0.290	1.740	16.0	1.1	99.0
MODEL 5	0.9770	0.332	1.705	16.0	0.0	99.0

Download

• 5 refined models [DOWNLOAD]

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prosa.services.came.sbg.ac.at/prosa.php

ProSA-web
Protein Structure Analysis

Please upload a structure in PDB format:
 No file chosen

Alternatively you can specify a structure by entering its PDB code, chain identifier and NMR model number:

PDB CODE:
PDB CHAIN ID:
PDB MODEL NUMBER:

If you leave the fields for chain id or model number blank, the first chain of the first model found in the PDB file will be analysed.

Results

Overall model quality

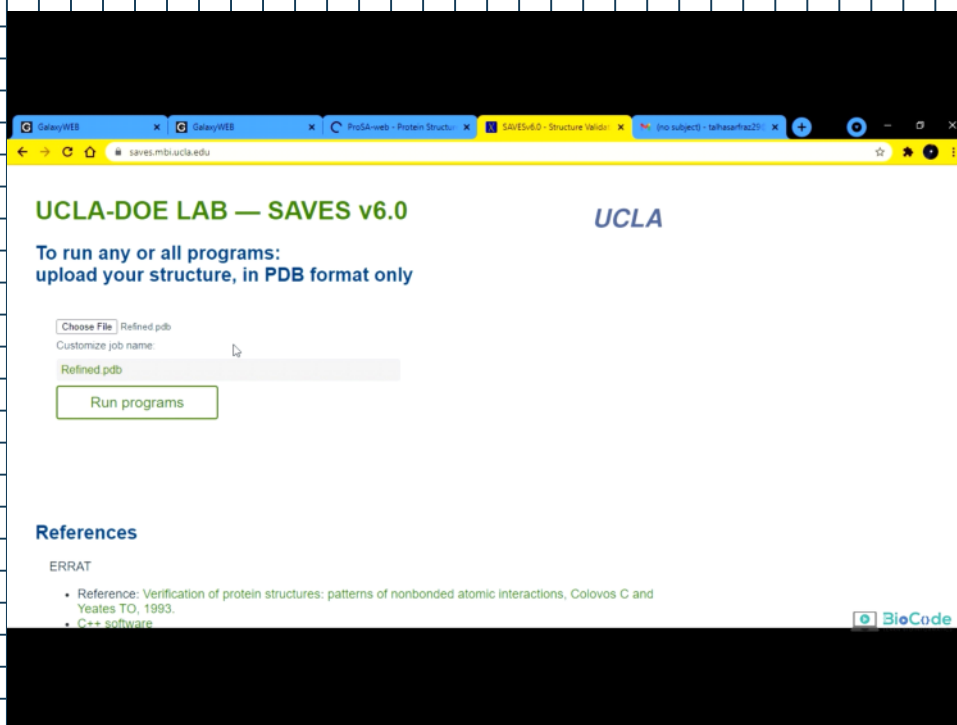
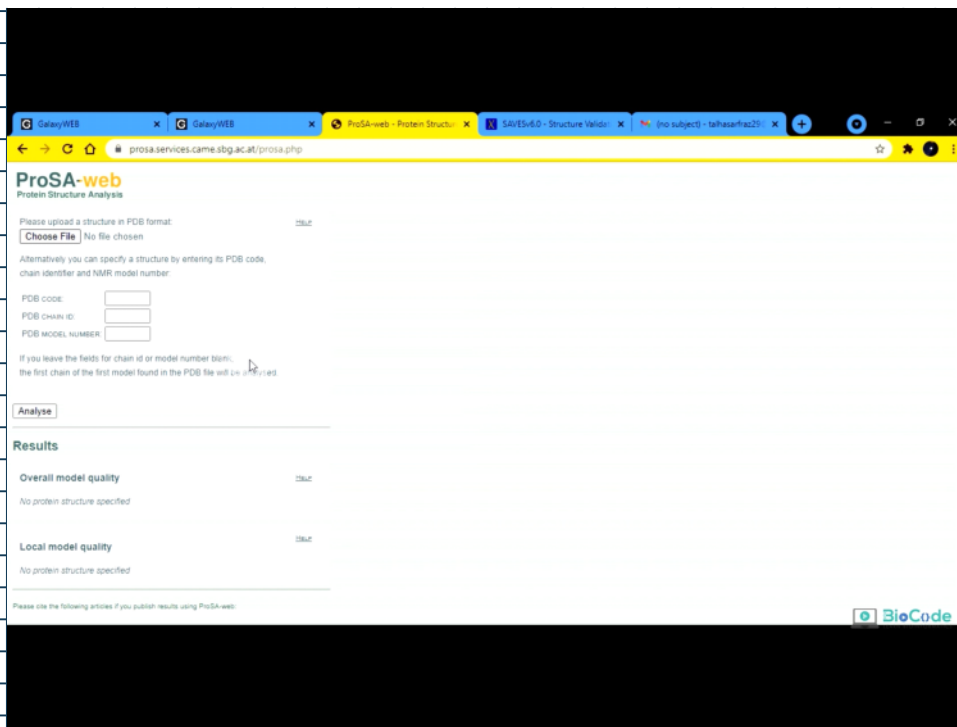
No protein structure specified

Local model quality

No protein structure specified

Please cite the following articles if you publish results using ProSA-web:

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VALIDATION, MOLECULAR MODELING, IMMUNE SIMULATION AND IN SILICO SIMULATION

Sunday, July 11, 2021 10:46 PM

The screenshot displays the UCLA-DOE LAB — SAVES v6.0 web interface. The top navigation bar includes the UCLA logo and the text "UCLA-DOE LAB — SAVES v6.0". Below this, a message states "To run any or all programs: upload your structure, in PDB format only". A file upload section shows a "Choose File" button, a "Refined.pdb" file selected, and a "Run programs" button. A "References" section lists an "ERRAT" reference. The bottom section, titled "job #724065: Refined.pdb [job link] [3D Viewer]", contains a grid of six validation tools: ERRAT, Verify3D, PROVE, WHATCHECK, PROCHECK, and CRYST. Each tool has a brief description and a "Start" button.

UCLA-DOE LAB — SAVES v6.0

UCLA

To run any or all programs:
upload your structure, in PDB format only

Choose File | Refined.pdb
Customize job name:
Refined.pdb
Run programs

References

ERRAT

- Reference: Verification of protein structures: patterns of nonbonded atomic interactions, Colovos C and Yeates TO, 1993.
- C++ software

Job 724065 has been created

New Job

job #724065: Refined.pdb [job link] [3D Viewer]

ERRAT Analyzes the statistics of non-bonded interactions between different atom types and plots the value of the error function versus position in a 9-residue sliding window, calculated by a comparison with statistics from highly refined structures. Start	Verify3D Determines the compatibility of an atomic model (3D) with its own amino acid sequence (1D) by assigning a structural class based on its location and environment (alpha, beta, loop, polar, nonpolar etc) and comparing the results to good structures. Start	PROVE Calculates the volumes of atoms in macromolecules using an algorithm which treats the atoms like hard spheres and calculates a statistical Z-score deviation for the model from highly resolved (2.0 Å or better) and refined (R-factor of 0.2 or better) PDB-deposited structures. Start
WHATCHECK Derived from a subset of protein verification tools from the WHATIF program (Vienn, 1990), this does extensive checking of many stereochemical parameters of the residues in the model. Start	PROCHECK Checks the stereochemical quality of a protein structure by analyzing residue-by-residue geometry and overall structure geometry. Start	CRYST This program searches the Protein Data Bank for entries that have a unit cell similar to your input file. CRYST1 record required. Use the standalone CRYST server for more options. Start

GalaxyWEB x GalaxyWEB x ProSA-web - Protein Structure x SAVES v6.0 - Structure Valid x (no subject) - RefineFact2 x

← → ↻ ⚙ sav.es.mbi.ucla.edu

UCLA-DOE LAB — SAVES v6.0

UCLA

Job 724065 has been created

[New Job](#)

job #724065: Refined.pdb [\[job link\]](#) [\[3D Viewer\]](#)

ERRAT Status: Started Time taken:	Verify3D Determines the compatibility of an atomic model (3D) with its own amino acid sequence (1D) by assigning a structural class based on its location and environment (alpha, beta, loop, polar, nonpolar etc) and comparing the results to good structures. Start	PROVE Calculates the volumes of atoms in macromolecules using an algorithm which treats the atoms like hard spheres and calculates a statistical Z-score deviation for the model from highly resolved (2.0 Å or better) and refined (R-factor of 0.2 or better) PDB-deposited structures. Start
WHATCHECK Derived from a subset of protein verification tools from the WHATIF program (Viend, 1990), this does extensive checking of many stereochemical parameters of the residues in the model	PROCHECK Checks the stereochemical quality of a protein structure by analyzing residue-by-residue geometry and overall structure geometry Start	CRYST This program searches the Protein Data Bank for entries that have a unit cell similar to your input file. CRYST1 record required. Use the standalone CRYST server for more options. Start

[BioCode](#)

GalaxyWEB x GalaxyWEB x ProSA-web - Protein Structure x SAVES v6.0 - Structure Valid x (no subject) - RefineFact2 x

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Job 724065 has been created

[New Job](#)

job #724065: Refined.pdb [\[job link\]](#) [\[3D Viewer\]](#)

ERRAT Complete Overall Quality Factor 80 Results	Verify3D Determines the compatibility of an atomic model (3D) with its own amino acid sequence (1D) by assigning a structural class based on its location and environment (alpha, beta, loop, polar, nonpolar etc) and comparing the results to good structures. Start	PROVE Calculates the volumes of atoms in macromolecules using an algorithm which treats the atoms like hard spheres and calculates a statistical Z-score deviation for the model from highly resolved (2.0 Å or better) and refined (R-factor of 0.2 or better) PDB-deposited structures. Start
WHATCHECK Derived from a subset of protein verification tools from the WHATIF program (Viend, 1990), this does extensive checking of many stereochemical parameters of the residues in the model	PROCHECK Checks the stereochemical quality of a protein structure by analyzing residue-by-residue geometry and overall structure geometry Start	CRYST This program searches the Protein Data Bank for entries that have a unit cell similar to your input file. CRYST1 record required. Use the standalone CRYST server for more options. Start

[BioCode](#)

GalaxyWEB x GalaxyWEB x ProSA-web - Protein Structure x SAVES v6.0 - Structure Valid x (no subject) - saveerrac20 x

← → ↻ ⚙ ⭐ 🔍

save.mbi.ucla.edu/results/job=724065&p=errac

UCLA-DOE LAB — SAVES v6.0

UCLA

← Control panel ↑ New Job

job #724065: Refined.pdb

ERRAT

ERRAT

Overall Quality Factor

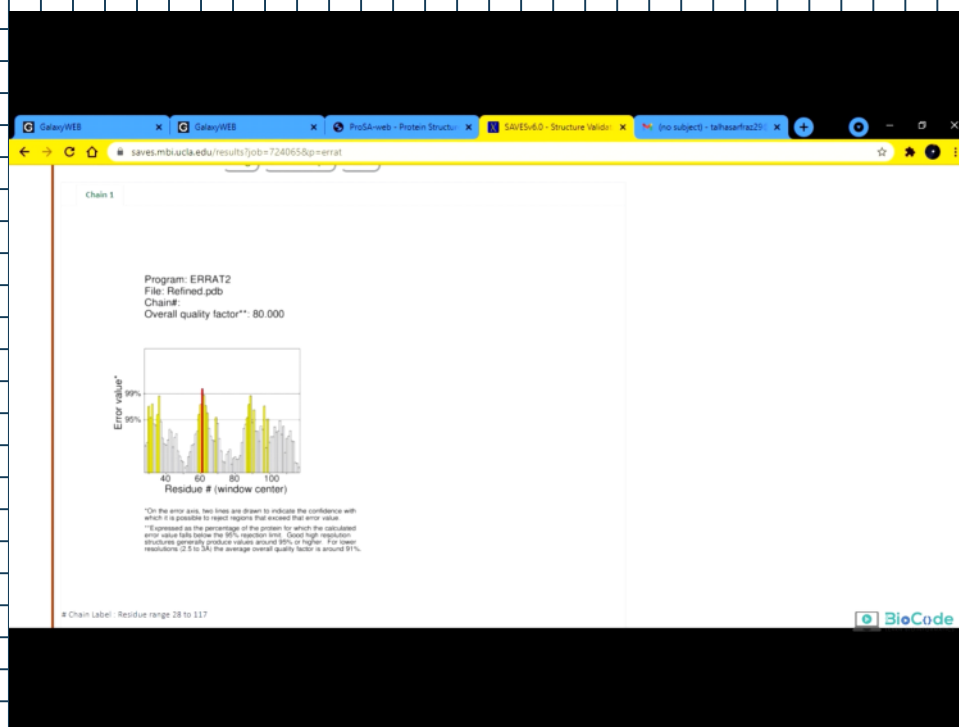
80

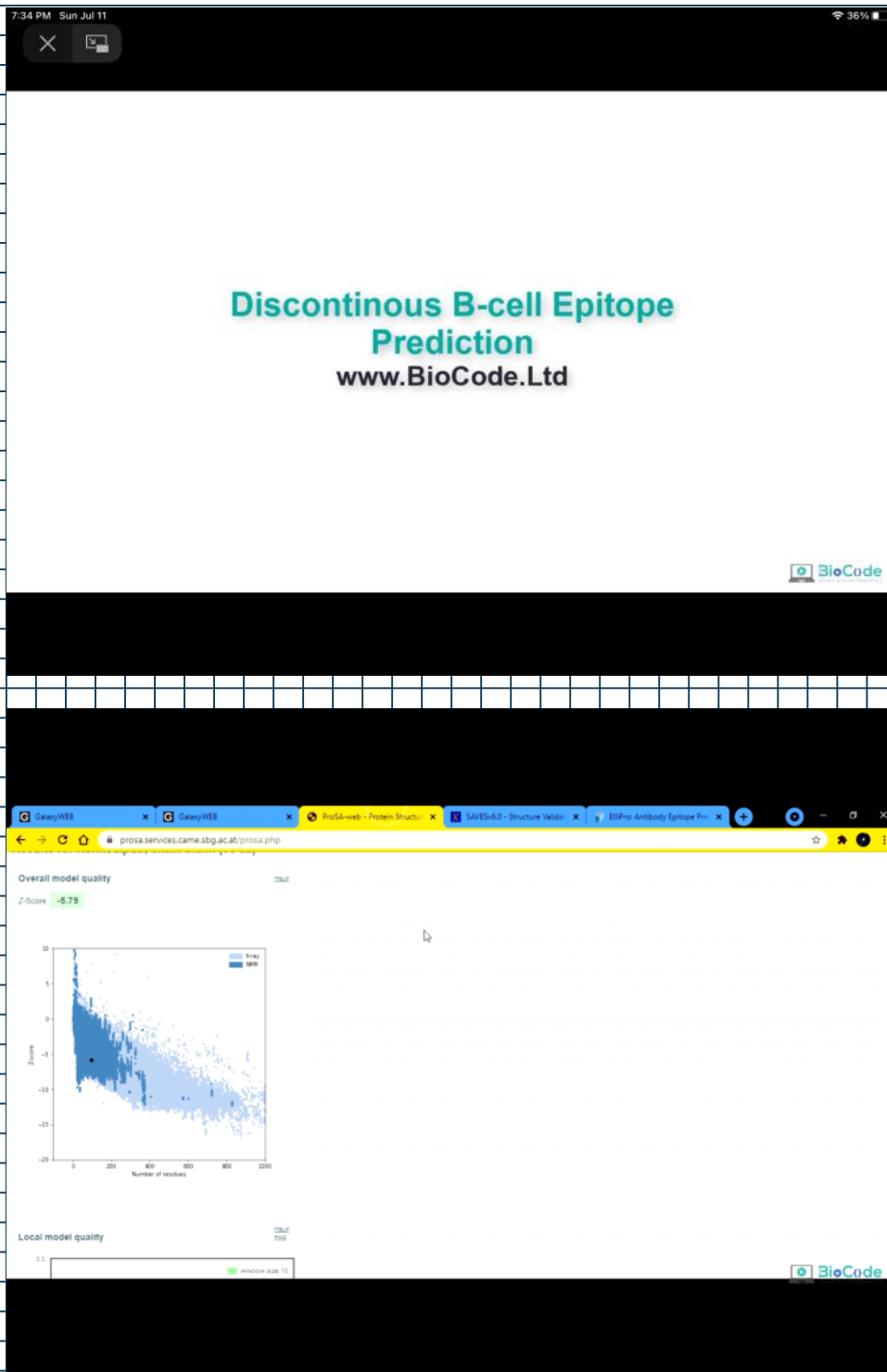
Log PostScript PDF

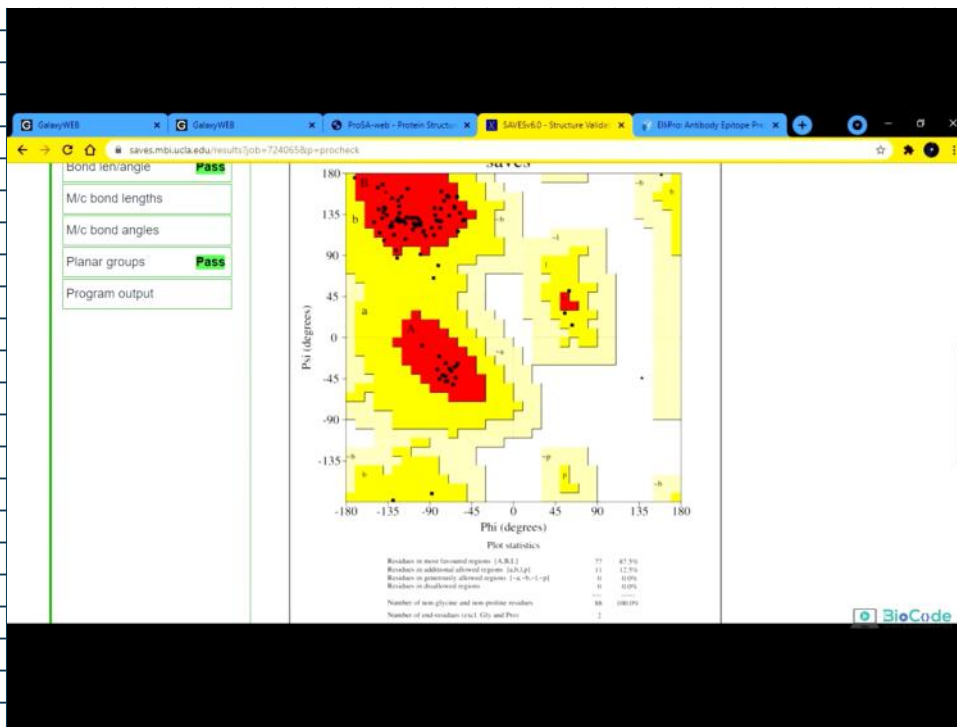
Chain 1

Program: ERRAT2

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IEDB Analysis Resource

Home | Help | Example | Reference | Download | Contact

ElliPro: Antibody Epitope Prediction

Specify Sequence(s)

Enter PDB ID(s) or upload PDB file:

Select Epitope Prediction Parameters

Minimum score: (Default is 0.5)

Maximum distance (Angstrom): (Default is 6)

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Supported by a contract from the [National Institute of Allergy and Infectious Diseases](#), a component of the National Institutes of Health in the Department of Health and Human Services.

GalaxyWEB x GalaxyWEB x ProSA-web - Protein Structu... x SAVES4.0 - Structure Valid... x ElliPro Prediction Result

tools.iedb.org/elliPro/result/predict/

IEDB Analysis Resource

Home Help Example Reference Download Contact

Input Sequences: filez6dwgr6y.pdb

Chain:
1 KQHWKTSQW YGRTPQPEV WLKYYVQDES KRLLNFRDS FAPTVKND EKGIVGSI
61 IPFVAMGDT YLGDSLSQ YELYNNASV KYGNFTL

Predicted Linear Epitope(s):

No.	Chain	Start	End	Peptide	Number of residues	Score	3D structure
1	-	73	79	DEKGI	6	0.835	View
2	-	50	55	DOESR	6	0.741	View
3	-	29	44	TKSDHWGRTPQPEIV	16	0.837	View
4	-	95	103	YLGDSLSQ	9	0.621	View
5	-	81	91	QSIFFPAAGE	11	0.586	View
6	-	61	66	KDSFAP	6	0.585	View

Predicted Discontinuous Epitope(s):

No.	Discontinues	Number of residues	Score	3D structure
1	_K24...Q25...Q50...Q51...E52...S53...K54...R55...D73...I74...E75...Q77...I78...G81...Y95...L96...G97...D98...L99...S100...L101...G102...Q103...H115...T120...L121	26	0.666	View
2	_K28...K30...S31...D32...V33...V34...I35...Q36...R37...T38...P39...V40...D41...P42...E43...V44...K51...D62...S63...F64...A65...P66...R67...F68...V69...A69	28	0.657	View
3	_A111...S112...V113	3	0.524	View

7:37 PM Sun Jul 11 biocode.org.uk - Private

+44 7518 945488 | support@biocode.org.uk

Search... Dashboard Log out

Steps:

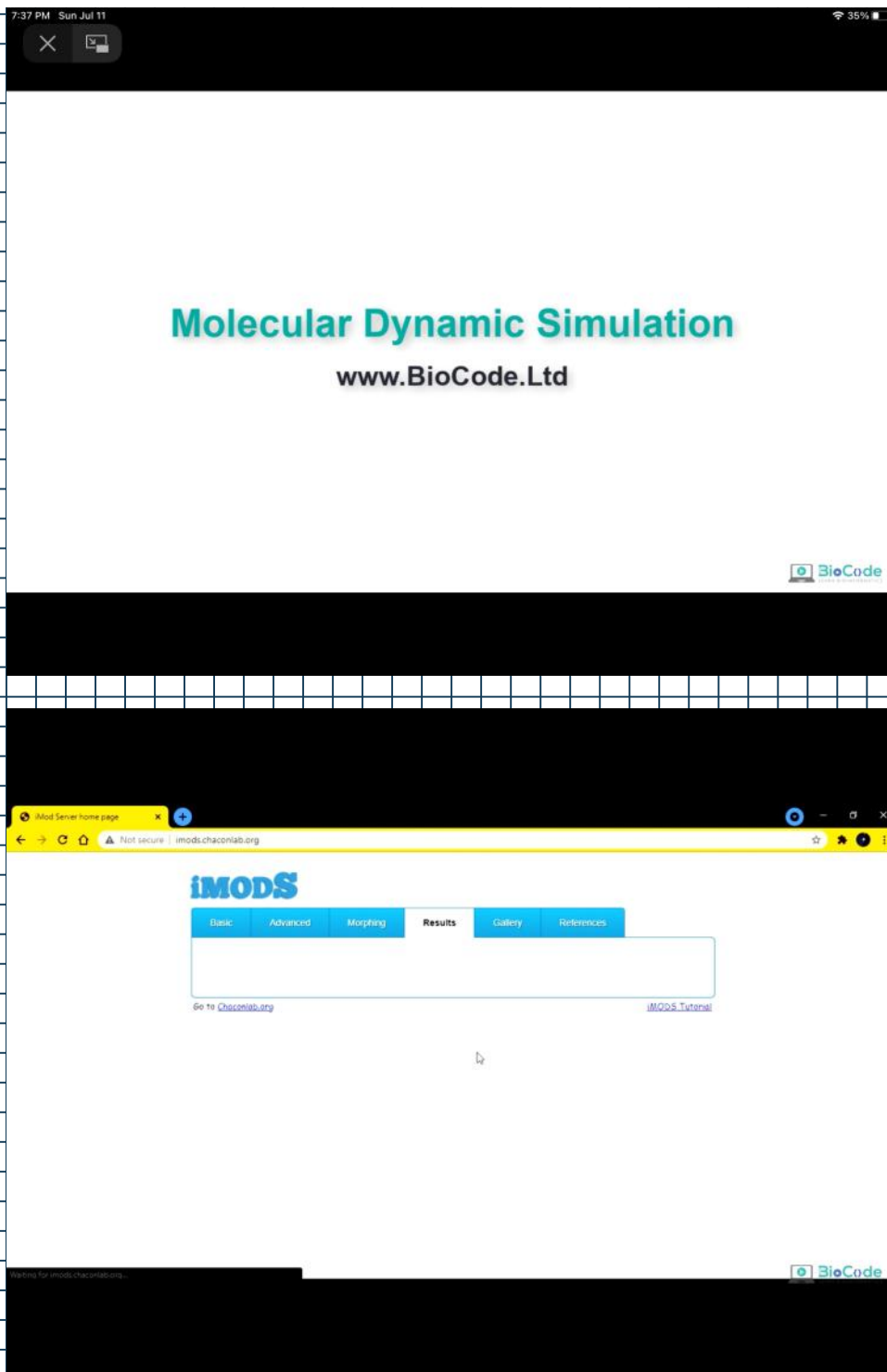
In this, results from ProSA-Web have been analyzed.

- Check the overall model quality, Z-score that shows goodness of a mode quality
- Check the PROCHECK results to see Rama Chandran plot that shows angles for the sheets and helices, phi, psi degrees. The results from this shows either the model is suitable for vaccine development or not.
- Use IEDB Analysis Resource for predicting the discontinues B-cell epitopes. This is done in ElliPro: Antibody Epitope Prediction tool available at IEDB.
- Upload the refined pdb structure of vaccine construct in ElliPro and enter on submit button.
- Results shows Predicted linear epitopes and discontinuous epitopes with their Peptide sequences, scores and 3D structures.
- Open the 3D structures of discontinuous epitopes one by one and check whether these amino acids are present in your actual structure or not by using any model visualization tool.
- The cross check helps whether the vaccine structure is stable or not and that is not true for further analysis.

Summary:

In this video tutorial, we learned how to check the refinement parameters for the validated and predicted model of vaccine construct just one click by using bioinformatics online refinement servers. The discontinues B-cell epitopes are also cross checked with the actual structure to know whether the exact amino acids are present or not. This helps in deciding whether the vaccine is useful for further analysis or not.

[Previous](#) [Next](#)



10:21 PM Sun Jul 11

38%

Immune Simulation

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Immune Simulation 10.1 Server

EMBOSS Sequence Form

ibiacuk/tools/dfs/emboss_secret

Input form

Web services

Help & Documentation

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above functions.

STEP 1 - Enter your input sequence

Enter or paste a set of

PROTEIN

sequences in any supported format:

Or, upload a file:

Choose File

 No file chosen

[Use a example sequence](#) | [Clear sequence](#) | [See more example inputs](#)

STEP 2 - Select Parameters

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[I agree, dismiss this banner](#)



EMBOSS Seqret - Sequence Format Converter

Input form | Web services | Help & Documentation | Bioinformatics Tools FAQ | Feedback | Share

PROTEIN

sequences in any supported format:

```
MIKLKFGVFFTVLLSSAYAHGTPQNTDLCAEYHNTQIYTLNDKIFSYESLAGKREMAITFKNGAIFQVEVPGSQHDSQKKAIERMKDTRLRIAYLTEAKVEKLCVWNN
KTPHAAIAISMANEAAKRSHDALTAQALS
AAYRRSHDALTAQALAAAYRVKLAFELVYGIAAYVKLAFELVYGIAAYVTAPLQTVAVKAAYVEGETLNLQLGYGPGGPGPGKTAVEALQIK
PTAAPGPGPLAFELVYGIKPLKVGPGGKLAFLVYGKPLKVGPGGQGFKKRLVGVGYRSKRRDRKRRSHDALKKIAISVTGEAPEKTDKKTSGQIKRKH
AYTSKKVGVGYRANVEGETL
```

Or, upload a file: No file chosen

[Use a example sequence](#) | [Clear sequence](#) | [See more example inputs](#)

STEP 2 - Select Parameters

INPUT FORMAT: Unknown format

OUTPUT FORMAT: EMBL entry format

The default settings will fulfill the needs of most users.

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BioCode

EMBOSS Seqret - Sequence Format Converter

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INPUT FORMAT: Unknown format

OUTPUT FORMAT: FASTA format

The default settings will fulfill the needs of most users.

(Click here, if you want to view or change the default settings.)

STEP 3 - Submit your job

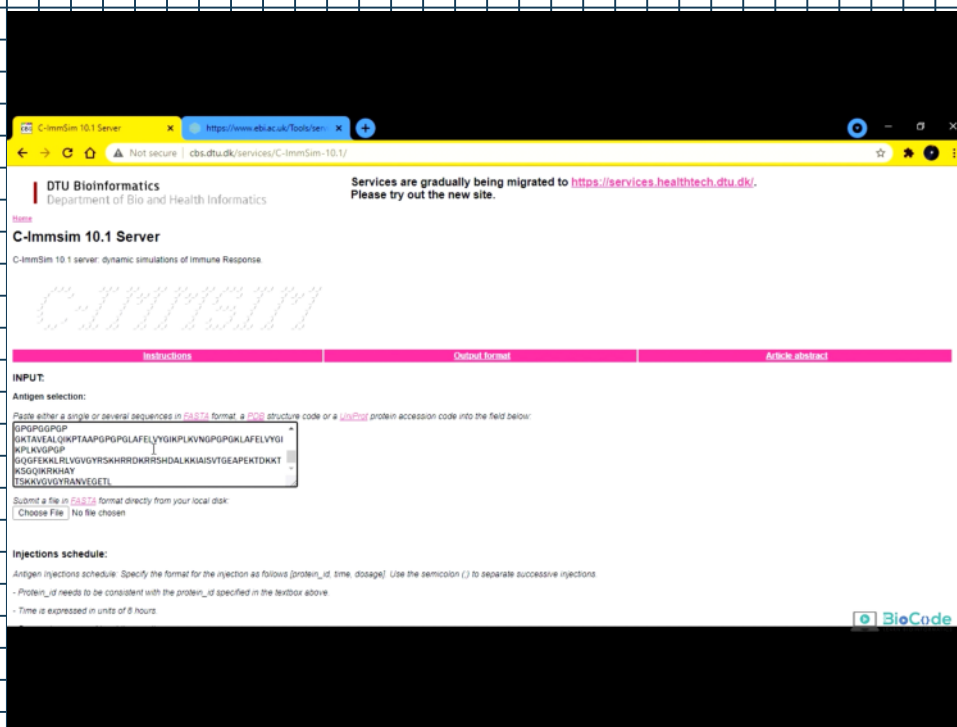
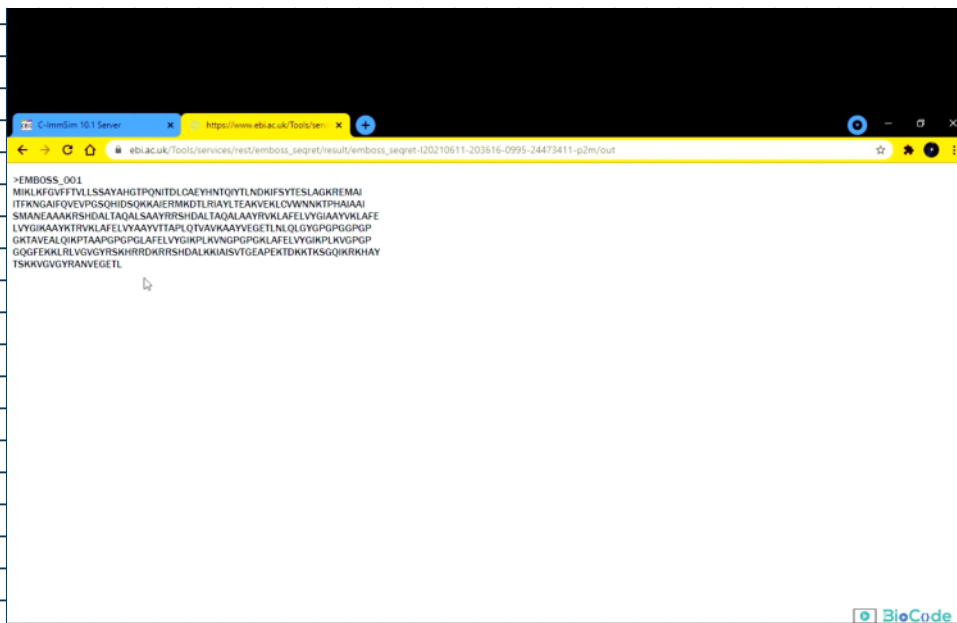
☐ Be notified by email (Tick this box if you want to be notified by email when the results are available)

If you use this service, please consider citing the following publication: **The EMBL-EBI search and sequence analysis tools APIs in 2019**

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BioCode



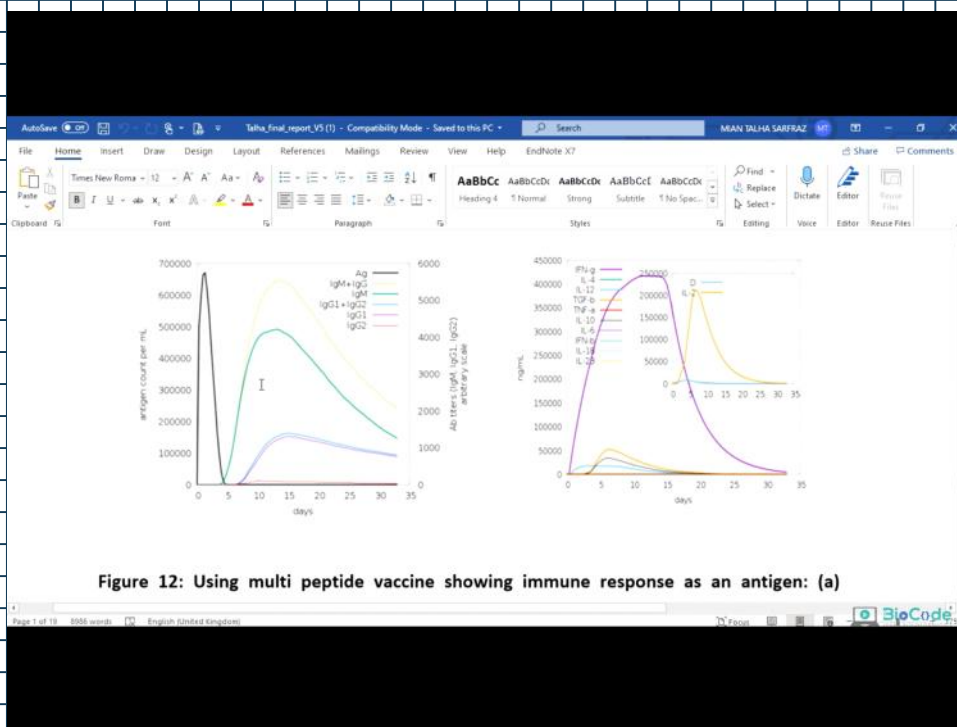
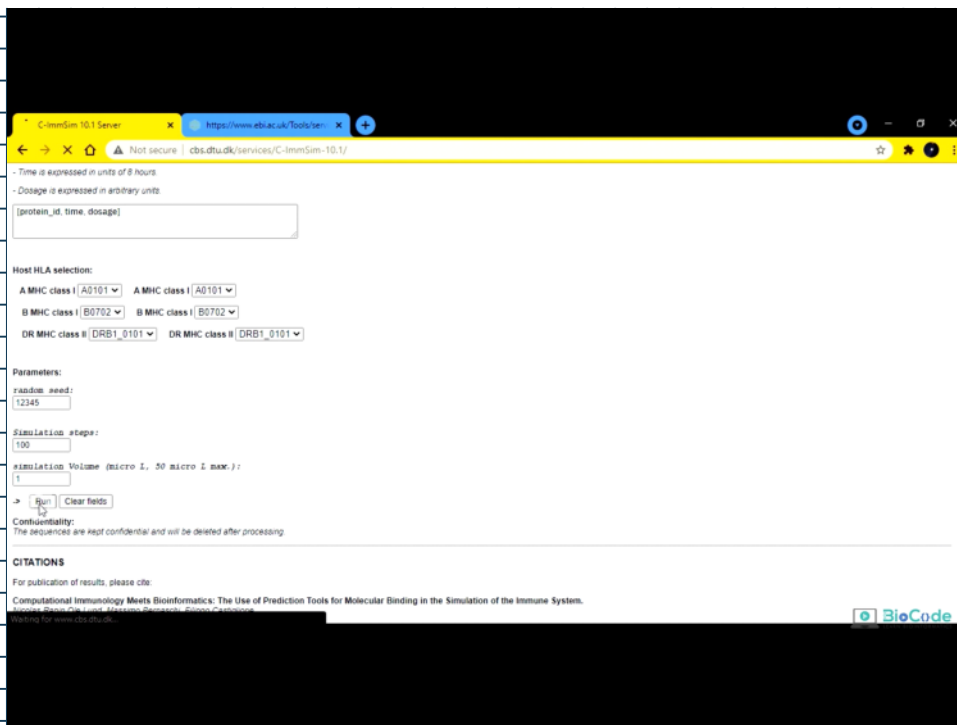
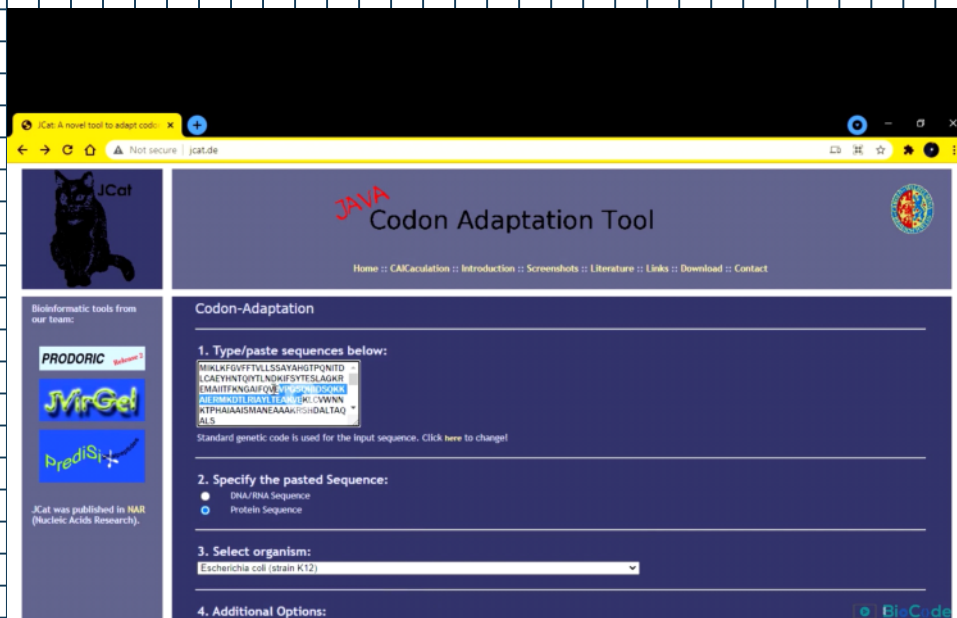
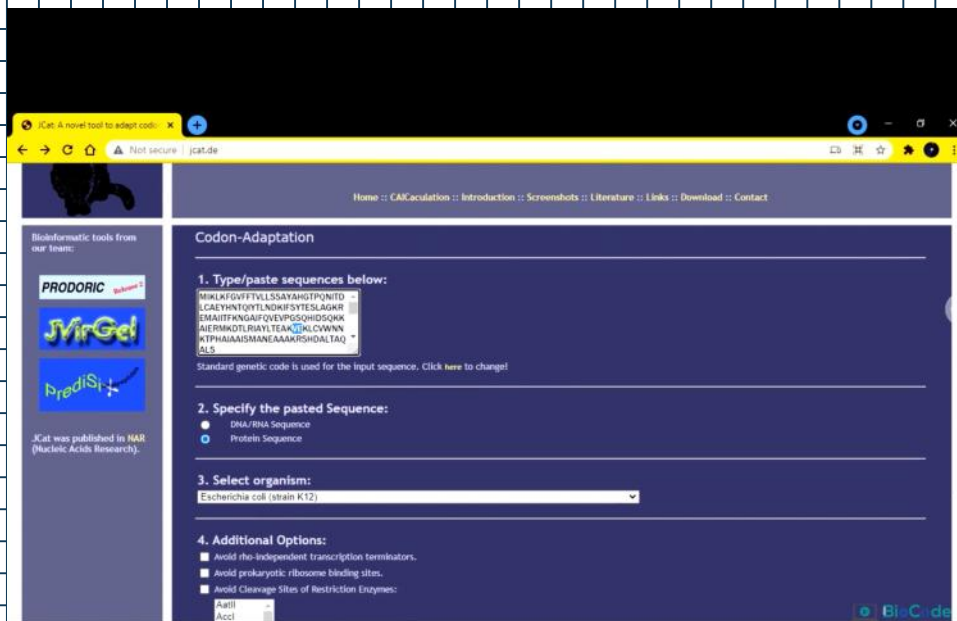
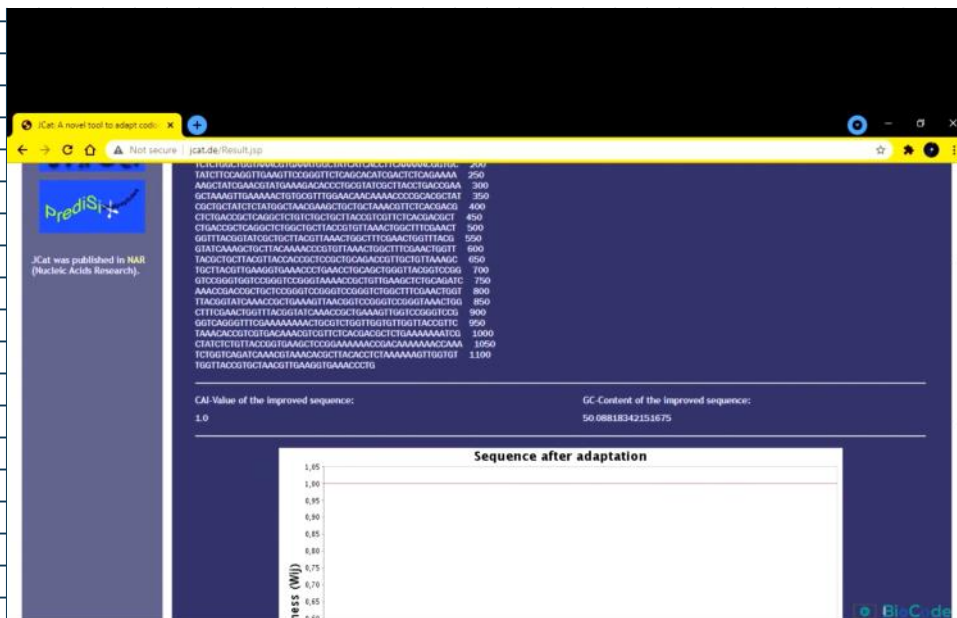


Figure 12: Using multi peptide vaccine showing immune response as an antigen: (a)

CODON OPTIMIZATION, DISULFIDE ENGINEERING AND RECEPTOR/PROTEIN BINDING PREDICTION

Sunday, July 11, 2021 10:49 PM



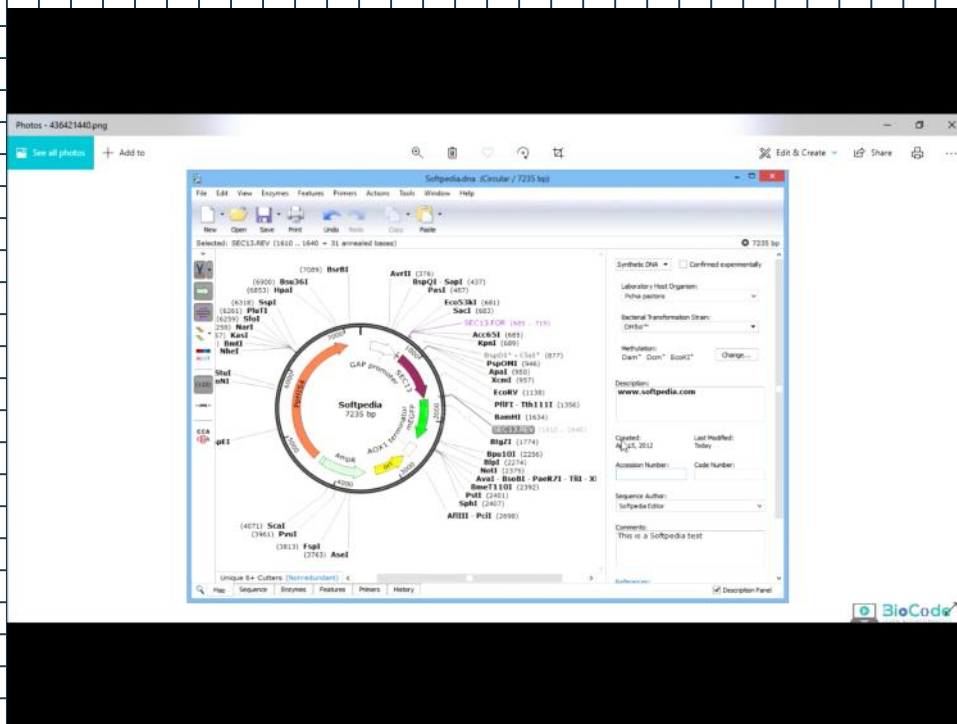


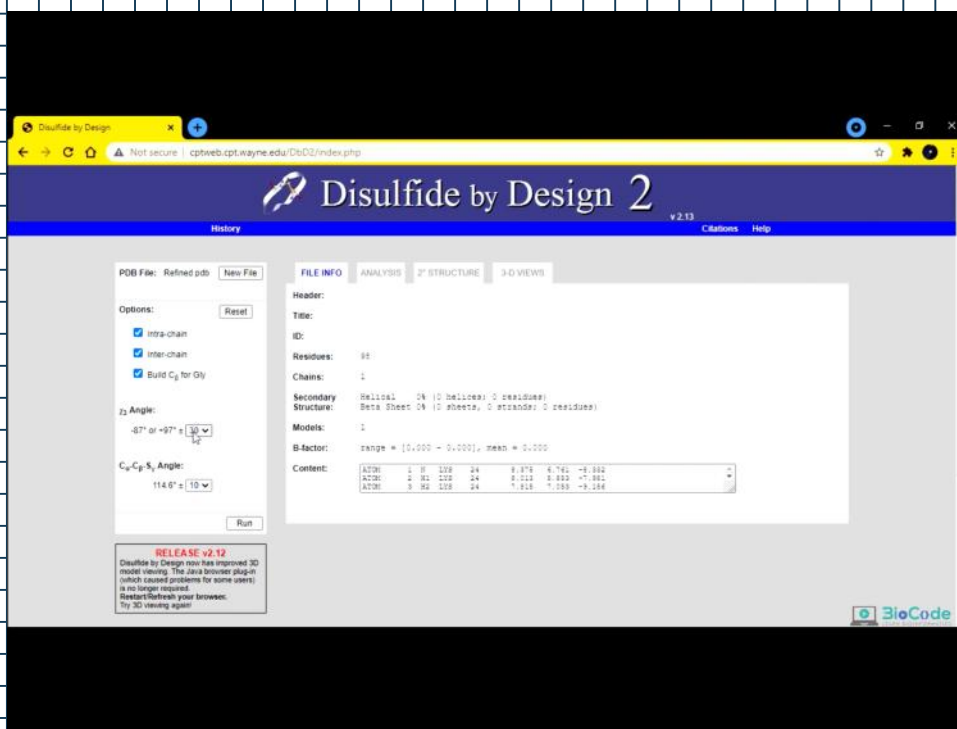
Photos - Haddock stats.PNG

Stats of the receptor TLR4 and vaccine complex.

Parameters	Values
HADDOCK-v.2.2 score	-74.6 ±2.2
Cluster-size	33
RMSD from the overall lowest energy structure	37.5 ±0.5
Van-der-Waals energy	-79.7 ±7.7
Electrostatic energy	-259.8 ±6.6
Desolvation energy	-59.9 ±21.9
Restraints violation energy	1170.4 ±149.76
Buried Surface Area	2415.3 ±164.9
Z-Score	-1.9

BioCode





Disulfide by Design 2 v2.13

History Citations Help

PDB File: Refined.pdb New File

Options: ☐ Intra-chain ☒ Inter-chain ☒ Build C_β for Gly

β₂ Angle: -87° or +97° ± 30°

C_α-C_β-S_γ Angle: 114.6° ± 10°

Run

RELEASE v2.12
Disulfide by Design now has improved 3D model viewing. The Java browser plugin (which caused problems for some users) is no longer required. Restart/Refresh your browser. Try 3D viewing again!

FILE INFO ANALYSIS 2nd STRUCTURE 3-D VIEWS

Click column headers to sort

Residue 1				Residue 2				Bond		
Chain	Seq #	AA	Struct	Chain	Seq #	AA	Struct	β ₂	kcal/mol	± B-factor
☐	27	VAL		☐	47	VAL		-90.67	3.04	0.00
☐	29	THR		☐	117	GLY		+92.17	2.39	0.00
☐	41	ASP		☐	44	TRP		+118.23	3.19	0.00
☐	54	LYS		☐	97	GLY		+98.27	2.27	0.00
☐	63	SER		☐	89	ALA		-105.38	4.11	0.00
☐	106	LEU		☐	117	GLY		-104.96	6.81	0.00
☐	108	MET		☐	115	VAL		-95.90	7.71	0.00

Create/View Mutant Save Results

BioCode

HADDOCK 2.4

Input data Input parameters Docking parameters

POWERED BY

JOHN J. HENDRICKS

Number of molecules: 2

Molecule 1 - Input

Where is the structure provided? *

I am submitting it

Which chain of the structure must be modelled? *

All

PDB structure to submit *

Choose File No file chosen

Search parameter...

EOSC-hub

EUROPEAN OPEN SCIENCE CLOUD

bioExcel

EGI

NWO

This work is co-funded by the Horizon 2020 projects EOSC-hub (grant number 777536) and BioExcel (grant numbers 823830 and 675728) and by a computing grant from NWO-ENW

BioCode

Predicted B-cell epitope

Thursday, October 14, 2021

10:34 PM



Predicted
B-cell epit...

Rank	Sequence	Start position	Score
1	RCGHIGALNEEAAK	368	0.91
2	SLAQLLLDPFFRTI	334	0.89
3	TITDYYKQLESSGW	281	0.88
4	EKNGRKRCWDRFNW	102	0.87
5	ATKQLIGSCKFRS	157	0.86
5	WKINDFNVDYRYCD	132	0.86
6	KCGRPVNIIDVFAF	85	0.85
6	NFLIAKDKDCQDLY	67	0.85
6	DEQLMNLIAKTNPS	205	0.85
6	FARQGIGIADKHLW	119	0.85
6	CWDRFNWQIEFARQ	109	0.85
7	LDACQRTKTITDYY	273	0.84
8	SDSGHSSTVSSGAQ	545	0.83
8	SDGWDRTSQTVSLA	323	0.83
9	IDVFAFENREKNGR	93	0.82
9	VTIATLEHIGVLEA	506	0.82
9	EKGNLSAIGTYLTI	45	0.82
9	SNIVGTIHITTSHF	10	0.82
10	VVWTAMYNRFDTG	484	0.81
10	PLTGFTARCEDEQ	194	0.81
11	VLEKDWLGFCHKF	352	0.80
12	QANKFSYLANVDLH	466	0.79
12	SAVVHCSDGWDRTS	317	0.79
13	LAKRTLSTLSYIDD	440	0.78
13	IFRADNSSKEIWVA	24	0.78
13	QKGYEDIKNYPNM	239	0.78
14	YQIMRLQPRAFEFN	395	0.77
14	RFHFFDIENIHVMR	253	0.77
14	NPSYQSLCLIDTRP	216	0.77
14	NVDYRYCDTYPEYL	138	0.77
15	KDRKDLNLAKRSL	433	0.76
15	LIGSCKFRSRARLP	162	0.76
16	STVSSGAQPTPLQ	551	0.75
16	QMASLQAVTFLAPS	520	0.75
17	APSFLEFMKMSGVM	531	0.74
17	NYPNMRFHFFDIEN	248	0.74
17	KFRSRARLPVLTW	167	0.74
18	HIGVLEAQMASLQA	513	0.73
18	VANALIGNIEKGNL	36	0.73
18	NIHVMRSSLNKLDD	261	0.73
19	YLCTPSSATKQILI	150	0.72
20	RSLLDCSAFLADLV	299	0.71
20	HRPNFGRFCRCAQP	181	0.71
21	KMISGVMSDSGHSS	538	0.70
21	FLADLVSRGISAVV	307	0.70
22	WSYIDDHDDYINP	448	0.69
22	FEFNERYLIDIEH	405	0.69
22	FRTRGFQVLEKD	344	0.69
23	YNRFDTGILLPREST	490	0.68
23	HKFDDRCGHIGALN	363	0.68
24	LHPSAIVVWTAMYN	478	0.67
24	RFCRCAQPLTGFTA	187	0.67
25	DLYETLLKCGRPVN	78	0.66
25	GALNEEAAKEVSPI	373	0.66

26	QAVTFLAPSLFMK	525	0.63
26	MMNKMQGKGYEDIK	234	0.63
26	ARLPVLTYWHRPNF	172	0.63
27	AKEVSPIFTQFLDC	380	0.62
27	RTSQTVSLAQLLLD	328	0.62
28	NSSKEIWVANALIG	29	0.58
29	HHDDYINPLYEFQA	454	0.57
30	GAQPTTPLQSLPEL	556	0.56
30	TIHITTSHFIFRAD	15	0.56
31	IDTRPLVNAMMNKM	225	0.53
32	HEHAYSCQYGTFLG	416	0.51

- 04/17/21

Saturday, April 17, 2021

5:34 PM

NCBI → "Gene" LCT Search

orthology* → genes in different species, that evolved from a common ancestral gene by speciation, and generally, orthologs retain the same function during the course of evolution

↓
Generally available in databases such as Homologene, eGenes, SGD, and MGD with further links to Flybase and Wormbase.

the degree of homology* → How similar two or more related genes between (cross) species or individuals.

Sequence Analysis & Retrieval

FASTA files are available through the "nucleotide" database for genes such as LCT (Homo Sapiens LCT (LCT) gene, intron 1, exon 2, and partial CDS.

[Link to FASTA File](#)

Genes in this database have a "Locus" which is an "accession number" or ID. If you type that on the NCBI search bar

which is an accession number or ID.
 If you type that on the NCBI search bar
 it'll take you to the gene "NM_002299.4"
 locus for Homo Sapiens lactase (LCT), mRNA.
 For example:

[Homo sapiens LCT \(LCT\) gene, intron 1, exon 2 and partial cds](#)

LOCUS: DQ109677

5761 bp

DNA: linear

PR1: 26-JUL-2016

DEFINITION: Homo sapiens LCT (LCT) gene, intron 1, exon 2 and partial cds.

length

date of test

Sequence of gene that codes for an enzyme/protein

[Homo sapiens lactase \(LCT\), mRNA](#)

LOCUS: NM_002299

6273 bp

mRNA: linear

PR1: 16-FEB-2021

DEFINITION: Homo sapiens lactase (LCT), mRNA.

Reference Seq: NM_002299.4

has been updated 4 times

FEATURES

source

Location/Qualifiers

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/chromosome="2"

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gene

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misc feature

10..12

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/note="upstream in-frame stop codon"

Taxon number

Chromosome #2
map position

ref # on other databases

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} Enzyme commission number
to expasy database

* when searching, molecs & items that
appear as a result, have a format type

Homo Sapiens, Lactase (LCT), mRNA

↑
Taxonomical
name
of
organism

↑
molecule
name

↑
molecule
Type

Click Here

NCBI Resources How To Sign In to NCBI

Nucleotide Nucleotide Advanced Search Help

GenBank

Homo sapiens lactase (LCT), mRNA

NCBI Reference Sequence: NM_002299.4

FASTA Graphics

Go to

LOCUS NM_002299 6273 bp mRNA linear PRI 02 2001

DEFINITION Homo sapiens lactase (LCT), mRNA.

ACCESSION NM_002299

VERSION NM_002299.4

KEYWORDS RefSeq; MANE Select.

SOURCE Homo sapiens (human)

ORGANISM Homo sapiens

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.

REFERENCE 1 (bases 1 to 6273)

AUTHORS Muendlein A, Leisher A, Zach C, Brandtner EM, Fraunberger P, Drexler H and Geiger K.

TITLE Real-time PCR based detection of the lactase non-persistence associated genetic variant LCT-13910C>T directly from whole blood

JOURNAL Mol. Biol. Rep. 46 (2), 2379-2385 (2019)

PUBMED 30720118

REMARK GeneRIF: The LCT-13910C>T polymorphism shows complete correlation with the lactase persistence/non-persistence phenotype

REFERENCE 2 (bases 1 to 6273)

AUTHORS Viissers LET, Sluijs I, van der Schouw YT, Forouhi NG, Imamura F, Burgess S, Barricarte A, Boeing H, Bonet C, Chirlaque MD, Fagherazzi G, Franks PW, Freisling H, Gunter MJ, Quiros JR, Ibsen DB, Kaaks R, Key T, Khaw KT, Kuhn T, Mokoroa O, Nilsson PM, Overvad K, Pala V, Palli D, Panico S, Sacerdote C, Spijkerman AMM, Tjonneland A, Tumino R, Rodriguez-Barranco M, Rolandsson O, Riboli

Send to: Complete Record Coding Sequences Gene Features

Download features. Format FASTA Nucleotide Create File

placements & positions of exons & introns

mRNA Sequence in

AUTHORS Vissers LET, Sluijs I, van der Schouw YT, Forouhi NG, Imamura F, Burgess S, Barricarte A, Boeing H, Bonet C, Chirlaque MD, Fagherazzi G, Franks PW, Freisling H, Gunter MJ, Quiros JR, Ibsen DB, Kaaks R, Key T, Khaw KT, Kuhn T, Mokoroa O, Nilsson PM, Overvad K, Pala V, Palli D, Panico S, Sacerdote C, Spijkerman AMW, Tjonneland A, Tumino R, Rodriguez-Barranco M, Rolandsson O, Riboli E, Sharp SJ, Langenberg C and Wareham NJ.
TITLE Dairy Product Intake and Risk of Type 2 Diabetes in EPIC-InterAct: A Mendelian Randomization Study
JOURNAL Diabetes Care 42 (4), 568-575 (2019)
PUBMED 38728219

mkRNA
Pathways for the LCT gene
Dietary intake and dairy carbohydrates
Metabolism of carbohydrates
Fasta Format
See all...

- 04/18/21

Saturday, April 17, 2021 8:42 PM

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[Translate](#)

No. of AA with each alphabet: A =4, G=3, L=2, P=2, T=3, all others 1

AA Name	3-L code	1-L code	AA Name	3-L code	1-L code
Ala nine	Ala	A	Leu cine	Leu	L
Arg inine	Arg	R	Lys ine	Lys	K
Asp aragine	Asn	N	Meth ionine	Met	M
Asp artic acid	Asp	D	Phenyl alanine	Phe	F
Cys teine	Cys	C	Pro line	Pro	P
Glu tamic acid	Glu	E	Ser ine	Ser	S
Glu tamine	Gln	Q	Thre onine	Thr	T
Gly cine	Gly	G	Tryp tophan	Trp	W
His tidine	His	H	Tyr osine	Tyr	Y
Ile ucine	Ile	I	Val ine	Val	V

ExPASy -
Translate ..

Translate



Translate is a tool which allows the translation of a nucleotide (DNA/RNA) sequence to a protein sequence.

DNA or RNA sequence

aac agt tcc tag aaa atg gag ctg tct tgg cat gta gtc ttt att gcc ctg cta agt ttt
 att tatt gcoot gota agt ttt tto atg otg ggg gto ag
 aot ggg agt otg atagaaa ttt to attt oac ogt ggt
 oot taacca atg actt gct gca aac ogt agt ggt
 oot ggg agacc agt tota aot ttt gtagc aggg gaca
 aag acat gtat gttt gto acc agc oot gccc att to
 otg ccaga ata act to agc agt to cat ggc agt cag at
 cacc oatt ata aggt att ttt gto atg ggc ac agt oot
 tocca gac gga gac acc agaat cca gac gac gaaa aca

Output format

- ☐ Verbose: Met, Stop, spaces between residues
☐ Compact: M, -, no spaces
☒ Includes nucleotide sequence
☐ Includes nucleotide sequence, no spaces

DNA strands

- ☒ forward ☐ reverse

Genetic codes - See NCBI's genetic codes

Standard

reset

TRANSLATE!

Results of translation

- Open reading frames are highlighted in red
- Select your initiator on one of the following frames to retrieve your amino acid sequence

Download all the translated frames

5' Frame 1

aac agt tcc tag aaa atg gag ctg tct tgg cat gta gtc ttt att gcc ctg cta agt ttt
 N S S - K M E L S W H V V F I A L L S F
 tca tgc tgg ggg tca gac tgg gag tct gat aga aat ttc att tcc acc gct ggt cct cta
 S C W G S D W E S D R N F I S T A G P L
 acc aat gac ttg ctg cac aac ctg agt ggt ctc ctg gga gac cag agt tct aac ttt gta
 T N D L L H N L S G L L G D Q S S N F V
 gca ggg gac aaa gac atg tat gtt tgt cac cag cca ctg ccc act ttc ctg cca gaa tac
 A G D K D M Y V C H Q P L P T F L P E Y
 ttc agc agt ctc cat gcc agt cag atc acc cat tat aag gta ttt ctg tca tgg gca cag
 F S S L H A S Q I T H Y K V F L S W A Q
 ctc ctc cca gca gga agc acc cag aat cca gac gag aaa aca gtg cag tgc tac cgg cga
 L L P A G S T Q N P D E K T V Q C Y R R
 ctc ctc aag gcc ctc aag act gca cgg ctt cag ccc atg gtc atc ctg cac cac cag acc
 L L K A L K T A R L Q P M V I L H H Q T
 ctc cct gcc agc acc ctc cgg aga acc gaa gcc ttt gct gac ctc ttc gcc gac tat gcc

N = asparagine

<https://web.expasy.org/translate/>

1/10

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L P A S T L R R T E A F A D L F A D Y A
aca ttc gcc ttc cac tcc ttc ggg gac cta gtt ggg atc tgg ttc acc ttc agt gac ttg
T F A F H S F G D L V G I W F T F S D L
gag gaa gtg atc aag gag ctt ccc cac cag gaa tca aga gcg tca caa ctc cag acc ctc
E E V I K E L P H Q E S R A S Q L Q T L
agt gat gcc cac aga aaa gcc tat gag att tac cac gaa agc tat gct ttt cag ggc gga
S D A H R K A Y E I Y H E S Y A F Q G G
aaa ctc tct gtt gtc ctg cga gct gaa gat atc cgg gag ctc ctg cta gaa cca ccc ata
K L S V V L R A E D I P E L L L E P P I
tct gcg ctt gcc cag gac acg gtc gat ttc ctc tct ctt gat ttg tct tat gaa tgc caa
S A L A Q D T V D F L S L D L S Y E C Q
aat gag gca agt ctg cgg cag aag ctg agt aaa ttg cag acc att gag cca aaa gtg aaa
N E A S L R Q K L S K L Q T I E P K V K
gtt ttc atc ttc aac cta aaa ctc cca gac tgc ccc tcc acc atg aag aac cca gcc agt
V F I F N L K L P D C P S T M K N P A S
ctg ctc ttc agc ctt ttt gaa gcc ata aat aaa gac caa gtg ctc acc att ggg ttt gat
L L F S L F E A I N K D Q V L T I G F D
att aat gag ttt ctg agt tgt tca tca agt tcc aag aaa agc atg tct tgt tct ctg act
I N E F L S C S S S S K K S M S C S L T
ggc agc ctg gcc ctt cag cct gac cag cag gac cac gag acc acg gac tcc tct cct
G S L A L Q P D Q Q Q D H E T T D S S P
gcc tct gcc tat cag aga atc tgg gaa gca ttt gcc aat cag tcc agg gcg gaa agg gat
A S A Y Q R I W E A F A N Q S R A E R D
gcc ttc ctg cag gat act ttc cct gaa ggc ttc ctc tgg ggt gcc tcc aca gga gcc ttt
A F L Q D T F P E G F L W G A S T G A F
aac gtg gaa gga ggc tgg gcc gag ggt ggg aga ggg gtg agc atc tgg gat cca cgc agg
N V E G G W A E G G R G V S I W D P R R
ccc ctg aac acc act gag ggc caa gcg acg ctg gag gtg gcc agc gac agt tac cac aag
P L N T T E G Q A T L E V A S D S Y H K
gta gcc tct gac gtc gcc ctg ctt tgc ggc ctc cgg gct cag gtg tac aag ttc tcc atc
V A S D V A L L C G L R A Q V Y K F S I
tcc tgg tcc cgg atc ttc ccc atg ggg cac ggg agc agc ccc agc ctc cca ggc gtt gcc
S W S R I F P M G H G S S P S L P G V A
tac tac aac aag ctg att gac agg cta cag gat gcg ggc atc gag ccc atg gcc acg ctg
Y Y N K L I D R L Q D A G I E P M A T L
ttc cac tgg gac ctg cct cag gcc ctg cag gat cat ggt gga tgg cag aat gag agc gtg
F H W D L P Q A L Q D H G G W Q N E S V
gtg gat gcc ttc ctg gac tat gcg gcc ttc tgc ttc tcc aca ttt ggg gac cgt gtg aag
V D A F L D Y A A F C F S T F G D R V K
ctg tgg ctg acc ttc cat gag cgg tgg gtg atg agc tac gca ggc tat ggc acc ggc cag
L W V T F H E P W V M S Y A G Y G T G Q
cac cct ccc ggc atc tct gac cca gga gtg gcc tct ttt aag gtg gct cac ttg gtc ctc
H P P G I S D P G V A S F K V A H L V L
aag gct cat gcc aga act tgg cac cac tac aac agc cat cat cgc cca cag cag cag ggg
K A H A R T W H H Y N S H H R P Q Q Q G
cac gtg ggc att gtg ctg aac tca gac tgg gca gaa ccc ctg tct cca gag agg cct gag
H V G I V L N S D W A E P L S P E R P E
gac ctg aga gcc tct gag cgc ttc ttg cac ttc atg ctg ggc tgg ttt gca cac ccc gtc
D L R A S E R F L H F M L G W F A H P V
ttt gtg gat gga gac tac cca gcc acc ctg agg acc cag atc caa cag atg aac aga cag
F V D G G D Y P A T L R T Q I Q Q M N R Q
tgc tcc cat cct gtg gct caa ctc ccc gag ttc aca gag gca gag aag cag ctc ctg aaa
C S H P V A Q L P E F T E A E K Q L L K
ggc tct gct gat ttt ctg ggt ctg tgg cat tac acc tcc cgc ctc atc agc aac gcc cca
G S A D F L G L S H Y T S R L I S N A P
caa aac acc tgc atc cct agc tat gat acc att gga ggc ttc tcc caa cac gtg aac cat
Q N T C I P S Y D T I G G F S Q H V N H
gtg tgg ccc cag acc tca tcc tct tgg att cgt gtg gtg ccc tgg ggg ata agg agg ctg
V W P Q T S S S W I R V V P W G I R R L
ttg cag ttt gta tcc ctg gaa tac aca aga gga aaa gtt cca ata tac ctt gcc ggg aat
L Q F V S L E Y T R G K V P I Y L A G N
ggc atg ccc ata ggg gaa agt gaa aat ctc ttt gat gat tcc tta aga gta gac tac ttc
G M P I G E S E N L F D D S L R V D Y F
aat caa tat atc aat gag gtg ctc aag gct atc aag gaa gac tct gtg gat gtt cgt tcc
N Q Y I N E V L K A I K E D S V D V R S
tac att cgt tcc ctc att gat ggc ttc gaa ggc cct tct ggt tac agc cag cgg ttt
Y I A R S L I D G F E G P S G Y S Q R F
ggc ctg cac cag gtc aac ttc agc gac agc aag tca agg act ccc agg aaa tct gcc
G L H H V N F S D S S K S R T P R K S A
tac ttt ttc act agc atc ata gaa aag aac ggt ttc ctc acc aag ggg gca aaa aga ctg
Y F T S I I E K N G F L T K G A K R L
cta cca cct aat aca gta aac ctc ccc tcc aaa gtc aga gcc ttc act ttt cca tct gag
L P P N T V N L P S K V R A F T F P S E
gtg ccc tcc aag gct aaa gtc gtt tgg gaa aag ttc tcc agc caa ccc aag ttc gaa aga
V P S K A K V V W E K F S S Q P K F E R

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gat ttg ttc tac cac ggg acg ttt cgg gat gac ttt ctg tgg ggc gtg tcc tct tcc gct
D L F Y H G T F R D D F L W G V S S S A
tat cag att gaa ggc gcg tgg gat gcc gat ggc aaa ggc ccc agc atc tgg gat aac ttt
Y Q I E G A W D A D G K G P S I W D N F
acc cac aca cca ggg agc aat gtg aaa gac aat gcc act gga gac atc gcc tgt gac agc
T H T P G S N V K D N A T G D I A C D S
tat cac cag ctg gat gcc gat ctg aat atg ctc cga gct ttg aag gtg aag gcc tac cgc
Y H Q L D A D L N M L R A L K V K A Y R
ttc tct atc tcc tgg tct cgg att ttc cca act ggg aga aac agc tct atc aac agt cat
F S I S W S R I F P T G R N S S I N S H
ggg gtt gat tat tac aac agg ctg atc aat ggc ttg gtg gca agc aac atc ttt ccc atg
G V D Y Y N R L I N G L V A S N I F P M
gtg aca ttg ttc cat tgg gac ctg ccc cag gcc ctc cag gat atc gga ggc tgg gag aat
V T L F H W D L P Q A L Q D I G G W E N
cct gcc ttg att gac ttg ttt gac agc tac gca gac ttt tgt ttc cag acc ttt ggt gat
P A L I D L F D S Y A D F C F Q T F G D
aga gtc aag ttt tgg atg act ttt aat gag ccc atg tac ctg gca tgg cta ggt tat ggc
R V K F W M T F N E P M Y L A W L G Y G
tca ggg gaa ttt ccc cca ggg gtg aag gac cca ggc tgg gca cca tat agg ata gcc cac
S G E F P P G V K D P G W A P Y R I A H
gcc gtc atc aaa gcc cat gcc aga gtc tat cac acg tac gat gag aaa tac agg cag gag
A V I K A H A R V Y H T Y D E K Y R Q E
cag aag ggg gtc atc tgg ctg agc ctc agt aca cac tgg gca gag ccc aag tca cca ggg
Q K G V I S L S L S T H W A E P K S P G
gtc ccc aga gat gtg gaa gcc gct gac cga atg ctg cag ttc tcc ctg ggc tgg ttt gct
V P R D V E A A D R M L Q F S L G W F A
cac ccc att ttt aga aac gga gac tat cct gac acc atg aag tgg aaa gtg ggg aac agg
H P I F R N G D Y P D T M K W K V G N R
agt gaa ctg cag cac tta gcc acc tcc cgc ctg cca agc ttc act gag gaa gag aag agg
S E L Q H L A T S R L P S F T E E E K R
ttc atc agg gcg acg gcc gac gtc ttc tgc ctc aac acg tac tac tcc aga atc gtg cag
F I R A T A D V F C L N T Y Y S R I V Q
cac aaa aca ccc agg cta aac cca ccc tcc tac gaa gac gac cag gag atg gct gag gag
H K T P R L N P P S Y E D D Q E M A E E
gag gac cct tgg tgg cct tcc acg gca atg aac aga gct gcg ccc tgg ggg acg cga agg
E D P S W P S T A M N R A A P W G T R R
ctg ctg aac tgg atc aag gaa gat ggt gac atc ccc att tac atc acc gaa aac gga
L L N W I K E E Y G D I P I Y I T E N G
gtg ggg ctg acc aat ccg aac acg gag gat act gat agg ata ttt tac cac aaa acc tac
V G L T N P N T E D T D R I F Y H K T Y
atc aat gag gct ttg aaa gcc tac agg ctc gat ggt ata gac ctt cga ggg tat gtc gcc
I N E A L K A Y R L D G I D L R G Y V A
tgg tct ctg atg gac aac ttt gag tgg cta aat ggc tac acg gtc aag ttt gga ctg tac
W S L M D N F E W L N G Y T V K F G L Y
cat gtt gat ttc aac aac acg aac agg cct cgc aca gca aga gcc tcc gcc agg tac tac
H V D F N N T N R P R T A R A S A R Y Y
aca gag gtc att acc aac aac ggc atg cca ctg gcc agg gag gat gag ttt ctg tac gga
T E V I T N N G M P L A R E D E F L Y G
cgg ttt cct gag ggc ttc atc tgg agt gca gct tct gct gca tat cag att gaa ggt gcg
R F P E G F I W S A A S A A Y Q I E G A
tgg aga gca gat ggc aaa gga ctc agc att tgg gac acg ttt tct cac aca cca ctg agg
W R A D G K G L S I W D T F S H T P L R
gtt gag aac gat gcc att gga gac gtg gcc tgt gac agt tat cac aag att gct gag gat
V E N D A I G D V A C D S Y H K I A E D
ctg gtc acc ctg cag aac ctg ggc gtg tcc cac tac cgt ttt tcc atc tcc tgg tct cgc
L V T L Q N L G V S H Y R F S I S W S R
atc ctc cct gat gga acc acc agg tac atc aat gaa gcg ggc ctg aac tac tac gtg agg
I L P D G T T R Y I N E A G L N Y Y V R
ctc atc gat aca ctg ctg gcc gcc agc atc cag ccc cag gtg acc att tac cac tgg gac
L I D T L L A A S I Q P Q V T I Y H W D
cta cca cag acg ctc caa gat gta gga ggc tgg gag aat gag acc atc gtg cag cgg ttt
L P Q T L Q D V G G W E N E T I V Q R F
aag gag tat gca gat gtg ctc ttc cag agg ctg gga gac aag gtg aag ttt tgg atc acg
K E Y A D V L F Q R L G D K V K F W I T
ctg aat gag ccc ttt gtc att gct tac cag ggc tat ggc tac gga aca gca gct cca gga
L N E P F V I A Y Q G Y G T A A P G
gtc tcc aat agg cct ggc act gcc ccc tac att gtt ggc cac aat cta ata aag gct cat
V S N R P G T A P Y I V G H N L I K A H
gct gag gcc tgg cat ctg tac aac gat gtg tac cgc gcc agt caa ggt ggc gtg att tcc
A E A W H L Y N D V Y R A S Q G G V I S
atc acc atc agc agt gac tgg gct gaa ccc aga gat ccc tct aac cag gag gat gtg gag
I T I S S D W A E P R D P S N Q E D V E
gca gcc agg aga tat gtt cag ttc atg gga ggc tgg ttt gca cat cct att ttc aag aat
A A R R Y V Q F M G G W F A H P I F K N
gga gat tac aat gag gtg atg aag acg cgg atc cgt gac agg agc ttg gct gca ggc ctc

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G D Y N E V M K T R I R D R S L A A G L
aac aag tct cgg ctg cca gaa ttt aca gag agt gag aag agg agg atc aac ggc acc tat
N K S R L P E F T E S E K R R I N G T Y
gac ttt ttt ggg ttc aat cac tac acc act gtc ctc gcc tac aac ctc aac tat gcc act
D F F G F N H Y T T V L A Y N L N Y A T
gcc atc tct tct ttt gat gca gac aga gga gtt gct tcc atc gca gat cgc tcg tgg cca
A I S S F D A D R G V A S I A D R S W P
gac tct ggc tcc ttc tgg ctg aag atg acg cct ttt ggc ttc agg agg atc ctg aac tgg
D S G S F W L K M T P F G F R R I L N W
tta aag gag gaa tac aat gac cct cca att tat gtc aca gag aat gga gtg tcc cag cgg
L K E E Y N D P P I Y V T E N G V S Q R
gaa gaa aca gac ctc aat gac act gca agg atc tac tac ctt cgg act tac atc aat gag
E E T D L N D T A R I Y Y L R T Y I N E
gcc ctc aaa gct gtg cag gac aag gtg gac ctt cga gga tac aca gtt tgg agt gcg atg
A L K A V Q D K V D L R G Y T V W S A M
gac aat ttt gag tgg gcc aca ggc ttt tca gag aga ttt ggt ctg cat ttt gtg aac tac
D N F E W A T G F S E R F G L H F V N Y
agt gac cct tct ctg cca agg atc ccc aaa gca tca gcg aag ttc tac gcc tct gtg gtc
S D P S L P R I P K A S A K F Y A S V V
cga tgc aat ggc ttc cct gac ccc gct aca ggg cct cac gct tgt ctc cac cag cca gat
R C N G F P D P A T G P H A C L H Q P D
gct gga ccc acc atc agc ccc gtg aga cag gag gag gtg cag ttc ctg ggg cta atg ctc
A G P T I S P V R Q E E V Q F L G L M L
ggc acc aca gaa gca cag aca gct ttg tac gtt ctc ttt tct ctt gtg ctt ctt gga gtc
G T T E A Q T A L Y V L F S L V L L G V
tgt ggc ttg gca ttt ctg tca tac aag tac tgc aag cgc tct aag caa ggg aaa aca caa
C G L A F L S Y K Y C K R S K Q G K T Q
cga agc caa cag gaa ttg agc cgg gtg tct tca ttc tga tga gtt acc acc tca agt tct
R S Q Q E L S P V S S F - - V T T S S S
atg aag cag gcc tag ttt ctt cat cta tgt tta ccg gcc acc aac ctt agg gtc tta
M K Q A - F L H L C L P A T K H L R V L
gac tct gct gat act gga ctt ctc cat aaa gtc ctg ctg cac cgt tag aga tga ctt taa
D S A D T G L L H K V L L H R - R - L -
tct tga atg att tcg act tgc tga gta aaa tgg aaa tat ctc cat ctt gct cca gta tca
S - M I S T C - V K W K Y L H L A P V S
gag ttc att tgg gca ttt gag aag caa gta gct ctt gcg gaa acg tgt aga tac tgg tct
E F I W A F E K Q V A L A E T C R Y W S
agt ggg tct gtg aac cac tta att gaa ctt aac agg gct gtt tta agt ttc aga gtt gtt
S G S V N H L I E L N R A V L S F R V V
aag ggt tgt taa ggg agc aaa aac cgt aaa aat cct tcc tat aag aag aaa tca act cca
K G C - G S K N R K N P S Y K K K S T P
ttg cat aga ctg caa tat cat ctc ctg ccc ttc tgc aag ctc tcc cta gct tca cat ctt
L H R L Q Y H L L P F C K L S L A S H L
gtg ttt tcc aga aaa taa aaa cag cag act gtc
V F S R K - K Q Q T V

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5'3' Frame 2

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aaca gtt cct aga aaa tgg agc tgt ctt ggc atg tag tct tta ttg ccc tgc taa gtt ttt
T V P R K W S C L G M - S L L P C - V F
cat gct ggg ggt cag act ggg agt ctg ata gaa att tca ttt cca ccg ctg gtc ctc taa
H A G G Q T G S L I E I S F P P L V L -
cca atg act tgc tgc aca acc tga gtg gtc tcc tgg gag acc aga gtt cta act ttg tag
P M T C C T T - V V S W E T R V L T L -
cag ggg aca aag aca tgt atg ttt gtc acc agc cac tgc cca ctt tcc tgc cag aat act
Q G T K T C M F V T S H C P L S C Q N T
tca gca gtc tcc atg cca gtc aga tca ccc att ata agg tat ttc tgt cat ggg cac agc
S A V S M P V R S P I I R Y F C H G H S
tcc tcc cag cag gaa gca ccc aga atc cag acg aga aaa cag tgc agt gct acc ggc gac
S S Q Q E A P R I Q T R K Q C S A T G D
tcc tca agg ccc tca aga ctg cac ggc ttc agc cca tgg tca tcc tgc acc acc aga ccc
S S R P S R L H G F S P W S S C T T R P
tcc ctg cca gca ccc tcc gga gaa ccg aag cct ttg ctg acc tct tcg ccg act atg cca
S L P A P S G E P K P L L T S S P T M P
cat tcg cct tcc act cct tcg ggg acc tag ttg gga tct ggt tca cct tca gtg act tgg
H S P S T P S G T - L G S G S P S V T W
agg aag tga tca agg agc ttc ccc acc agg aat caa gag cgt cac aac tcc aga ccc tca
R K - S R S F P T R N Q E R H N S R P S
gtg atg ccc aca gaa aag cct atg aga ttt acc acg aaa gct atg ctt ttc agg gcg gaa
V M P T E K P M R F T T K A M L F R A E
aac tct ctg ttg tcc tgc gag ctg aag ata tcc ccg agc tcc tgc tag aac cac cca tat
N S L L S C E L K I S R S S C - N H P Y
ctg cgc ttg ccc agg aca cgg tcg att tcc tct ctc ttg att tgt ctt atg aat gcc aaa
L R L P R T R S I S S L L I C L M N A K
atg agg caa gtc tgc ggc aga agc tga gta aat tgc aga cca ttg agc caa aag tga aag
M R Q V C G R S - V N C R P L S Q K - K

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ttt tca tct tca acc taa aac tcc cag act gcc cct cca cca tga aga acc cag cca gtc
F S S S T - N S Q T A P P - R T Q P V
tgc tct tca gcc ttt ttg aag cca taa ata aag acc aag tgc tca cca ttg ggt ttg ata
C S S A F L K P - I K T K C S P L G L I
tta atg agt ttc tga gtt gtt cat caa gtt cca aga aaa gca tgt ctt gtt ctc tga ctg
L M S F - V V H Q V P R K A C L V L - L
gca gcc tgg ccc ttc agc ctg acc agc agc acc acg aga cca cgg act cct ctc ctg
A A W P F S L T S S R T T R P R T P L L
cct ctg cct atc aga gaa tct ggg aag cat ttg cca atc agt cca ggg cgg aaa ggg atg
P L P I R E S G G K H L P I S P G R K G M
cct tcc tgc agg ata ctt tcc ctg aag gct tcc tct ggg gtg cct cca cag gag cct tta
P S C R I L S L K A S S G V P P Q E P L
acg tgg aag gag gct ggg cgg agg gtg gga gag ggg tga gca tct ggg atc cac gca ggc
T W K E A G P R V G E G - A S G I H A G
ccc tga aca cca ctg agg gcc aag cga cgc tgg agg tgg cca gcg aca gtt acc aca agg
P - T P L R A K R R W R W P A T V T T R
tag cct ctg acg tgc ccc tgc ttt gcg gcc tcc ggg ctc agg tgt aca agt tct cca tct
- P L T S P C F A A S G L R C T S S P S
cct ggt ccc gga tct tcc cca tgg ggc acg gga gca gcc cca gcc tcc cag gcg ttg cct
P G P G S S P W G T G A A P A S Q A L P
act aca aca agc tga ttg aca ggc tac agg atg cgg gca tgc agc cca tgg cca cgc tgt
T T T S - L T G Y R M R A S S P W P R C
tcc act ggg acc tgc ctc agg ccc tgc agg atc atg gtg gat ggc aga atg aga gcg tgg
S T G T C L R P C R I M V D G R M R A W
tgg atg cct tcc tgg act atg cgg cct tct gct tct cca cat ttg ggg acc gtg tga agc
W M P S W T M R P S A S P H L G T V - S
tgt ggg tga cct tcc atg agc cgt ggg tga gct acg cag gct atg gca cgg gcc acc
C G - P S M S R G - A T Q A M A P A S
acc ttc cgg gca tct ctg acc cag gag tgg cct ctt tta agg tgg ctc act tgg tcc tca
T L P A S L T Q E W P L L R W L T W S S
agg ctc atg cca gaa ctt ggc acc act aca aca gcc atc atc gcc cac agc agc agg ggc
R L M P E L G T T T T A I I A H S S R G
acg tgg gca ttg tgc tga act cag act ggg cag aac ccc tgt ctc cag aga ggc ctg agg
T W A L C - T Q T G Q N P C L Q R G L R
acc tga gag cct ctg agc gct tct tgc act tca tgc tgg gct ggt ttg cac acc cgg tct
T - E P L S A S C T S C W A G L H T P S
ttg tgg atg gag act acc cag cca ccc tga gga ccc aga tcc aac aga tga aca gag agt
L W M E T T Q P P - G P R S N R - T D S
gct ccc atc ctg tgg ctc aac tcc cgg agt tca cag agg cag aga agc agc tcc tga aag
A P I L W L N S P S Q R Q R S S S - K
gct ctg ctg att ttc tgg gtc tgt cgc att aca cct ccc gcc tca tca gca acg ccc cac
A L L I F W V C R I T P P A S S A T P H
aaa aca cct gca tcc cta gct atg ata cca ttg gag gct tct ccc aac acg tga acc atg
K T P A S L A M I P L E A S P N T - T M
tgt ggc ccc aga cct cat cct gga ttc gtg tgg tgc cct ggg gga taa gga ggc tgt
C G P R P H P L G F V W C P G G - G G C
tgc agt ttg tat ccc tgg aat aca caa gag gaa aag ttc caa tat acc ttg cgg gga atg
C S L Y P W N T Q E E K F Q Y T L P G M
gca tgc cca tag ggg aaa gtg aaa atc tct ttg atg att cct taa gag tag act act tca
A C P - G K V K I S L M I P - E - T T S
atc aat ata tca atg agg tgc tca agg cta tca agg aag act ctg tgg atg ttc gtt cct
I N I S M R C S R L S R K T L W M F V P
aca ttg ctc gtt ccc tca ttg atg gct tgc aag gcc ctt ctg gtt aca gcc agc ggt ttg
T L L V P S L M A S K A L L V T A S G L
gcc tgc acc acg tca act tca ggc aca gca gca agt caa gga ctc cca gga aat ctg cct
A C T T S T S A T A A S Q G L P G N L P
act ttt tca cta gca tca tag aaa aga acg gtt tcc tca cca agg ggg caa aaa gac tgc
T F S L A S - K R T V S S P R G Q K D C
tac cac cta ata cag taa acc tcc cct cca aag tca gag cct tca ctt ttc cat ctg agg
Y H L I Q - T S P P K S E P S L F H L R
tgc cct cca agg cta aag tct ttt ggg aaa agt tct cca gcc aac cca agt tgc aaa gag
C P P R L K S F G K S S P A N P S S K E
att tgt tct acc acg gga cgt ttc ggg atg act ttc tgt ggg cgg tgt cct ctt cgg ctt
I C S T T G R F G M T F C G A C P L P L
atc aga ttg aag gcg cgt ggg atg cgg atg gca aag gcc cca gca tct ggg ata act tta
I R L K A R G M P M A K A P A S G I T L
ccc aca cac cag gga gca atg tga aag aca atg cca ctg gag aca tgc cct gtg aca gct
P T H Q G A M - K T M P L E T S P V T A
atc acc agc tgg atg cgg atc tga ata tgc tcc gag ctt tga agg tga agg cct acc gct
I T S W M P I - I C S E L - R - R P T A
tct cta cct cct ggt ctc gga tct tcc caa ctg gga gaa aca gct cta tca aca gtc atg
S L S P G L G F S Q L G E T A L S T V M
ggg ttg att att aca aca ggc tga tca atg gct tgg tgg caa gca aca tct ttc cca tgg
G L I I T T G - S M A W W Q A T S F P W
tga cat tgt tcc att ggg acc tgc ccc agg ccc tcc agg ata tgc gag gct ggg aga atc

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- H C S I G T C P R P S R I S E A G R I
ctg cct tga ttg act tgt ttg aca gct acg cag act ttt gtt tcc aga cct ttg ttg ata
L P - L T C L T A T Q T F V S R P L V I
gag tca agt ttt gga tga ctt tta atg agc cca tgt acc tgg cat ggc tag gtt atg gct
E S S F G - L L M S P C T W H G - V M A
cag ggg aat ttc ccc cag ggg tga agg acc cag gct ggg cac cat ata gga tag ccc acg
Q G N F P Q G - R T Q A G H H I G - P T
ccg tca tca aag ccc atg cca gag tct atc aca cgt acg atg aga aat aca ggc agg agc
P S S K P M P E S I T R T M R N T G R S
aga agg ggg tca tct cgc tga gcc tca gta cac act ggg cag agc cca agt cac cag ggg
R R G S S R - A S V H T G Q S P S H Q G
tcc cca gag atg tgg aag cgc ctg acc gaa tgc tgc agt tct ccc tgg gct ggt ttg ctc
S P E M W K P L T E C C S S P W A G L L
acc cca ttt tta gaa acg gag act atc ctg aca cca tga agt gga aag tgg gga aca gga
T P F L E T E T I L T P - S G K W G T G
gtg aac tgc agc act tag cca cct ccc gcc tgc caa gct tca ctg agg aag aga aga ggt
V N C S T - P P P A C Q A S L R K R R G
tca tca ggg cga cgg ccg acg tct tct gcc tca aca cgt act act cca gaa tgc tgc agc
S S G R R P T S S A S T R T T P E S C S
aca aaa cac cca ggc taa acc cac cct cct acg aag acg acc agg aga tgg ctg agg agg
T K H P G - T H P P T K T T R R W L R R
agg acc ctt cgt ggc ctt cca cgg caa tga aca gag ctg cgc cct ggg gga cgc gaa ggc
R T L R G L P R Q - T E L R P G G R E G
tgc tga act gga tca agg aag agt atg gtg aca tcc cca ttt aca tca cgc aaa acg gag
C - T G S R K S M V T S P F T S P K T E
tgg ggc tga cca atc cga aca cgg agg ata ctg ata gga tat ttt acc aca aaa cct aca
W G - P I R T R R I L I G Y F T T K P T
tca atg agg ctt tga aag cct aca ggc tgc atg gta tag acc ttc gag ggt atg tgc cct
S M R L - K P T G S M V - T F E G M S P
ggt ctc tga tgg aca act ttg agt ggc taa atg gct aca cgg tca agt ttg gac tgt acc
G L - W T T L S G - M A T R S S L D C T
atg ttg att tca aca aca cga aca ggc ctc gca cag caa gag cct cgc cca ggt act aca
M L I S T T R T G L A Q Q E P P P G T T
cag agg tca tta cca aca acg gca tgc cac tgg cca ggg agg atg agt ttc tgt acg gac
Q R S L P T T A C H W P G R M S F C T D
ggt ttc ctg agg gct tca tct gga gtg cag ctt ctg ctg cat atc aga ttg aag gtg cgt
G F L R A S S G V Q L L L H I R L K V R
gga gag cag atg gca aag gac tca gca ttt ggg aca cgt ttt ctc aca cac cac tga ggg
G E Q M A K D S A F G T R F L T H H - G
ttg aga acg atg cca ttg gag acg tgg cct gtg aca gtt atc aca aga ttg ctg agg atc
L R T M P L E T W P V T V I T R L L R I
tgg tca ccc tgc aga acc tgg ggc tgt ccc act acc gtt ttt cca tct cct ggt ctc gca
W S P C R T W A C P T T V F P S P G L A
tcc tcc ctg atg gaa cca cca ggt aca tca atg aag cgg gcc tga act act acg tga ggc
S S L M E P P G T S M K R A - T T T - G
tca tgc ata cac tgc tgg ccg cca gca tcc agc ccc agg tga cca ttt acc act ggg acc
S S I H C W P P A S S P R - P F T T G T
tac cac aga cgc tcc aag atg tag gag gct ggg aga atg aga cca tgc tgc agc ggt tta
Y H R R S K M - E A G R M R P S C S G L
agg agt atg cag atg tgc tct tcc aga ggc tgg gag aca agg tga agt ttt gga tca cgc
R S M Q M C S S R G W E T R - S F G S R
tga atg agc cct ttg tca ttg ctt acc agg gct atg gct acg gaa cag cag ctc cag gag
- M S P L S L L T R A M A T E Q Q L Q E
tct cca ata ggc ctg gca ctg ccc cct aca ttg ttg gcc aca atc taa taa agg ctc atg
S P I G L A L P P T L L A T I - - R L M
ctg agg cct ggc atc tgt aca acg atg tgt acc ggc cca gtc aag gtg ggc tga ttt cca
L R P G I C T T M C T A P V K V A - F P
tca cca tca gca gtg act ggg ctg aac cca gag atc cct cta acc agg agg atg tgg agg
S P S A V T G L N P E I P L T R R M W R
cag cca gga gat atg ttc agt tca tgg gag gct ggt ttg cac atc cta ttt tca aga atg
Q P G D M F S S W E A G L H I L F S R M
gag att aca atg agg tga tga aga cgc gga tcc gtg aca gga gct tgg ctg cag gcc tca
E I T M R - - R R G S V T G A W L Q A S
aca agt ctc ggc tgc cag aat tta cag aga gtg aga aga gga gga tca acg gca cct atg
T S L G C Q N L Q R V R R G G S T A P M
act ttt ttg ggt tca atc act aca cca ctg tcc tgc cct aca acc tca act atg cca ctg
T F L G S I T T P L S S P T T S T M P L
cca tct ctt ttg atg cag aca gag gag ttg ctt cca tgc cag atc gct cgt ggc cag
P S L L L M Q T E E L L P S Q I A R G Q
act ctg gct cct tct ggc tga aga tga cgc ctt ttg gct tca gga gga tcc tga act ggt
T L A P S G - R - R L L A S G G S - T G
taa agg agg aat aca atg acc ctc caa ttt atg tca cag aga atg gag tgt ccc agc ggc
- R R N T M T L Q F M S Q R M E C P S G
aag aaa cag acc tca atg aca ctg caa gga tct act acc ttc gga ctt aca tca atg agg
K K Q T S M T L Q G S T T F G L T S M R

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ccc tca aag ctg tgc agg aca agg tgg acc ttc gag gat aca cag ttt gga gtg cga tgg
P S K L C R T R W T F E D T Q F G V R W
aca att ttg agt ggg cca cag gct ttt cag aga gat ttg gtc tgc att ttg tga act aca
T I L S G P Q A F Q R D L V C I L - T T
gtg acc ctt ctg tgc caa gga tcc cca aag cat cag cga agt tct acg cct ctg tgg tcc
V T L L C Q G S P K H Q R S S T P L W S
gat gca atg gct tcc ctg acc cgg cta cag ggc ctc acg ctt gtc tcc acc agc cag atg
D A M A S L T P L Q G L T L V S T S Q M
ctg gac cca cca tca gcc cgg tga gac agg agg agg tgc agt tcc tgg ggc taa tgc tgc
L D P P S A P - D R R R C S S W G - C S
gca cca cag aag cac aga cag ctt tgt acg ttc tct ttt ctg tgc ttc ttg gag tct
A P Q K H R L C T F S F L L C F L E S
gtg gct tgg cat ttc tgt cat aca agt act gca agc gct cta agc aag gga aaa cac aac
V A W H F C H T S T A S A L S K G K H N
gaa gcc aac agg aat tga gcc cgg tgt ctt cat tct gat gag tta cca cct caa gtt cta
E A N R N - A R C L H S D E L P P Q V L
tga agc agg cct agt ttc atc tat gtt tac cgg cca cca aac acc tta ggg tct tag
- S R P S F F I Y V Y R P P N T L G S -
act ctg ctg ata ctg gac ttc tcc ata aag tcc tgc tgc acc gtt aga gat gac ttt aat
T L L I L D F S I K S C C T V R D D F N
ctt gaa tga ttt cga ctt gct gag taa aat gga aat atc tcc atc ttg ctg cag tat cag
L E - F R L A E - N G N I S I L L Q Y Q
agt tca ttt ggg cat ttg aga agc aag tag ctc ttg cgg aaa cgt gta gat act ggt cta
S S F G H L R S K - L L R K R V D T G L
gtg ggt ctg tga acc act taa ttc aac tta aca ggg ctg ttt taa gtt tca gag ttg tta
V G L - T T - L N L T G L F - V S E L L
agg gtt gtt aag gga gca aaa acc gta aaa atc ctt cct ata aga aat caa ctc cat
R V V K G A K T V K I L P I R R N Q L H
tgc ata gac tgc aat atc atc tcc tgc cct tct gca agc tct ccc tag ctt cac atc ttg
C I D C N I I S C P S A S S P - L H I L
tgt ttt cca gaa aat aaa aac agc aga ctg
C F P E N K N S R L

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5'3' Frame 3

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aacag ttc cta gaa aat gga gct gtc ttg gca tgt agt ctt tat tgc cct gct aag ttt ttc
Q F L E N G A V L A C S L Y C P A K F F
atg ctg ggg gtc aga ctg gga gtc tga tag aaa ttt cat ttc cac cgc tgg tcc tct aac
M L G V R L G V - - K F H F H R W S S N
caa tga ctt gct gca caa cct gag tgg tct cct ggg aga cca gag ttc taa ctt tgt agc
Q - L A A Q P E W S P G R P E F - L C S
agg gga caa aga cat gta tgt ttg tca cca gcc act gcc cac ttt cct gcc aga ata ctt
R G Q R H V C L S P A T A H F P A R I L
cag cag tct cca tgc cag tca gat cac cca tta taa ggt att tct gtc atg ggc aca gct
Q Q S P C Q S D H P L - G I S V M G T A
cct ccc agc agg aag cac cca gaa tcc aga cga gaa aac agt gca gtg cta ccg gcg act
P P S R K H P E S R R E N S A V L P A T
cct caa ggc cct caa gac tgc acg gct tca gcc cat ggt cat cct gca cca cca gac cct
P Q G P Q D C T A S A H G H P A P P D P
ccc tgc cag cac cct ccg gag aac cga agc ctt tgc tga cct ctt cgc cga cta tgc cac
P C Q H P P E N R S L C - P L R R L C H
att cgc ctt cca ctc ctt cgg gga cct agt tgg gat ctg gtt cac ctt cag tga ctt gga
I R L P L L R G P S W D L V H L Q - L G
gga agt gat caa gga gct tcc cca gga atc aag agc gtc aca act cca gac cct cag
G S D Q G A S P P G I K S V T T P D P Q
tga tgc cca cag aaa agc cta tga gat tta cca cga aag cta tgc ttt tca ggg cgg aaa
- C P Q K S L - D L P R K L C F S G R K
act ctc tgt tgt cct gcg agc tga aga tat ccc gga gct cct gct aga acc acc cat atc
T L C C P A S - R Y P G A P A R T T H I
tgc gct tgc cca gga cac ggt cga ttt cct ctc tct tga ttt gtc tta tga atg cca aaa
C A C P G H G R F P L S - F V L - M P K
tga ggc aag tct gcg gca gaa gct gag taa att gca gac cat tga gcc aaa agt gaa agt
- G K S A A E A E - I A D H - A K S E S
ttt cat ctt caa cct aaa act ccc aga ctg ccc ctc cac cat gaa gaa ccc agc cag tct
F H L Q P K T P R L P L H E E P S Q S
gct ctt cag cct ttt tga agc cat aaa taa aga cca agt gct cac cat tgg gtt tga tat
A L Q P F - S H K - R P S A H H W V - Y
taa tga gtt tct gag ttg ttc atc aag ttc caa gaa aag cat gtc ttg ttc tct gac tgg
- - V S E L F I K F Q E K H V L F S D W
cag cct ggc cct tca gcc tga gca gca gga cca cga gac cac gga ctc ctc tcc tgc
Q P G P S A - P A A G P R D H G L L S C
ctc tgc cta tca gag aat ctg gga agc att tgc caa tca gtc cag ggc gga aag gga tgc
L C L S E N L G S I C Q S V Q G G K G C
ctt cct gca gga tac ttt ccc tga agg ctt cct ctg ggg tgc ctc hac agg agc ctt taa
L P A G Y F P - R L P L G C L H R S L -
cgt gga agg agg ctg ggc cga ggg tgg gag agg ggt gag cat ctg gga tcc acg cag gcc

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R G R R L G R G W E R G E H L G S T Q A
cct gaa cac cac tga ggg cca agc gac gct gga ggt ggc cag cga cag tta cca caa ggt
P E H H - G P S D A G G G Q R Q L P Q G
agc ctc tga cgt cgc cct gct ttg cgg cct ccg ggc tca ggt gta caa gtt ctc cat ctc
S L - R R P A L R P P G S G V Q V L H L
ctg gtc ccg gat ctt ccc cat ggg gca cgg gag cag ccc cag cct ccc agg cgt tgc cta
L V P D L P H G A R E Q P P P R R C L
cta caa caa gct gat tga cag gct aca gga tgc ggg cat cga gcc cat ggc cac gct gtt
L Q Q A D - Q A T G C G H R A H G H A V
cca ctg gga cct gcc tca ggc cct gca gga tca tgg tgg atg gca gaa tga gag cgt ggt
P L G P A S G P A G S W W M A E - E R G
gga tgc ctt cct gga cta tgc ggc ctt ctg ctt ctc cac att tgg gga ccg tgt gaa gct
G C L P G L C G L L L L H I W G P C E A
gtg ggt gac ctt cca tga gcc gtg ggt gat gag cta cgc agg cta tgg cac cgg cca gca
V G D L P - A V G D E L R R L W H R P A
ccc tcc cgg cat ctc tga ccc agg agt ggc ctc ttt taa ggt ggc tca ctt ggt cct caa
P S R H L - P R S G L F - G G S L G P Q
ggc tca tgc cag aac ttg gca cca cta caa cag cca tca tgc ccc aca gca gca ggg gca
G S C Q N L A P L Q Q P S S P T A A G A
cgt ggg cat tgt gct gaa ctc aga ctg ggc aga acc cct gtc tcc aga gag gcc tga gga
R G H C A E L R L G R T P V S R E A - G
cct gag agc ctc tga gcg ctt ctt gca ctt cat gct ggg ctg gtt tgc aca ccc cgt ctt
P E S L - A L L A L H A G L V C T P R L
tgt gga tgg aga cta ccc agc cac cct gag gac cca gat cca aca gat gaa cag aca gtg
C G W R L P S H P E D P D P T D E Q T V
ctc cca tcc tgt ggc tca act ccc cga gtt cac aga ggc aga gaa gca gct cct gaa agg
L P S C G S T P R R V H R G R E A A P E R
ctc tgc tga ttt tct ggg tct gtc gca tta cac ctc ccg cct cat cag caa cgc ccc aca
L C - F S G S V A L H L P P H Q Q R P T
aaa cac ctg cat ccc tag cta tga tac cat tgg agg ctt ctc cca aca cgt gaa cca tgt
K H L H P - L - Y H W R L L P T R E P C
gtg gcc cca gac ctc atc ctc ttg gat tgc ggt gcc ctg ggg gat aag gag gct gtt
V A P D L I L L D S C G A L G D K E A V
gca gtt tgt atc cct gga ata cac aag agg aaa agt tcc aat ata cct tgc cgg gaa tgg
A V C I P G I H K R K S S N I P C R E W
cat gcc cat agg gga aag tga aaa tct ctt tga tga ttc ctt aag agt aga cta ctt caa
H A H R G K - K S L - - F L K S R L L C Q
tca ata tat caa tga ggt gct caa ggc tat caa gga aga ctc tgt gga tgt tgc ttc cta
S I Y Q - G A Q G Y Q G R L C G C S F L
cat tgc tgc ttc cct cat tga tgg ctt cga agg ccc ttc tgg tta cag cca gcg gtt tgg
H C S F P H - W L R R P F W L Q P A V W
cct gca cca cgt caa ctt cag cga cag cag caa gtc aag gac tcc cag gaa atc tgc cta
P A P R Q L Q R Q Q V K D S Q E I C L
ctt ttt cac tag cat cat aga aaa gaa cgg ttt cct cac caa ggg ggc aaa aag act gct
L F H - H H R K E R F P H Q G G K K T A
acc acc taa tac agt aaa cct ccc ctc caa agt cag agc ctt cac ttt tcc atc tga ggt
T T - Y S K P P L Q S Q S L H F S I - G
gcc ctc caa ggc taa agt cgt ttg gga aaa gtt ctc cag cca acc caa gtt cga aag aga
A L Q G - S R L G K V L Q P T Q V R K R
ttt gtt cta cca cgg gac gtt tgc gga tga ctt tct gtg ggg cgt gtc ctc ttc cgc tta
F V L P R D V S G - L S V G R V L F R L
tca gat tga agg cgc gtg gga tgc cga tgg caa agg ccc cag cat ctg gga taa ctt tac
S D - R R V G C R W Q R P Q H L G - L Y
cca cac acc agg gag caa tgt gaa aga caa tgc cac tgg aga cat cgc ctg tga cag cta
P H T R E Q C E R Q C H W R H R L - Q L
tca cca gct gga tgc cga tct gaa tat gct ccg agc ttt gaa ggt gaa ggc cta ccg ctt
S P A G C R S E Y A P S F E G E G L P L
ctc tat ctc ctg gtc tgc gat ttt ccc aac tgg gag aaa cag ctc tat caa cag tca tgg
L Y L L V S D F P N W E K Q L Y Q Q S W
ggt tga tta tta caa cag gct gat caa tgg ctt ggt ggc aag caa cat ctt tcc cat ggt
G - L L Q Q A D Q W L G G K Q H L S H G
gac att gtt cca ttg gga cct gcc cca ggc cct cca gga tat cgg agg ctg gga gaa tcc
D I V P L G P A P G P P G Y R R L G E S
tgc ctt gat tga ctt gtt tga cag cta cgc aga ctt ttg ttt cca gac ctt tgg tga tag
C L D - L V - Q L R R L L F P D L W - -
agt caa gtt ttg gat gac ttt taa tga gcc cat gta cct ggc atg gct agg tta tgg ctc
S Q V L D D F - - A H V P G M A R L W L
agg gga att tcc ccc agg ggt gaa gga ccc agg ctg ggc acc ata tag gat agc cca cgc
R G I S P R G E G P R L G T I - D S P R
cgt cat caa agc cca tgc cag agt cta tca cac gta cga tga gaa ata cag gca gga gca
R H Q S P C Q S L S H V R - E I Q A G A
gaa ggg ggt cat ctc gct gag cct cag tac aca ctg ggc aga gcc caa gtc acc agg ggt
E G G H L A E P Q Y T L G R A Q V T R G
ccc cag aga tgt gga agc cgc tga ccg aat gct gca gtt ctc cct ggg ctg gtt tgc tca
P Q R C G S R - P N A A V L P G L V C S

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ccc cat ttt tag aaa cgg aga cta tcc tga cac cat gaa gtg gaa agt ggg gaa cag gag
P H F - K R R L S - H H E V E S G E Q E
tga act gca gca ctt agc cac ctc cgg cct gcc aag ctt cac tga gga aga gaa gag gtt
- T A A L S H L P P A K L H - G R E E V
cat cag ggc gac ggc cga cgt ctt ctg cct caa cac gta cta ctc cag aat cgt gca gca
H Q G D G R R L L P Q H V L L Q N R A A
caa aac acc cag gct aaa ccc acc ctc cta cga aga cga cca gga gat ggc tga gga gga
Q N T Q A K P T L L R R R P G D G - G G
gga ccc ttc gtg gcc ttc cac ggc aat gaa cag agc tgc gcc ctg ggg gac gcg aag gct
G P F V A F H G N E Q S C A L G G D A K A
gct gaa ctg gat caa gga aga gta tgg tga cat ccc cat tta cat cac cga aaa cgg agt
A E L D Q G R V W - H P H L H H R K R S
ggg gct gac caa tcc gaa cac gga gga tac tga tag gat att tta cca caa aac cta cat
G A D Q S E H G G Y - - D I L P Q N L H
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Q - G F E S L Q A R W Y R P S R V C R L
gtc tct gat gga caa ctt tga gtg gct aaa tgg cta cac ggt caa gtt tgg act gta cca
V S D G Q L - V A K W L H G Q V W T V P
tgt tga ttt caa caa cac gaa cag gcc tgc cac agc aag agc ctc cgc cag gta cta cac
C - F Q Q H E Q A S H S K S L R Q V L H
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V S - G L H L E C S F C C I S D - R C V
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- E R C H W R R G L - Q L S Q D C - G S
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P P - W N H Q V H Q - S G P E L L R E A
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H R Y T A G R Q H P A P G D H L P L G P
acc aca gac gct cca aga tgt agg agg ctg gga gaa tga gac cat cgt gca gcg gtt taa
T T D A P R C R R L G E - D H R A A V -
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G V C R C A L P E A G R Q G E V L D H A
gaa tga gcc ctt tgt cat tgc tta cca ggg cta tgg cta cgg aac agc agc tcc agg agt
E - A L C H C L P G L W L R N S S S R S
ctc caa tag gcc tgg cac tgc ccc cta cat tgt tgg cca caa tct aat aaa ggc tca tgc
L Q - A W H C P L H C W P Q S N K G S C
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- G L A S V Q R C V P R Q S R W R D F H
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R L Q - G D E D A D P - Q E L G C R P Q
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Q V S A A R I Y R E - E E E D Q R H L -
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L F W V Q S L H H C P R L Q P Q L C H C
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L W L L L A E D D A F W L Q E D P E L V
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K G G I Q - P S N L C H R E W S V P A G
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R N R P Q - H C K D L L P S D L H Q - G
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M Q W L P - P R Y R A S R L S P P A R C
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W T H H Q P R E T G G G A V P G A N A R
cac cac aga agc aca gac agc ttt gta cgt tct ctt ttc tgt gct tct tgg agt ctg
H H R S T D S F V R S L F S C A S W S L
tgg ctt ggc att tct gtc ata caa gta ctg caa gcg ctc taa gca agg gaa aac aca acg

```

```

W L G I S V I Q V L Q A L - A R E N T T
aag cca aca gga att gag ccc ggt gtc ttc att ctg atg agt tac cac ctc aag ttc tat
K P T G I E P G V F I L M S Y H L K F Y
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L C - Y W T S P - S P A A P L E M T L I
ttg aat gat ttc gac ttg ctg agt aaa atg gaa ata tct cca tct tgc tcc agt atc aga
L N D F D L L S K M E I S P S C S S I R
gtt cat ttg ggc att tga gaa gca agt agc tct tgc gga aac gtg tag ata ctg gtc tag
V H L G I - E A S S S C G N V - I L V -
tgg gtc tgt gaa cca ctt aat tga act taa cag ggc tgt ttt aag ttt cag agt tgt taa
W V C E P L N - T - Q G C F K F Q S C -
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G L L R E Q K P - K S F L - E E I N S I
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A - T A I S S P A L L Q A L P S F T S C
gtt ttc cag aaa ata aaa aca gca gac tgt
V F Q K I K T A D C

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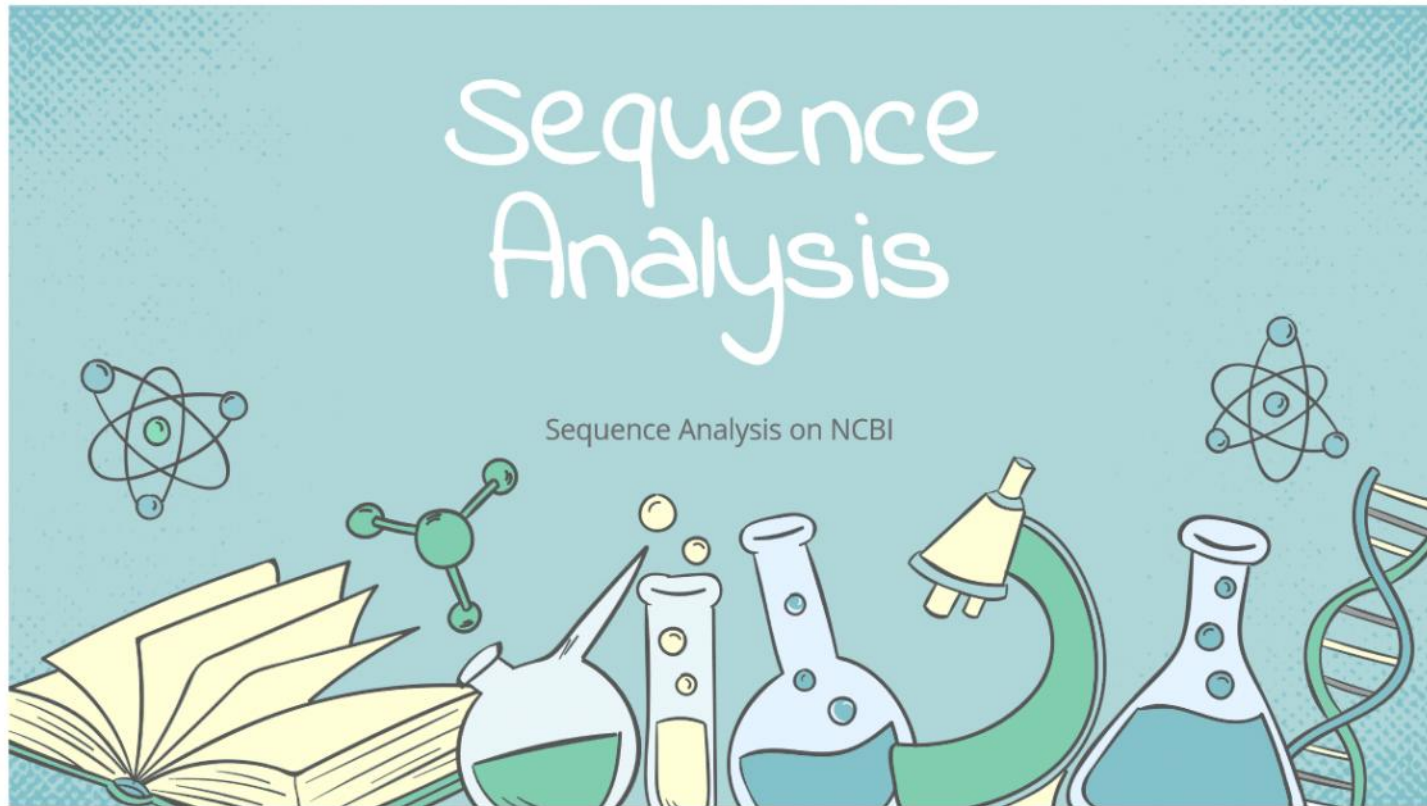
[Back to the top](#)

- Sequence Analysis on NCBI

Saturday, April 17, 2021 12:29 PM



Sequence
Analysis o...



Content

- Basic Description for Sequence Analysis.
- Objectives of Sequence Analysis.
- Possible Input & Output.
- Purpose of Sequence Analysis.
- Conclusion regarding the main topic.



11

NCBI Develop and maintain molecular and bibliographic databases for searching, retrieval and analysis of various sequences and provide web access point for data and software.”

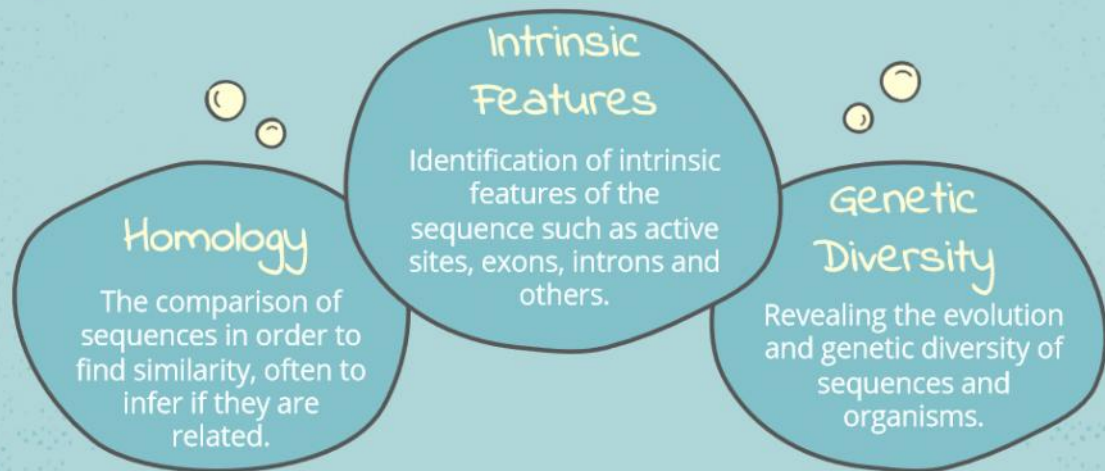


Introduction

NCBI introduces various Sequence databases for analysis of nucleotide, protein sequences or molecular structures. The most common database for this purpose is GenBank, which provides sequence and annotation information for efficient analysis.



why Sequence Analysis?



Input & Output

Input

The database requires to search the query through the name, keyword or accession number in the relevant source database.



Output

Different sort of databases contains different data. It provides the query results in the relevant source databases along with the sequence information in FASTA or GenBank formats.

Sequence Analysis on NCBI



GenPept/GenBank & Fasta Formats

NCBI provides sequence analysis through GenBank/GenPept pages for DNA/Protein sequences as well as in Fasta format.



Information in GenBank/GenPept Formats

These formats contains various information in three sections as Header, Features & Sequence.



Information in FASTA Format

Fasta Format contains one line description of the query as well as the sequence.

Sequence Annotation

The feature section in GenBank format provide the sequence annotation information as:



Source

Provides the source
of query sequence

Gene

Locate the position
of the genes in
sequence

Exon

Position of exons in
the gene

CDS

CDS section provides
the position of coding
DNA

Conclusion



NCBI introduces an efficient retrieval system to retrieve sequences for query organisms and provides various sort of sequence analysis and annotation information.

- Table Browser

Monday, June 21, 2021 7:41 PM

<https://genome.ucsc.edu/cgi-bin/hgTables>

Choose Service | BioCode x | BioCode Learning Dashboard x | Table Browser x | GTF File Extension - What is x | You job 1624324292 is fin x | +

genome.ucsc.edu/cgi-bin/hgTables

Apps | Genomes | Genome Browser | Tools | Mirrors | Downloads | My Data | Projects | Help | About Us

Table Browser

Use this tool to retrieve and export data from the Genome Browser annotation track database. You can limit retrieval based on data attributes and intersect or merge with data from another track, or retrieve DNA sequence covered by a track. [More...](#)

Select dataset

clade: genome: assembly:

group: track:

table: [describe table schema](#)

Define region of interest

region: ☐ genome ☒ position [lookup](#) [define regions](#)

identifiers (names/accessions): [upload list](#)

Optional: Subset, combine, compare with another track

filter: [create](#)

intersection: [create](#)

Retrieve and display data

output format:

Note: Table Browser GTF files contain transcripts, but no gene identifiers or symbols. If you are looking for fully formatted gene model files for use in genome analysis pipelines, check the [bigZinc/genes](#) directory on our download server.

Send output to ☐ Galaxy ☐ GREAT

output filename: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

[get output](#) [summary/statistics](#)

uniprot-proteo...fasta ^ uniprot-proteo...fasta ^ [Show All](#) x